

PROJECTIVE POLYNOMIAL OBSERVATIONS: FINITE SPECTRAL CALIBRATION AND DIVISORIAL ENTRANCES

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ABSTRACT. We study unordered polynomial spectra observed through unknown stochastic channels and a scalar normalization. For the full-rank coprime binary model of fixed degree, we prove that a finite collection of scalar constrained observation costs determines the complete leading spectral fibre, with all channel and mixture parameters free. The costs recover a positive definite quotient form, and the multiplicity pattern decodes its cone section into a joint root set. At degree d , at most $d(2d+1)$ scalar calibrations suffice; perturbations of those values give explicit Hausdorff and diameter bounds. For an isolated Nash remote observation germ, we identify the entrance exponent as the maximal residual-to-time order ratio over accessible real divisors and identify its leading coefficient by a resolved metric-unit maximum. A Puiseux residue theorem relates the real weighted fibre to coordinate real tropicalization, while finite weighted jets imply filtered naturality without an assumed convergence of conjugates. The complete multiplicity law, rank-deficient cubic fibre, inverse through channel-rank loss, weight wall, endpoint wall, and their exact critical-shell calculations are retained. The general semialgebraic atlas and entrance asymptotics remain available beyond the explicit binary normal forms.

1. INTRODUCTION

A normalized polynomial experiment can be locally identifying and yet poorly resolve its roots. It can also admit a remote exact representative which enters a small observation ball while the local inverse remains coercive. These are different phenomena. The first depends on the joint geometry of root splitting and the observation metric; the second depends on contact between a moving observation and a remote image. This article determines both for stochastic polynomial observations.

Our concrete experiment is

$$K(z) = \sum_{a=1}^2 \alpha_a f_a(z) u_a v_a^\top, \quad P_j = K(T_j) / (\mathbf{1}^\top K(T_j) \mathbf{1}),$$

with monic degree- d real-rooted components, roots in $[0, D]$, strictly positive centre weights, and unknown binary stochastic channels. The target is the unordered root multiset of $f_1 f_2$. The observation radius t is weighted Hellinger distance; the spectral modulus is the *diameter of the entire feasible set* in the bottleneck metric, not only its maximal distance from the central spectrum.

Finite information determining a joint root fibre. For coprime components, invertible strictly positive channels and at least $2d+1$ exterior clocks, normalizer recovery and a matrix-pencil inverse control all coefficient and stochastic errors by observation error. Real-root cluster geometry then leaves only two leading regimes: an interior repeated root splits at order $\sqrt{t}$, while simple and one-sided boundary roots move at order t .

Date: September 20, 2026.

2020 Mathematics Subject Classification. 14P10, 35R30, 15A22, 62G05, 62G15.

Key words and phrases. Polynomial inverse problems, real algebraic geometry, spectral fibres, finite calibration, real tropicalization, identification walls.

Theorem 4.1 identifies a finite scalar datum which retains the joint metric information lost by scalar orders. On each marked multiplicity pattern there are $k \leq 2d$ retained sum or variance coordinates. For v in their feasible cone, constrain those coordinates to equal tv , minimize over the remaining roots and all stochastic parameters, and set

$$c_\theta(v) = \lim_{t \downarrow 0} \frac{4}{t^2} \min h_w(F_T(\xi), P_\theta)^2.$$

These limits are not formal probes of an arbitrary compact set. They are costs of actual polynomial-root competitors in the full model. We prove that $c_\theta(v) = v^\top Q_\theta v$, with $Q_\theta > 0$, and hence that the $k(k+1)/2$ values

$$c_\theta(e_i), \quad c_\theta(e_i + e_j) \quad (i < j)$$

recover the entire quotient form. The multiplicity pattern decodes $\{v : v^\top Q_\theta v \leq 4\}$ into the complete joint leading root set. Corollary 4.3 gives quantitative metric stability from errors in these finitely many scalars. This finite sufficiency result is the structural starting point of the article.

The underlying multiplicity law is preserved in full. If an interior cluster is repeated, its sharp whole-model modulus is

$$\omega_\theta(t) = C(\theta)\sqrt{t} + O_H(t), \quad C(\theta) = \max_j 2\sqrt{(m_j - 1)/m_j} \kappa_j^{-1/4}.$$

Otherwise the modulus is linear. The complete root sets, endpoint constraints, optimal coefficient orders and compact-uniform errors are in Theorem 3.2. The calibration theorem strengthens their information content without replacing the root-expansion proof by a general definability argument.

Remote geometry and the two cubic walls. The rank-deficient cubic family $f = z(z - r)^2$, $g = (z - s)^3$ has an exact fibre controlled by

$$B = (3D - r)s - 2rD, \quad E = (1 - \alpha)(r + 3s)(2r - 3s)^2 - 4\pi r^3.$$

Its identifying locus is $\{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}$. The complete classification, inverse through rank loss, oblique local Fisher geometry and both identification walls are given by the direct pencil and feasible-chart arguments below. The weight wall and endpoint wall have different nonidentified-side openings; neither is removed by passing to an abstract tangent relation. The exact equality-shell and signed second-order calculations are retained.

For a more general isolated remote germ, let $R(x, \delta) = f(x) - p(\delta)$ and $d(\delta) = \min_x \|R(x, \delta)\|$. In the nonexact Nash case, Theorem 17.1 identifies

$$\rho = \max_{E \text{ real accessible}} \frac{\text{ord}_E(R_1, \dots, R_M)}{\text{ord}_E \delta}, \quad d(\delta) = M^{-1} \delta^\rho + o(\delta^\rho).$$

Here only divisors with positive time order enter the maximum, and M is the maximum of the continuous resolved extension of $\delta^\rho / \|R\|$ on the accessible zero-time preimage. The exponent is thus a relative real contact invariant of a residual ideal and a time ideal, not an invariant of the remote image alone. The coefficient requires their observation-metric units. The formula recovers the singular contact n/m and its sharp coefficient in the existing polynomial probability example, and recovers order one and the cone-projection coefficient in the regular regime. The continuous-semialgebraic family theorem, its rational order lists, its uniform remainders and its exact threshold decisions remain in Theorem 16.1.

Which real weighted object is being measured? The general weighted relation retains real feasibility before imposing a norm and decoding roots. Theorem 13.1 identifies it with the residue of its bounded weighted points over the real algebraic Puiseux field:

$$T_w^+ S = \text{res}\{t^{-w}y : y \in S(\mathbb{R}\langle t \rangle), \text{val } y \geq w\}.$$

Its nonzero coordinate signs and valuations give the corresponding fibre of coordinate real tropicalization [19]; the residue magnitudes, observation-unit section and spectral decoder are retained before that projection. Positive real scale is distinguished from positive coordinate orthant, and the precise sufficient smooth lifting test is stated alongside the comparison with positive tropical lifting [20]. Theorem 13.3 derives the required conjugate limits from finite weighted jets of a Nash isomorphism and its inverse.

The earlier algebraic initial-fibre comparison, scalar distance-function reconstruction, sharp-marginal separation and full binary valuative separation are all retained. The distance-function identity is a general metric fact; it is not the finite model-specific theorem above. Likewise, ordinary valuation data discard positive metric units; the binary separation example makes that mechanism explicit rather than claiming that the mechanism itself is new. The residue comparison positions the object against real tropical geometry without claiming a new tropical fundamental theorem.

Organization and dependencies. The model inverse, multiplicity law, finite calibration, cubic fibre and two walls come first. Their proofs are direct. The relative real atlas, weighted comparisons and singular entrance theory follow, with the complete foundational arguments in the appendix. Definition C.1 specifies the finite complexity datum underlying all fixed-format assertions. Common monomial presentations concern a finite chosen diagram, not every scalar polynomial simultaneously. Appendix C states exactly which concrete proofs do and do not require the general atlas. All proofs and finite calculations from the preceding article are retained; its full text and historical archive also remain in the separate archival volume.

2. THE EXPERIMENT AND THE SPECTRAL METRIC

Fix $D > 0$, a degree $d \geq 1$, and $0 < \pi < 1/2$. A closed binary model point consists of two monic real polynomials f_1, f_2 of degree d , all of whose roots lie in $[0, D]$, weights $\alpha_1 + \alpha_2 = 1$ with $\alpha_a \geq \pi$, and column-stochastic matrices $U = (u_1, u_2)$ and $V = (v_1, v_2)$ with nonnegative entries. Set

$$K(z) = \sum_{a=1}^2 \alpha_a f_a(z) u_a v_a^\top, \quad q(z) = \mathbf{1}^\top K(z) \mathbf{1} = \sum_{a=1}^2 \alpha_a f_a(z), \quad P_j = \frac{K(T_j)}{q(T_j)}. \quad (2.1)$$

The clocks satisfy $D < T_1 < \dots < T_\ell$, and $w_j > 0$, $\sum_j w_j = 1$. Consequently $q(T_j) > 0$, including at every closed-model competitor. The target polynomial is $A = f_1 f_2$. The model is compact in monic coefficients and stochastic entries. Component permutations are identified. The matrices, roots, clocks and design weights are allowed to vary when a family of centres is considered, but every competitor uses its centre's fixed design.

For probability arrays define

$$h_w(P, Q)^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{Q_{je}})^2.$$

For two multisets $Z = \{z_i\}_{i=1}^n$ and $Z' = \{z'_i\}_{i=1}^n$, put

$$d_\infty(Z, Z') = \min_{\sigma \in \mathfrak{S}_n} \max_i |z_i - z'_{\sigma(i)}|.$$

For real roots this is the maximum discrepancy of their increasing orderings. All diameters in this article are diameters of sets of multisets, not maximal distances from the centre. Write

$$\omega_\theta(t) = \text{diam}\{Z(A_\xi) : h_w(F_T(\xi), P_\theta) \leq t\}. \quad (2.2)$$

A local modulus $\omega_{\theta, \text{loc}}$ restricts the competitors to a specified coefficient neighbourhood of the centre's component pair. This restriction is never implicit.

At a strictly positive datum the Fisher inner product on zero-sum arrays is

$$\langle H, H' \rangle_{I, \theta} = \sum_{j, e} \frac{w_j}{P_{\theta, je}} H_{je} H'_{je}.$$

For component increments p_a of degree below d and stochastic increments $\eta = (\dot{\alpha}_1, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2)$, where channel increments have entry sum zero, define

$$\begin{aligned} \dot{K} &= \sum_a \{ \alpha_a p_a u_a v_a^\top + f_a (\dot{\alpha}_a u_a v_a^\top + \alpha_a \dot{u}_a v_a^\top + \alpha_a u_a \dot{v}_a^\top) \}, \\ \dot{q} &= \sum_a (\alpha_a p_a + \dot{\alpha}_a f_a), \quad S_\theta(p_1, p_2, \eta)_j = \frac{\dot{K}(T_j) - P_{\theta, j} \dot{q}(T_j)}{q(T_j)}. \end{aligned} \quad (2.3)$$

Here $\dot{\alpha}_2 = -\dot{\alpha}_1$. These rational formulas define every Fisher program below in the original experiment coordinates.

Lemma 2.1 (Normalizer recovery). *Suppose f_1, f_2 are coprime, V is invertible, the weights are positive, and $\ell \geq 2d + 1$. The observed array determines q and K uniquely among all closed-model competitors. Their coefficients are locally Lipschitz functions of the observations, with uniform constants on compact sets of such centres. No rank assumption on U is required.*

Proof. Let e have degree below d , and let H be a matrix polynomial of degree at most d satisfying $H(T_j) = e(T_j)P_j$. Interpolation of the polynomial $qH - eK$, of degree at most $2d$, gives $qH = eK$. Its second marginal, multiplied by $\text{diag}(\alpha_1, \alpha_2)^{-1}V^{-1}$, gives $q \mid ef_1$ and $q \mid ef_2$. Coprimality implies $q \mid e$, hence $e = 0$ and $H = 0$.

More explicitly, let Π_T project onto the orthogonal complement of the degree- d Vandermonde column space. The map

$$N_\theta : e \longmapsto \Pi_T(e(T_j)P_{\theta, j})_j, \quad \deg e < d,$$

is injective. For a competitor, subtraction of the normalization equations yields $\|N_\theta(q' - q)\| \leq C \max_j \|P'_j - P_j\|$; the competitor normalizer coefficients are bounded by compactness of the root interval. The smallest singular value of N_θ is positive. A bounded left inverse recovers $q' - q$, and Vandermonde interpolation recovers $K' - K$. The singular values and the coefficient bounds have positive uniform margins on compact sets of the stated centres. $\square$

Lemma 2.2 (Full-rank coefficient inverse). *Under Lemma 2.1, suppose also that U is invertible. Then every sufficiently close observation has, up to one component permutation, coefficient and stochastic parameter error bounded by a constant times its observation error. Every exact competitor has the same component polynomials and stochastic parameters. The constants are uniform on compact subsets of these centres.*

Proof. Choose two clocks such that $f_1(T_j)/f_1(T_i) \neq f_2(T_j)/f_2(T_i)$. Otherwise the nonzero polynomial $f_2(T_i)f_1 - f_1(T_i)f_2$, of degree at most d , would vanish at all $2d + 1$ clocks. The matrix $K(T_j)K(T_i)^{-1}$ has those two distinct eigenvalues and eigenvectors u_1, u_2 . Normalize the eigenvectors to have entry sum one. Simple-eigenvalue inversion is analytic in a neighbourhood of the centre. Applying the recovered U^{-1} to $K(z)$ and taking row sums recovers $\alpha_a f_a$; leading coefficients and normalized rows recover α_a, V . This construction applies to arbitrary closed-model competitors because their $K'(T_i)$ stays invertible. It also proves exact uniqueness. A finite

neighbourhood cover gives compact-uniform constants. Equivalently, the derivative in monic coefficient and stochastic coordinates has a bounded left inverse. $\square$

3. AN EXPLICIT MULTIPLICITY AND BOUNDARY ATLAS

Fix d, D, π and $\ell \geq 2d + 1$. Let $\mathfrak{M}_d$ be the centres of (2.1) with strict weights, strictly positive invertible U, V , and coprime component polynomials. No restriction is imposed on root multiplicities or on contact with $0, D$. Competitors still range over the entire closed model. Partition $\mathfrak{M}_d$ by the component assignment, multiplicity and endpoint/interior status of each distinct root. At fixed d this is a finite semialgebraic partition. On a pattern stratum write r_j, m_j for its distinct roots and multiplicities, and $a(j) \in \{1, 2\}$ for their component assignments.

For every cluster let s_j denote its first-order sum displacement. It is free at an interior root, nonnegative at 0 , and nonpositive at D . For every repeated interior cluster let $X_j \geq 0$. Define

$$p_a(z) = - \sum_{j:a(j)=a} s_j \frac{f_a(z)}{z - r_j} - \sum_{\substack{j:a(j)=a \\ r_j \in (0,D), m_j \geq 2}} X_j \frac{f_a(z)}{(z - r_j)^2}. \quad (3.1)$$

Insert these polynomials into (2.3). Let L contain the free interior sum and stochastic columns. Let D_0 contain the nonnegative endpoint sum columns (reverse the sign at D) and all X_j columns. With $W = \text{diag}(w_j/P_{je})$, define the projected Gram matrix

$$Q = D_0^\top W D_0 - D_0^\top W L (L^\top W L)^{-1} L^\top W D_0. \quad (3.2)$$

For a repeated interior cluster j , let

$$\kappa_j = \min\{v^\top Q v : v \geq 0, v_{X_j} = 1\}. \quad (3.3)$$

All matrix entries are rational in the root, channel, weight and design coordinates. Empty collections of free or cone columns have their usual zero-dimensional meanings.

Lemma 3.1 (Ordered cluster diameter). *For $m \geq 2$, the set*

$$\{y_1 \leq \dots \leq y_m : \sum_i y_i = 0, \frac{1}{2} \sum_i y_i^2 \leq X\}$$

has diameter $\sqrt{2X(m-1)/m}$ in the maximum coordinate metric.

Proof. Fix a coordinate j . Minimizing the squared norm under its value, ordering and zero sum gives

$$\max y_j = \sqrt{\frac{2X(j-1)}{m(m-j+1)}}, \quad \min y_j = -\sqrt{\frac{2X(m-j)}{mj}}.$$

For the maximum, the last $m-j+1$ coordinates are equal and the first $j-1$ are equal; the minimum is analogous. This also covers the zero endpoint extrema when $j=1$ or $j=m$. Put $a=j-1, b=m-j$. For $a, b \geq 1$, the assertion that the resulting coordinate width is at most $\sqrt{2X(m-1)/m}$ is equivalent to

$$ab(a+b+2)^2 \geq 4(a+1)(b+1).$$

Writing $S = a+b \geq 2$ and $ab \geq 1$, its difference is $ab(S^2 + 4S) - 4S - 4 \geq S^2 - 4 \geq 0$. For $a=0$ or $b=0$ equality holds directly. An extreme vector with one coordinate of absolute value $\sqrt{2X(m-1)/m}$ and the other $m-1$ coordinates equal attains the endpoint width against the zero vector. The diameter of a subset in the maximum metric is its largest coordinate width. This proves the formula, including possible simultaneous constraints on other clusters when it is used below. $\square$

Theorem 3.2 (Multiplicity and boundary laws). *Every centre in $\mathfrak{M}_d$ is exactly identifying. On each root pattern stratum the following whole-model assertions hold uniformly on compact subsets H .*

(a) *If there is a repeated interior cluster, the exponent is $1/2$. The complete leading root set consists of zero clusters at the simple and boundary roots and ordered vectors $y_j \in \mathbb{R}^{m_j}$ at the repeated interior roots satisfying*

$$\sum_i y_{j,i} = 0, \quad X_j = \frac{1}{2} \sum_i y_{j,i}^2, \quad \min_{\text{endpoint sums} \geq 0} v^\top Q v \leq 4. \quad (3.4)$$

The X_j in v are the displayed variances. After grouping and scaling roots by $\sqrt{t}$, the observation ball is within $O_H(\sqrt{t})$ of this set in Hausdorff distance. In particular

$$\omega_\theta(t) = C(\theta)\sqrt{t} + O_H(t), \quad C(\theta) = \max_j 2\sqrt{\frac{m_j - 1}{m_j}} \kappa_j^{-1/4}. \quad (3.5)$$

Every leading point, including diameter-extremizing pairs, is realized by stochastic alternatives.

(b) *If no interior root is repeated, the exponent is 1. For each simple interior cluster take a free real root increment, and for each boundary cluster take an ordered vector of increments with all entries nonnegative at 0 or nonpositive at D . Let s_j be their sums and use (3.1) without X terms. The compact set of these root increments with*

$$\min_\eta \|S_\theta(p_1, p_2, \eta)\|_{L, \theta}^2 \leq 4 \quad (3.6)$$

is the complete leading set $\mathcal{L}_{\text{lin}, \theta}$. Its diameter $C_{\text{lin}}(\theta)$ is positive, and

$$\omega_\theta(t) = C_{\text{lin}}(\theta)t + O_H(t^2).$$

The scaled-root Hausdorff error is $O_H(t)$.

(c) *The optimal coefficient orders of a cluster of multiplicity m , with coefficient weight k , are*

$$\beta_k = \begin{cases} 1, & r \in (0, D), \ k = 1, \\ k/2, & r \in (0, D), \ 2 \leq k \leq m, \\ k, & r \in \{0, D\}, \ 1 \leq k \leq m. \end{cases} \quad (3.7)$$

The real order list, boundary and multiplicity pattern, and Fisher active supports admit a common finite semialgebraic refinement. On its square-root strata C^4 is a selected rational function of the root and experiment coordinates; the positive constants on the linear strata are algebraic semialgebraic functions. Both are Nash after refinement. Exact algebraic evaluation uses finite programs.

Proof. Lemma 2.2 gives exact identification and a uniform inverse in coefficients and stochastic parameters. The partial fractions $1/(z - r_j)$ and $1/(z - r_j)^2$ occurring in (3.1) are linearly independent at distinct roots, with their indicated pole orders. The coefficient derivative, and hence its score, is injective on these variables and the free stochastic coordinates. Thus L has full column rank and Q is positive definite whenever it is nonempty. The constrained minima exist, are positive, and have uniformly bounded minimizers on H .

We first obtain coordinates for *all* nearby competitors. Coefficient recovery and analytic coprime cluster factorization give an $O_H(t)$ error in each cluster coefficient and in the stochastic parameters. For an interior cluster, write its exact roots as

$$r_j + ts_j/m_j + \sqrt{t} y_{j,i}, \quad \sum_i y_{j,i} = 0.$$

The first coefficient bounds s_j , and the second bounds $\sum_i y_{j,i}^2$, since for centred real roots their second elementary symmetric function equals $-\frac{1}{2} \sum_i y_{j,i}^2$. At an endpoint all root displacements have the same sign. Their sum is $O_H(t)$, so each is $O_H(t)$. These observations prove the upper bounds in (3.7). For the lower bounds, translate an interior cluster by a nonzero order- t amount for $k = 1$; for $k \geq 2$ split it by $\sqrt{t}$ times a zero-sum real vector with nonzero k th elementary symmetric function. The vector with $m - 1$ equal nonzero coordinates and the last their negative sum has this property for every $k \geq 2$. Its first nonzero coefficient is of order t and the coefficient inverse makes the observation distance comparable to t . At an endpoint displace all roots by the same allowed order- t amount. This proves all lower orders, including the one-sided boundary case. In root-arc parameter ε , the interior split has $\text{ord } G = 4$, $\text{ord } H_k = 2k$, whereas translation or an endpoint arc has orders $2, 2k$. These are admissible witnesses for the real order data, and (3.7) rules out any smaller candidate.

For part (a), expansion of the exact root coordinates gives precisely (3.1) at order t . Terms involving three centred root displacements are $O_H(t^{3/2})$; all higher ones and the bounded mean and endpoint terms satisfy the same bound. Consequently

$$F_T(\xi) = P_\theta + tS_\theta + O_H(t^{3/2}), \quad h_w(F_T(\xi), P_\theta) = \frac{t}{2} \|S_\theta\|_{I,\theta} + O_H(t^{3/2}). \quad (3.8)$$

This follows by Taylor expansion of the square-root probability vector and the reverse triangle inequality; it does not require a nonzero score. Membership in the ball implies $\|S_\theta\| \leq 2 + M\sqrt{t}$. Multiply all sum, variance and stochastic first-order variables by $\rho = 2/(2 + M\sqrt{t})$ and each centred root vector by $\sqrt{\rho}$. This gives a point of (3.4) within $O_H(\sqrt{t})$ of the scaled competitor. Endpoint and simple clusters contribute only $O_H(\sqrt{t})$ in these scaled coordinates.

Conversely take a point of (3.4) and bounded minimizing endpoint and free variables. Use the displayed interior root arcs; at an endpoint realize its sum by equal one-sided displacements, and use the prescribed order- t stochastic increments. Scale first-order variables by $\rho = 1 - M'\sqrt{t}$ and centred roots by $\sqrt{\rho}$. Strict weight and channel margins and cluster separation on H make these genuine closed-model alternatives. For a sufficiently large M' , (3.8) puts them inside the observation ball. This is the reverse Hausdorff inclusion, with the same uniform error.

Homogeneity gives $X_{j,\max} = 2/\sqrt{\kappa_j}$. The marginal projection of the leading set at cluster j includes every ordered zero-sum vector with squared norm at most $2X_{j,\max}$: scale a minimizing feasible vector for $X_j = 1$ and choose the desired shape in that cluster. Lemma 3.1 therefore gives its diameter $2\sqrt{(m_j - 1)/m_j \kappa_j^{-1/4}}$. The diameter in the maximum cluster metric is the largest marginal diameter; no independence of the clusters is used. Cluster separation excludes cross-cluster optimal matchings. Diameters differ by at most twice Hausdorff distance, proving (3.5).

For part (b), every root displacement is $O_H(t)$ by the same sum argument at the endpoints and the simple-root inverse in the interior. Divide the exact displacements by t . They belong to the stated polyhedral sign and ordering cone. Taylor expansion now has remainder $O_H(t^2)$. The first-order score controls their sums and all stochastic variables; on the one-sided endpoint cones the sums control the individual displacements. Hence the root set in (3.6) is compact. The radial argument with $\rho = 1 - Mt$ proves both Hausdorff inclusions with error $O_H(t)$. An allowed nonzero displacement gives positive diameter. This proves (b).

For (c), enumerate the subsets of the cone coordinates other than a fixed X_j . On a support A , put $v_A = -Q_{AA}^{-1}Q_{Aj}$ and require $v_A \geq 0$ and $Q_{bj} + Q_{bA}v_A \geq 0$ off the support. Its value is $Q_{jj} - Q_{jA}Q_{AA}^{-1}Q_{Aj}$. Convexity proves necessity and sufficiency. All tests and selected values are rational, and

$$C^4 = \max_j \frac{16(m_j - 1)^2}{m_j^2 \kappa_j}.$$

There are finitely many patterns and supports. For the linear law, (3.6) is an explicit quadratic inequality on a polyhedral root cone after projecting the free stochastic variables. Its coordinate maxima and minima, and hence its diameter, have finite real algebraic descriptions by (10.5) or by face optimization. Quantifier elimination gives the algebraic value at algebraic input. Stratify the resulting semialgebraic functions and refine with the order lists of Theorem 10.4. Positivity and compact-uniform coercivity ensure continuity within each fixed pattern; Nash refinement gives the final assertion. No continuity of C is asserted across a change of scaling exponent. $\square$

Example 3.3 (A multiplicity crossover). Let $x \in [0, b]$, $b < 1/2$, take $F(x) = (1/2 + x, 1/2 - x)$, and target $A_x(z) = (z - r)^2 - x$. At a fixed centre $a \in (0, b)$, differentiation of the square-root probability vector gives

$$h(F(a + u), F(a)) = \frac{|u|}{\sqrt{1 - 4a^2}} + O_a(u^2).$$

The admissible x interval therefore has endpoints $a \pm t\sqrt{1 - 4a^2} + O_a(t^2)$, and its spectral diameter is

$$\omega_a(t) = \frac{\sqrt{1 - 4a^2}}{\sqrt{a}} t + O_a(t^2).$$

At $a = 0$ it instead has leading law $\sqrt{t}$. More precisely, uniformly for λ in compact subsets of $[0, \infty)$,

$$\frac{\omega_{\lambda t}(t)}{\sqrt{t}} \longrightarrow \sqrt{\lambda + 1} - \sqrt{\max(0, \lambda - 1)}. \quad (3.9)$$

Indeed the feasible interval endpoints are $t \max(0, \lambda - 1) + O(t^2)$ and $t(\lambda + 1) + O(t^2)$, with truncation at zero, and $x \mapsto Z(A_x)$ has ordered roots $r \pm \sqrt{x}$. This example belongs to the general real-graph class of Section 9; it is not a claim about an unobserved restriction of the closed binary model. Here the linear constant really diverges at a multiplicity wall. Section 6 gives a different mechanism.

4. FINITE CALIBRATION OF THE JOINT SPECTRAL FIBRE

The full-rank inverse imposes a restriction on the possible leading fibres which is absent for general compact semialgebraic sets. On each multiplicity stratum a finite list of scalar, constrained observation costs determines the entire joint spectral fibre, not merely its coordinate radii. The constraints specify root sums and variances; mixture weights and both stochastic channels are minimized over.

Throughout this section the centre belongs to $\mathfrak{M}_d$ of Section 3. Distinct clusters are marked by their limiting root and component; their multiplicities and endpoint status form the pattern $\mathcal{P}$. The marking is unambiguous near the centre, up to the already identified component permutation. Fix a small closed cluster neighbourhood B_θ containing the centre in its relative interior. No stochastic coordinate is held fixed in this neighbourhood. Lemma 2.2 implies that every sufficiently small-cost competitor in the whole closed model belongs to this neighbourhood, after the component permutation.

For the roots $r_j + u_{j,i}$ in cluster j put

$$s_j = \sum_{i=1}^{m_j} u_{j,i}, \quad X_j = \frac{1}{2} \sum_{i=1}^{m_j} (u_{j,i} - s_j/m_j)^2.$$

Write $\epsilon_j = 1$ at 0 and $\epsilon_j = -1$ at D ; thus $\epsilon_j s_j \geq 0$ at an endpoint. There are two retained coordinate systems.

If an interior cluster is repeated, retain the X_j at all repeated interior clusters and the $\epsilon_j s_j$ at all endpoint clusters. Denote their number by k and their cone by $\mathcal{K} = \mathbb{R}_+^k$. All interior

sums and all stochastic increments are free variables. If no interior cluster is repeated, retain all cluster sums, with the same sign reversal at D . Then $\mathcal{K} = \mathbb{R}^{k_{\text{int}}} \times \mathbb{R}_+^{k_{\text{end}}}$, and only the stochastic increments are free. In both cases $k \leq 2d$. Let $\gamma = 1/2$ in the first case and $\gamma = 1$ in the second.

For $v \in \mathcal{K}$ and sufficiently small $t > 0$, let $B_\theta(t; v)$ be the competitors in B_θ whose retained coordinates are exactly tv . In particular, prescribing $X_j = tv_j$ does not prescribe the cluster mean or its individual root spacings. Define the scalar calibration cost

$$c_\theta(v) = \lim_{t \downarrow 0} \frac{4}{t^2} \min_{\xi \in B_\theta(t; v)} h_w(F_T(\xi), P_\theta)^2. \quad (4.1)$$

Existence, independence of a sufficiently small B_θ , and the fact that these are actual closed-model alternatives are included in the theorem below. The normalization 4 agrees with (3.8).

Insert the retained and free variables into the score (2.3), using the polynomial increments (3.1) in the repeated regime and its first-sum terms in the linear regime. Write the resulting score as $La + Dv$, with a free. Set

$$Q_\theta = D^\top W D - D^\top W L (L^\top W L)^{-1} L^\top W D, \quad W = \text{diag}(w_j / P_{\theta, j_e}). \quad (4.2)$$

Thus the score is $La + Dv$. An empty free block contributes zero. In the repeated regime this is exactly (3.2), with the same signs and the same free stochastic directions.

Theorem 4.1 (Finite scalar determination). *For every centre in $\mathfrak{M}_d$ the limit (4.1) exists and*

$$c_\theta(v) = \min_a \|La + Dv\|_{L, \theta}^2 = v^\top Q_\theta v, \quad Q_\theta > 0. \quad (4.3)$$

The convergence in (4.1) is uniform for v in compact subsets of $\mathcal{K}$ and for centres in compact subsets of a fixed pattern stratum. Its error is $O(\sqrt{t})$ in the repeated regime and $O(t)$ in the linear regime.

The finite marked scalar datum

$$\mathfrak{C}_\theta = (\mathcal{P}; c_\theta(e_i), c_\theta(e_i + e_j) : 1 \leq i < j \leq k) \quad (4.4)$$

contains $k(k+1)/2$ numbers and determines the complete leading spectral fibre, with its bottleneck metric. It also determines its joint leading cluster-polynomial coefficient image. All mixture weights and channel parameters remain unknown in these calibrations.

More explicitly,

$$(Q_\theta)_{ii} = c_\theta(e_i), \quad (Q_\theta)_{ij} = \frac{c_\theta(e_i + e_j) - c_\theta(e_i) - c_\theta(e_j)}{2}. \quad (4.5)$$

Let $\mathcal{V}_\theta = \{v \in \mathcal{K} : v^\top Q_\theta v \leq 4\}$. In the repeated regime decode v into all ordered, zero-sum vectors y_j with $\frac{1}{2} \sum_i y_{j,i}^2 = v_{X_j}$ at the repeated interior clusters, and zero vectors at every other cluster. Endpoint coordinates of v are existentially quantified. In the linear regime decode interior sums into the corresponding single root increments; at an endpoint use all ordered one-sided increments with signed sum v_j . The union of these decoded vectors is precisely the leading fibre $\mathcal{L}_\theta$ of the whole observation ball, scaled by $t^{-\gamma}$.

Proof. The coefficient inverse gives an injective derivative in monic coefficient and stochastic coordinates. Within either retained/free system the polynomial directions are independent: after division by f_a , their partial fractions have distinct simple poles, and distinct double poles exactly at the repeated interior clusters. Hence the combined score matrix (L, D) is injective. Orthogonal projection off $\text{ran } L$ shows that (4.2) is positive definite and proves the algebraic minimum in (4.3). This also bounds its minimizing free variable a uniformly on the indicated compact sets.

We next prove that this quadratic program is the limit of the actual constrained minima, not a substitute experiment. For a repeated interior cluster and a prescribed $X \geq 0$, choose an ordered

vector y with $\sum_i y_i = 0$ and $\frac{1}{2} \sum_i y_i^2 = X$. Such a vector exists for every X : for example take one entry $-(m-1)b$ and the other $m-1$ entries b , where $b^2 = 2X/(m(m-1))$. Use the exact roots

$$r_j + ta_j/m_j + \sqrt{t} y_{j,i}.$$

At a simple interior root use its order- t sum. At an endpoint realize the prescribed signed sum by equal one-sided displacements. Give the weights and channels their prescribed order- t stochastic increments. Strict interior margins for these parameters and separation of the clusters make all these alternatives feasible for small t . They satisfy the retained constraints exactly. Their observation expansion is

$$F_T(\xi) - P_\theta = t(La + Dv) + O(t^{1+\gamma}), \quad \frac{4}{t^2} h_w(F_T(\xi), P_\theta)^2 = \|La + Dv\|_{l,\theta}^2 + O(t^\gamma), \quad (4.6)$$

where in the linear case all displacements, including the individual endpoint displacements, are $O(t)$. Taking the minimizing a gives the required upper bound.

The constrained sets are closed in the compact cluster neighbourhood and are nonempty by this construction, so their minima are attained. The upper bound makes every minimizing observation error $O(t)$. Lemma 2.2 then makes all coefficient and stochastic errors $O(t)$, uniformly on compact sets of centres and retained vectors. Analytic coprime cluster factorization gives the same bound on cluster coefficients. The first coefficient bounds every free interior sum by $O(t)$. At an endpoint the displacements have one sign, so their prescribed sum controls every displacement. At a repeated interior root the retained centred variance bounds each centred displacement by $O(\sqrt{t})$. Thus every minimizer has bounded free first-order variables and satisfies (4.6), with exactly its prescribed v . Its leading score norm is no smaller than the displayed quadratic minimum. This proves the lower bound, the limit and the uniform errors. The same argument forces every minimizer into every smaller cluster neighbourhood for small t ; the germ and limit do not depend on its initial choice. No channel or weight was fixed during either bound.

Each e_i and $e_i + e_j$ belongs to the relevant cone, including in the linear regime. Polarization therefore recovers the whole matrix from legal scalar calibrations. Theorem 3.2, whose direct root-expansion and recovery argument was used above, identifies the whole-model leading sets by precisely the stated decoders and the constraint $v^\top Q_\theta v \leq 4$. This proves sufficiency for the joint set, not just its diameter. Conversely, the decoder explicitly recovers every leading point and every pair of points. Replacing each root block by $\prod_i (z - y_{j,i})$ gives its joint coefficient image; continuity of the fixed-degree root map gives the inverse on ordered real-rooted blocks. The bottleneck metric is the maximum coordinate metric on these ordered blocks, since separated limiting clusters cannot be matched across one another. This proves the metric assertion. $\square$

Remark 4.2 (The marking and the scope of invariance). The datum is intrinsic to the observation germ with its fixed spectral coordinate and marked multiplicity pattern. Reparametrizing free channel or mean variables leaves $\text{ran } L$ unchanged. Adding a free score column to a retained column leaves its quotient class unchanged. An isometry of the observation tangent metric transports all these classes isometrically. These operations preserve the calibration datum and its decoded metric fibre. A permutation of the marking permutes the datum. An arbitrary change of target coordinate must transport the spectral decoder and its metric; no invariance under an untransported change of units is asserted. The datum need not be minimal: existential elimination of endpoint coordinates may identify two different matrices at the level of the spectral image. We assert sufficiency, not the false converse that equality of spectral images always implies equality of all the calibration numbers.

Corollary 4.3 (Quantitative finite reconstruction). *Use the same pattern and decoder for two positive definite matrices Q, Q' , and suppose $Q \geq \lambda I$ and $\|Q' - Q\|_{\text{op}} \leq \eta < \lambda$. Put $\vartheta = \eta/\lambda$ and*

$\beta = \gamma/2$. Then

$$(1 + \vartheta)^{-\beta} \mathcal{L}(Q) \subseteq \mathcal{L}(Q') \subseteq (1 - \vartheta)^{-\beta} \mathcal{L}(Q). \quad (4.7)$$

If $R = \max_{y \in \mathcal{L}(Q)} \|y\|_\infty$, their Hausdorff distance is at most

$$R \max\{1 - (1 + \vartheta)^{-\beta}, (1 - \vartheta)^{-\beta} - 1\}.$$

Their diameters satisfy the same two multiplicative bounds as (4.7). If each number in (4.4) is known with error at most e , the matrix reconstructed by (4.5) has operator-norm error at most $3ke/2$.

Proof. The hypotheses give $(1 - \vartheta)Q \leq Q' \leq (1 + \vartheta)Q$. The corresponding cone sections therefore lie between the radial rescalings by $(1 + \vartheta)^{-1/2}$ and $(1 - \vartheta)^{-1/2}$. Their root decoders are homogeneous of degree γ : variances scale as squared roots in the repeated regime, and sums scale as roots in the linear regime. This proves (4.7). Compare each point with its radial rescaling to obtain the Hausdorff estimate; diameter is monotone under set inclusion and homogeneous under dilation. Finally a diagonal entry has error at most e and an off-diagonal entry at most $3e/2$. The maximal absolute row sum bounds the operator norm by $3ke/2$. $\square$

The preceding theorem is not the distance-function reconstruction of an arbitrary compact set. It first derives a quadratic quotient from the full binary coefficient inverse, then realizes a finite collection of its probes by feasible polynomial-root experiments, and finally decodes that quotient into a constrained joint spectrum. The metric reconstruction Theorem 12.3 remains useful for leading sets outside this class; no quadratic shape is assumed in the general real atlas.

5. A RANK-DEFICIENT CUBIC STRATUM

Let $\mathfrak{F}$ denote the degree-three centres with

$$f = z(z - r)^2, \quad g = (z - s)^3, \quad 0 < s < r < D,$$

strict weights $\pi < \alpha < 1 - \pi$, strictly positive U, V , $\det V \neq 0$, and at least seven clocks. The first channel may have either rank. Put $\gamma = 1 - \alpha$ and

$$A = (r + 3s)(2r - 3s)^2, \quad B = (3D - r)s - 2rD, \quad E = \gamma A - 4\pi r^3. \quad (5.1)$$

Here A is a scalar, not the target polynomial.

Theorem 5.1 (The entire cubic fibre). *The exact spectral fibre is a singleton precisely on*

$$\mathfrak{I} = \{\det U \neq 0\} \cup \{B < 0\} \cup \{E < 0\}. \quad (5.2)$$

On its complement, besides (f, g) all component-polynomial pairs are, up to permutation,

$$(f, h_c), \quad h_c = (1 - c)f + cg, \quad c_* \leq c \leq \min(c_D, \gamma/\pi), \quad (5.3)$$

where

$$c_* = \frac{4r^3}{A}, \quad b = \frac{rs}{3s - 2r}, \quad R(z) = \frac{f(z)}{g(z)}, \quad c_D = \frac{R(D)}{R(D) - 1}.$$

The interval is nonempty even at equality in either constraint, and every one of its pairs has a positive stochastic realization. Conditional on the component pair, that realization is unique.

Proof. Lemma 2.1 fixes q, K for every competitor. If U is invertible, Lemma 2.2 applies; in particular $\det K / \det K_3 = fg$ gives spectral identification.

If $\det U = 0$, stochasticity gives $u_1 = u_2 = u$. The second marginal $V \operatorname{diag}(\alpha, \gamma)(f, g)^\top$ has coefficient rank two. Every competitor therefore has invertible second channel and its two monic polynomials lie in the affine pencil h_c . Expansion gives

$$\operatorname{disc}(h_c) = c(c - 1)^2(r - s)^3(Ac - 4r^3), \quad h_c(0) = -cs^3, \quad (5.4)$$

and

$$\frac{R'(z)}{R(z)} = \frac{(2r-3s)z+rs}{z(z-r)(z-s)}, \quad A = 4r^3 - 27s^2(r-s) < 4r^3.$$

Nonnegative roots exclude $c < 0$, and the discriminant excludes $0 < c < 1$. For $c > 1$ there is a root in (s, r) and no root in $[0, s]$. The other two admissible roots must be above r , solving $R(z) = c/(c-1) > 1$. They exist only for $s > 2r/3$. In that case R increases to its maximum at b and then decreases to one. The double-root value is $c_* > 1$; the larger root increases from b to D as c increases from c_* to c_D . Thus the whole real-rooted pencil in $[0, D]$ is $\{0, 1\}$, with $[c_*, c_D]$ added exactly when $b \leq D$.

For a competing pair with pencil coordinates c_1, c_2 , the weights obey $\alpha'_1 c_1 + \alpha'_2 c_2 = \gamma \in (0, 1)$. One coordinate must be zero and the other is one or a value $c \geq c_*$. The latter weight is γ/c , so the floor imposes $c \leq \gamma/\pi$; the other weight is automatically at least $\alpha > \pi$. Conversely put

$$\beta_f = \alpha v_1 + \gamma(1 - 1/c)v_2, \quad \beta_h = \gamma v_2/c.$$

Both vectors are positive and $\beta_f f + \beta_h h_c = \alpha v_1 f + \gamma v_2 g$. Their entry sums are the stated weights; normalizing them and keeping first channel (u, u) gives the realization. Coefficient functionals dual to the independent polynomials recover the two rank-one coefficient matrices uniquely, hence their weights and normalized marginals. Finally $b \leq D$ is equivalent to $B \geq 0$ and, on this set, $c_* \leq \gamma/\pi$ is equivalent to $E \geq 0$. $\square$

5.1. Quantitative inversion across the rank locus. All coefficient norms below are Euclidean, including the leading coefficient when a monic polynomial is represented as a vector. Constants are specified from a fixed compact $K_0 \subset \mathfrak{F}$. Norm comparisons between these finite-dimensional spaces are included in these constants. Put

$$\delta = \max_j \|P'_j - P_{\theta,j}\|_F.$$

Use coefficient order $(1, z, z^2, z^3)$. The two components form $F_{\text{coef}} \in \mathbb{R}^{2 \times 4}$, with one polynomial per row, and the second-marginal coefficient matrix is $B_\theta = V \text{diag}(\alpha, \gamma) F_{\text{coef}} \in \mathbb{R}^{2 \times 4}$. Its two rows are the scalar coordinate polynomials of that marginal. Here σ_2 is its smaller singular value and matrix norms are Euclidean operator norms (or the corresponding fixed norm comparisons). Set

$$\beta_0 = \min_{K_0} \sigma_2(B_\theta) > 0, \quad d_{fg} = \min_{K_0} \|g - f\|_{\text{coef}} > 0.$$

Normalizer recovery supplies C_B with $\|B' - B_\theta\| \leq C_B \delta$. Let K_d bound $\|\text{diag}(\alpha') F'_{\text{coef}}\|$ on all closed-model competitors for the designs in K_0 , and set

$$W_0 = \frac{2K_d}{\pi\beta_0}, \quad C_f = \max_{K_0} \|f\|_{\text{coef}}, \quad K_{\text{aff}} = W_0 C_B (1 + C_f). \quad (5.5)$$

Proposition 5.2 (From the second marginal to the pencil). *If $\delta \leq \beta_0/(2C_B)$, every closed competitor has invertible V' and its component polynomials satisfy*

$$f'_b = h_{c_b} + e_b, \quad \deg e_b < 3, \quad \|e_b\|_{\text{coef}} \leq K_{\text{aff}} \delta. \quad (5.6)$$

Moreover $\|\text{diag}(\alpha')^{-1} V'^{-1}\| \leq W_0$. If the ordered component pair is within ρ of (f, g) , then

$$|c_1|, |c_2 - 1| \leq (\rho + K_{\text{aff}} \delta)/d_{fg}. \quad (5.7)$$

No lower bound on $|\det U|$ is used.

Proof. Since $B' = V' \text{diag}(\alpha') F'_{\text{coef}}$,

$$\beta_0/2 \leq \sigma_2(B') \leq \sigma_{\min}(V') K_d.$$

This proves the inverse bound, using $\alpha'_a \geq \pi$. Write C_θ for the matrix with coefficient rows f, g and put

$$T = \text{diag}(\alpha')^{-1} V'^{-1} V \text{diag}(\alpha, \gamma), \quad E_0 = \text{diag}(\alpha')^{-1} V'^{-1} (B' - B_\theta).$$

Then $F'_{\text{coef}} = TC_\theta + E_0$ and $\|E_0\| \leq W_0 C_B \delta$. If a row of T is (l, c) , monicity says $l + c + (E_0)_{\text{lead}} = 1$. Subtracting $(E_0)_{\text{lead}} f$ from that error row gives the exact decomposition $f'_b = (1 - c)f + cg + e_b$, with the bound in (5.5) and zero leading coefficient in e_b . Finally $h_c - f = c(g - f)$ and $h_c - g = (c - 1)(g - f)$ imply (5.7) by the triangle inequality. $\square$

Lemma 5.3 (Two robust isolation estimates). *There are explicit positive $r_f, r_g, \varepsilon_0, C_0, C_1$, depending only on K_0 , with the following property. If $h_c + e$ is monic with all roots in $[0, D]$, $\deg e < 3$ and $\|e\|_{\text{coef}} \leq \varepsilon$, then*

$$|c| \leq r_f, \varepsilon \leq \varepsilon_0 \implies |c| \leq C_0 \varepsilon, \quad |1 - c| \leq r_g, \varepsilon \leq \varepsilon_0 \implies |1 - c| \leq C_1 \varepsilon.$$

The constants require only root-separation and coefficient bounds on K_0 , not a sign or a nonzero margin for B, E .

Proof. Let $s_0 = \min_{K_0} s > 0$ and $d_0 = \min_{K_0} 4r^3(r - s)^3 > 0$. Choose L_D as a Lipschitz bound for the discriminant on a coefficient neighbourhood of $\{h_c : \theta \in K_0, |c| \leq 1\}$, and R_D such that

$$|\text{disc}(h_c) + 4r^3(r - s)^3 c| \leq R_D c^2 \quad (|c| \leq 1).$$

These constants are finite because the discriminant is polynomial. Take $r_f \leq \min(1, d_0/(2R_D))$, omitting the second bound if $R_D = 0$. The endpoint condition gives $-cs^3 + e(0) \leq 0$, hence $c \geq -\varepsilon/s_0^3$. For $0 \leq c \leq r_f$, nonnegative discriminant gives $0 \leq -d_0 c + R_D c^2 + L_D \varepsilon$, hence $c \leq 2L_D \varepsilon/d_0$. One may take $C_0 = \max(s_0^{-3}, 2L_D/d_0)$.

Near $c = 1$, put $v = 1 - c$ and translate $z = s + x$. There are exact identities

$$h_c(s + x) = x^3 + vA_2x^2 + vA_1x + vA_0, \quad A_2 = 3s - 2r, \quad A_1 = f'(s), \quad A_0 = f(s) = s(r - s)^2.$$

Let L_s bound coefficient translation of e on K_0 , set $E_s = L_s \varepsilon$, and write the perturbed coefficients as $u_i = vA_i + e_i$, $|e_i| \leq E_s$. Choose $M \geq \max_{K_0} (|A_1|, |A_2|)$ and $a_* = \min_{K_0} A_0 > 0$. Depressing the perturbed cubic by its own quadratic coefficient gives

$$p = u_1 - u_2^2/3, \quad q = u_0 - u_1u_2/3 + 2u_2^3/27.$$

For $u = |v| + E_s \leq 1$,

$$|p| \leq C_p u, \quad |q - A_0 v| \leq E_s + C_q u^2, \quad C_p = M + 1 + (M + 1)^2/3, \quad C_q = (M + 1)^2/3 + 2(M + 1)^3/27.$$

Real-rootedness is exactly $p \leq 0$ and $|q| \leq c_{\text{cub}}(-p)^{3/2}$ with $c_{\text{cub}} = 2/(3\sqrt{3})$. Thus

$$a_* |v| \leq E_s + C_q u^2 + c_{\text{cub}} C_p^{3/2} u^{3/2}. \quad (5.8)$$

Choose

$$\eta = \min\{1, a_*/(4C_q), (a_*/(4c_{\text{cub}} C_p^{3/2}))^2\}, \quad r_g = \eta/2, \quad \varepsilon_0 \leq \eta/(2L_s).$$

Then $u \leq \eta$, the last two terms in (5.8) are at most $a_* u/2$, and $|v| \leq (2/a_* + 1)E_s$. Take $C_1 = (2/a_* + 1)L_s$. Decrease ε_0 also to stay in the discriminant neighbourhood used in the first argument. Every threshold is determined by the stated positive lower margins and finite polynomial upper bounds. $\square$

Let d_{par} be Euclidean distance in monic coefficients and stochastic entries, minimized over the component permutation.

Lemma 5.4 (A local inverse through the walls). *For every compact $K_0 \subset \mathfrak{F}$, there exist $\rho, M, t_0 > 0$ such that a closed-model competitor whose component pair is within ρ of (f, g) , up to permutation, satisfies*

$$d_{\text{par}}(\xi, \theta) \leq M \max_j \|F_T(\xi)_j - P_{\theta,j}\|_F \quad (5.9)$$

whenever the observation error is below t_0 . These constants require no lower bound on $|\det U|$, $-B$, or $-E$. For a compact $H \subset \mathfrak{I}$ and small fixed ρ , the remote separation

$$\eta_H(\rho) = \min_{\theta \in H, \xi \text{ in the closed model } d_{\text{par}}(\xi, \theta) \geq \rho} h_w(F_T(\xi), P_\theta) \quad (5.10)$$

is positive if the minimizing set is nonempty. The whole-model inverse therefore holds uniformly on H .

Proof. Let $r_* = \min(r_f, r_g)$ from Lemma 5.3 and choose

$$\rho \leq d_{fg} r_*/4, \quad t_0 \leq \min\{\beta_0/(2C_B), \rho/K_{\text{aff}}, \varepsilon_0/K_{\text{aff}}\}.$$

Proposition 5.2 gives an error at most $\varepsilon = K_{\text{aff}}\delta$ in the affine pencil and pencil coordinates in the required neighbourhoods. The two isolation estimates then give $|c_1| + |c_2 - 1| \leq (C_0 + C_1)K_{\text{aff}}\delta$. Together with (5.6), this controls both polynomial coefficient vectors by a common multiple of δ .

Fix the component permutation placing the close ordered pair near (f, g) ; separation of f, g makes this choice unambiguous for a smaller ρ . To recover the stochastic parameters without dividing by $\det U$, choose coefficient functionals ℓ_1, ℓ_2 dual to f, g . Their norms are bounded by the inverse Gram matrix of this pair, uniformly on K_0 . The matrix $T'_{ij} = \ell_i(f'_j)$ is $I + O_{K_0}(\delta)$. Decrease t_0 so that $\|T' - I\| \leq 1/2$; then $\|(T')^{-1}\| \leq 2$. Apply the two functionals entrywise to K' . Inverting T' recovers $M'_a = \alpha'_a v'_a v'^{\top}_a$ with $O_{K_0}(\delta)$ error from M_a . Entry sums give the weights, and row and column sums divided by those weights give the two channels. These divisions have lower bound π . This proves (5.9) with constants constructed from normalizer, second-marginal, Gram, root and weight margins only. In particular no full-rank inverse has been continued through its singular determinant.

For the last assertion the feasible set in (5.10) is compact, allowing the centre's design to vary in H . A zero minimum would give an exact competitor separated from that centre. The entire fibre and conditional uniqueness in Theorem 5.1 exclude this on H . Small whole-model observation balls thus lie in the local neighbourhood. It is this exclusion radius, rather than a local coefficient constant, that may collapse at a remote wall. $\square$

5.2. The complete local Fisher fibre. Use $a, X, Y \geq 0$, free b, c , and five free stochastic coordinates η . The component jets are

$$p_f = -a(z - r)^2 - bz(z - r) - Xz, \quad p_g = -c(z - s)^2 - Y(z - s). \quad (5.11)$$

Let L be the seven free score columns and D_0 the three columns indexed by $0, x, y$ for (a, X, Y) . Define Q by (3.2) and

$$J(X, Y) = \min_{a \geq 0} (a, X, Y)Q(a, X, Y)^{\top}, \quad \kappa_x = \min_{Y \geq 0} J(1, Y), \quad \kappa_y = \min_{X \geq 0} J(X, 1). \quad (5.12)$$

Proposition 5.5 (Coercivity and finite supports). *Throughout $\mathfrak{F}$, L has rank seven, $Q_{00} > 0$, and κ_x, κ_y are positive and locally Lipschitz. For every compact $K_0 \subset \mathfrak{F}$,*

$$\|S_\theta(a, b, c, \eta, X, Y)\|_{I, \theta} \geq c_{K_0} \|(a, b, c, \eta, X, Y)\| \quad (a, X, Y \geq 0).$$

Each κ_i is given by four finite support tests. For $A \subseteq \{0, x, y\} \setminus \{i\}$ with Q_{AA} positive definite, put

$$v_A = -Q_{AA}^{-1}Q_{Ai}, \quad k_{i,A} = Q_{ii} - Q_{iA}Q_{AA}^{-1}Q_{Ai}.$$

The test is $v_A \geq 0$ and $Q_{ji} + Q_{jA}v_A \geq 0$ for inactive j . The empty support has value Q_{ii} . At least one test succeeds and all successful tests give κ_i .

Proof. Every feasible jet is realized by roots

$$ta, \quad r + tb/2 \pm \sqrt{tX}, \quad s + tc/3 + \sqrt{t}y_i, \quad \sum_i y_i = 0, \quad \frac{1}{2} \sum_i y_i^2 = Y,$$

and order- t stochastic increments. Their observation increment is $tS + O(t^{3/2})$. A zero score, combined with Lemma 5.4, forces the component jets and η to vanish. Coefficient comparison in (5.11) then forces every variable to vanish. Compactness of the feasible unit sphere gives coercivity, uniformly on K_0 . This also proves independence of the free columns, $Q_{00} > 0$, and existence, positivity and boundedness of the constrained minimizers.

Choose a minimizer with minimal positive support in the projected score columns. Those supported columns are independent: a nonzero dependence would allow a signed change of positive coefficients, preserving the represented vector, until one coefficient vanished. On an independent support, stationarity and the inactive inequalities are exactly the stated tests. Conversely their KKT conditions prove optimality by convexity. This handles a singular full Q without inverting it. Uniformly bounded minimizers and smooth score dependence give local Lipschitz dependence by evaluating each problem at a minimizer of the other problem. $\square$

Theorem 5.6 (Local normal form and whole-model restriction). *At every centre of $\mathfrak{F}$ the local leading root set is*

$$(\{0\}, \{-\sqrt{X}, \sqrt{X}\}, Z(z^3 - Yz - Z)), \quad X, Y \geq 0, \quad J(X, Y) \leq 4, \quad 27Z^2 \leq 4Y^3. \quad (5.13)$$

On compact subsets of $\mathfrak{F}$, scaled local observation balls converge to this set with Hausdorff error $O(\sqrt{t})$. Hence

$$\omega_{\theta, \text{loc}}(t) = C_{\text{loc}}(\theta)\sqrt{t} + O(t), \quad C_{\text{loc}} = \max\{\sqrt{2}\kappa_x^{-1/4}, 2\sqrt{2/3}\kappa_y^{-1/4}\}. \quad (5.14)$$

The error is compact-uniform, and C_{loc} is positive and locally Lipschitz on all of $\mathfrak{F}$. On compact subsets of $\mathfrak{J}$ these statements hold for the whole-model modulus. The support tests, (5.2), and $9\kappa_y - 16\kappa_x$ give a finite semialgebraic partition. On its constant choices,

$$C_{\text{loc}}^4 = \max\{4/\kappa_x, 64/(9\kappa_y)\}$$

is one rational function, and C_{loc} is Nash after refinement.

Proof. The local inverse bounds coefficients and stochastic increments by $O(t)$ for every local competitor. Its first component has exact roots $ta, r + tb/2 \pm \sqrt{tX}$ with bounded a, b, X and $a, X \geq 0$. Its second component has exact roots $s + tc/3 + \sqrt{t}y_i$ with $\sum_i y_i = 0$ and bounded squared sum. Put $Y = \frac{1}{2} \sum_i y_i^2$ and $Z = \prod_i y_i$. The real-cubic condition is exactly $Y \geq 0, 27Z^2 \leq 4Y^3$. Expansion gives $F = P + tS + O(t^{3/2})$ and $h_w = t\|S\|_I/2 + O(t^{3/2})$ uniformly on these bounded coordinates. Radial contraction of all first-order variables, together with $Z \mapsto \rho^{3/2}Z$, proves the outer Hausdorff inclusion. Conversely choose bounded minimizing free and endpoint variables and realize the displayed root arcs with $\rho = 1 - M\sqrt{t}$. They lie in the observation ball and give the inner inclusion. The inequality boundary and all joint cubic coefficient relations are therefore realized.

Homogeneity gives $X_{\text{max}} = 2/\sqrt{\kappa_x}$ and $Y_{\text{max}} = 2/\sqrt{\kappa_y}$. The double-cluster diameter is $\sqrt{X_{\text{max}}}$. Lemma 3.1 with $m = 3$ gives the triple diameter $2\sqrt{Y_{\text{max}}/3}$, attained also by $(-2h, h, h)$ and $(-h, -h, 2h)$ with $h = \sqrt{Y_{\text{max}}/3}$. The maximum of marginal diameters gives (5.14). Proposition 5.5, cluster separation and the positive remote exclusion in Lemma 5.4 give the remaining claims. The local formula remains valid at $B = 0$ or $E = 0$, but its identification with the whole-model modulus does not. $\square$

Example 5.7 (The earlier rational phase values). For $D = 1$, $\pi = 1/10$, $r = 3/5$, $s = 1/5$, $\alpha = 2/5$, $u_1 = u_2 = (1/3, 2/3)^\top$, $v_1 = (1/4, 3/4)^\top$, $v_2 = (3/4, 1/4)^\top$, clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ and equal weights, $B = -18/25$. The empty supports are admissible, the double cluster dominates, and exact rational evaluation gives

$$C^4 = \frac{78550423766145732273576408862601811503774391446}{385279214348552940073664077156875}, \quad 3778.707 < C < 3778.708.$$

Changing only u_2 to $(2/5, 3/5)^\top$ gives the triple-dominant phase, $59.970 < C < 59.971$. The exact arithmetic records for both values are retained in the archival companion. These large values are not evidence for an unspecified wall blow-up.

Corollary 5.8 (Centre-known local risk). *Let $H \subset \mathfrak{I}$ be compact, fix $0 < c \leq 1/(8\sqrt{2})$, and observe independent categorical samples at the design clocks with $n_j/N \rightarrow w_j$ uniformly on H (integer rounding suffices). For squared spectral loss on the specified neighbourhood $h_w(F_T(\xi), P_\theta) \leq cN^{-1/2}$, with the centre known, the minimax risk $\mathcal{R}_N(\theta, c)$ satisfies uniformly on H*

$$\frac{3c}{32}C_{\text{loc}}(\theta)^2 - O_H(N^{-1/4}) \leq \sqrt{N}\mathcal{R}_N(\theta, c) \leq cC_{\text{loc}}(\theta)^2 + O_H(N^{-1/4}).$$

Proof. A constant decision in the specified spectral ball has loss at most its squared diameter. A diameter pair in that ball has weighted Hellinger distance at most $2cN^{-1/2}$. Tensorization bounds its product Hellinger distance by $2c\sqrt{\max_j n_j/(Nw_j)}$. Uniform convergence and the positive minimum of the design weights on H make the maximum ratio at most 2 for all sufficiently large N , uniformly on H . Thus the product distance is at most $2c\sqrt{2} \leq 1/4$. Total variation is bounded by this Hellinger distance in the present convention. The two-point testing bound for squared metric loss is $d^2(1 - \text{TV})/8$, hence at least $3d^2/32$. Substitute (5.14). The centre and its neighbourhood are known; unknown-centre adaptation and honest confidence sets across a wall are not asserted. $\square$

6. A REMOTE BRANCH AT THE STOCHASTIC WEIGHT WALL

Fix r, s, D with $B > 0$, positive u, v_1, v_2 with $v_1 \neq v_2$, and $U = (u, u)$, $V = (v_1, v_2)$. Assume

$$\gamma_0 = \pi c_*, \quad \alpha_0 = 1 - \pi c_*, \quad \pi < \alpha_0 < 1 - \pi. \quad (6.1)$$

The design is fixed. Let θ_δ be the original (f, g) centre with weight $\alpha = \alpha_0 + \delta$. Then $E = -A\delta$, so $\delta > 0$ is the identified side. Put

$$a = \frac{4rs}{r + 3s}, \quad b = \frac{rs}{3s - 2r}, \quad h_* = (1 - c_*)f + c_*g = (z - a)(z - b)^2. \quad (6.2)$$

Substitution gives $s < a < r < b < D$. At $\delta = 0$ the two exact component pairs are (f, g) and (f, h_*) . The remote pair has weight π on h_* and second-channel columns

$$v_f^* = \frac{\alpha_0 v_1 + (\gamma_0 - \pi)v_2}{1 - \pi}, \quad v_h^* = v_2. \quad (6.3)$$

Let ξ_* be this point; its observation is $P_0 = P_{\theta_0}$. Its second channel is invertible because $v_f^* - v_2 = \alpha_0(v_1 - v_2)/(1 - \pi) \neq 0$. The sorted base and remote spectra are respectively

$$(0, s, s, s, r, r), \quad (0, a, r, r, b, b),$$

whose distance is

$$D_* = \max(r - s, b - r) > 0. \quad (6.4)$$

Thus a zero-length additional pencil interval already has a positive exact-fibre spectral diameter.

6.1. A complete real corner chart at the remote point. Use the eleven coordinates $v = (a_0, b_0, X, c_0, d_0, Y, k, \dot{u}_f, \dot{u}_h, \dot{v}_f, \dot{v}_h)$, where a binary channel increment is represented by its one free zero-sum coordinate. Set

$$\begin{aligned} p_f &= -a_0(z-r)^2 - b_0z(z-r) - Xz, \\ p_h &= -c_0(z-b)^2 - d_0(z-a)(z-b) - Y(z-a), \\ \mathcal{T}_* &= \{v : a_0, X, Y, k \geq 0\}. \end{aligned} \tag{6.5}$$

The remaining seven coordinates are free. Here k is the increment of the weight on h_* ; the weight on f has increment $-k$. Let J_*v be the full coefficient and stochastic vector with polynomial increments p_f, p_h and these stochastic increments.

Proposition 6.1 (Remote tangent cone and converse realization). *In a neighbourhood of ξ_* , modulo the one component permutation, the closed model has a Nash coordinate chart from a neighbourhood of zero in $\mathcal{T}_*$. Its derivative is the invertible linear map J_* . Consequently its coefficient–stochastic tangent cone is exactly $J_*(\mathcal{T}_*)$. Every normalized secant has a subsequential limit in this cone, and every vector of the cone is realized by a feasible arc. For bounded $v \in \mathcal{T}_*$ the arcs can be chosen with*

$$\xi(\tau, v) = \xi_* + \tau J_*v + O(\tau^2), \quad F_T(\xi(\tau, v)) = P_0 + \tau S_*v + O(\tau^2), \tag{6.6}$$

uniformly on bounded coordinate sets. Here S_ is the score (2.3) at the remote point. All assertions are uniform on compact subsets of the wall family with its stated strict margins.*

Proof. Factor each nearby polynomial into its separated simple and quadratic factors. Coprime factorization is analytic in coefficients. Write the simple root of the first component as u , the mean of its quadratic roots as $r + v_f/2$, and their squared half-separation as x . Write the corresponding coordinates of the second component as $a + w$, $b + v_h/2$ and y . Thus the exact polynomials are

$$(z-u)((z-r-v_f/2)^2-x), \quad (z-a-w)((z-b-v_h/2)^2-y).$$

These six coordinates are Nash functions of the coefficients near the remote pair. Conversely the displayed formulas are polynomial in the six coordinates. Their derivative has the three columns in (6.5) for each component. They are independent: for the first polynomial, evaluation at 0 and r first determines a_0 and X , and then the remaining coefficient determines b_0 ; for the second, evaluation at a and b determines c_0 and Y , and then d_0 . The inverse function theorem gives the asserted local coefficient chart.

The first simple root is feasible exactly when $u \geq 0$, locally. Its upper endpoint is inactive. All other cluster centres are strictly inside $(0, D)$, so their interval constraints are inactive in this neighbourhood. Real-rootedness of the two quadratic factors is exactly $x, y \geq 0$. The weight on h_* is at its floor, giving $k \geq 0$; the other floor has strict margin $1 - 2\pi > 0$. Channel entries are strictly positive, so only their linear column-sum equations remain active. This proves that the complete feasible chart is the indicated orthant times seven free coordinates, not merely an outer cone. Separated factors are unique. The alternative component labelling is a second, identical chart under the permutation quotient and introduces no extra directions.

For the secant assertion, let τ be the norm of a coefficient and stochastic displacement. Analytic factorization bounds the simple and mean shifts by $O(\tau)$. The quadratic coefficients bound x, y by $O(\tau)$ as well, since variance is mean squared minus the constant coefficient. Dividing the exact chart coordinates by τ gives a bounded vector in $\mathcal{T}_*$. Its subsequential limits give all normalized tangent secants. In the norm-normalized case J_*v has norm one in the limit, so no nonzero secant disappears in the chart.

Conversely set $u = \tau a_0$, $v_f = \tau b_0$, $x = \tau X$, $w = \tau c_0$, $v_h = \tau d_0$, $y = \tau Y$, choose weight $\pi + \tau k$ on h , and take the specified order- τ channel increments. The roots are

$$\tau a_0, \quad r + \tau b_0/2 \pm \sqrt{\tau X}, \quad a + \tau c_0, \quad b + \tau d_0/2 \pm \sqrt{\tau Y}.$$

They remain feasible for small τ , uniformly for bounded coordinates. They are analytic in $\sqrt{\tau}$; their polynomial coefficients are analytic in τ . In fact the exact identities are

$$f_\tau - f - \tau p_f = \tau^2 \{a_0 b_0 (z - r) + a_0 X + b_0^2 z/4\} - \tau^3 a_0 b_0^2/4, \quad (6.7)$$

$$h_\tau - h_* - \tau p_h = \tau^2 \{c_0 d_0 (z - b) + c_0 Y + d_0^2 (z - a)/4\} - \tau^3 c_0 d_0^2/4. \quad (6.8)$$

There is no order- $\tau^{3/2}$ coefficient term at these simple/double clusters. Taylor expansion of the rational observation map, whose normalizers stay positive, proves (6.6). Compact strict root, channel, weight and clock margins make all inverse chart and derivative bounds uniform as stated. $\square$

6.2. The constrained entrance program. Put $B_0 = \frac{d}{d\delta} P_{\theta_\delta} \Big|_{\delta=0}$ and use the Fisher norm at P_0 . Define

$$\mu_E = \frac{1}{2} \min_{v \in \mathcal{T}_*} \|S_* v - B_0\|_{I,0}. \quad (6.9)$$

This is an explicit finite-dimensional program on the cone just proved to be the full feasible tangent cone.

Lemma 6.2 (Positive entrance separation). *The score has no nonzero zero on $\mathcal{T}_*$, and B_0 is not in its image. Thus $\mu_E > 0$ and its minimum is attained. Eliminating seven free variables gives sixteen independent-support KKT tests on four nonnegative variables. For algebraic input μ_E is real algebraic.*

Proof. Suppose first that $S_* v = B_0$. The differentiated normalizer equations, using Lemma 2.1 at the coprime remote pair with invertible second channel, imply $\dot{q} = f - g$ and $\dot{K} = \frac{d}{d\delta} K_{\theta_\delta} \Big|_0$. The derivative of the second marginal belongs to $\text{span}(f, g)$. Its stochastic derivative terms already belong to that span; the remote second-channel weight matrix is invertible. Hence $p_f, p_h \in \text{span}(f, g)$. Since their degrees are below three, write $p_f = t_f(g - f)$ and $p_h = t_h(g - f)$.

Evaluation at 0 gives $t_f = a_0 r^2/s^3 \geq 0$, while evaluation at r gives $t_f = -Xr/(r - s)^3 \leq 0$. Therefore $t_f = 0$ and $p_f = 0$. At b the second polynomial gives

$$t_h = \frac{Y(b - a)}{f(b) - g(b)} \geq 0,$$

because $R(b) > 1$. Differentiation of the monic normalizer now gives

$$\pi t_h + c_* k = -1, \quad (6.10)$$

an impossibility. For a zero score, the same argument has right-hand side zero and implies $t_h = k = 0$. All polynomial jets vanish. Coefficient functionals dual to f, h_* then imply that each coefficient matrix has zero derivative; taking its row and column sums forces all four channel derivatives to vanish. The independent chart columns in Proposition 6.1 force $v = 0$.

The intersection of $\mathcal{T}_*$ with the unit sphere is compact, so $\|S_* v\|_{I,0} \geq c\|v\|$ on the cone. Thus the objective in (6.9) tends to infinity with $\|v\|$; its minimum is attained and positive, since equality $S_* v = B_0$ was excluded.

Let L_* be the seven free columns, D_* the four nonnegative columns, $W = \text{diag}(w_j/P_{0,je})$, and $\mathcal{P}$ the Fisher-orthogonal projection off $\text{im } L_*$. Put

$$Q_* = D_*^\top W \mathcal{P} D_*, \quad g_* = D_*^\top W \mathcal{P} B_0, \quad b_* = B_0^\top W \mathcal{P} B_0.$$

Coercivity implies full rank of the free columns, and

$$4\mu_E^2 = \min_{z \geq 0} \{z^\top Q_* z - 2g_*^\top z + b_*\}. \quad (6.11)$$

For each of the sixteen supports A with $Q_{*,AA}$ positive definite, set $z_A = Q_{*,AA}^{-1}g_{*,A}$, require $z_A \geq 0$ and $(Q_* z - g_*)_j \geq 0$ off A , and take value $b_* - g_{*,A}^\top Q_{*,AA}^{-1}g_{*,A}$. Include the empty support. The minimal-positive-support argument of Proposition 5.5 proves completeness, including singular Q_* . All entries are rational in the wall coordinates; take the positive square root after selecting the exact minimum. $\square$

Lemma 6.3 (Local coefficient control at the remote point). *In a coefficient and stochastic neighbourhood of ξ_* in the closed model,*

$$d_{\text{par}}(\xi, \xi_*) \leq C \|F_T(\xi) - P_0\|.$$

Every competitor at parameter distance $O(\delta)$ has bounded exact chart coordinates at scale δ and observation expansion $P_0 + \delta S_ v + O(\delta^2)$. Conversely each bounded cone vector has such a feasible realization.*

Proof. If the inverse failed, normalize a sequence of displacements by their norm. Proposition 6.1 gives a nonzero limit $J_* v$ with $v \in \mathcal{T}_*$ and zero score, contradicting Lemma 6.2. Equivalently in its chart, Taylor's theorem and cone coercivity give $\|F_T(\Phi(w)) - P_0\| \geq c\|w\| - C\|w\|^2 \geq c\|w\|/2$ for small feasible w . The chart is locally bi-Lipschitz in coefficients and stochastic entries, which proves the inverse and the stated bounded coordinate assertion. The expansions and converse are (6.6). This proof does not assume the inverse in order to identify the tangent cone. $\square$

6.3. The two-scale law with a quadratic entrance remainder. Choose disjoint small closed coefficient and stochastic neighbourhoods N_0, N_* of the two exact representatives at the wall, modulo component permutation, with ξ_* in the relative interior of N_* . Put

$$d_{\text{rem}}(\delta) = \min_{\xi \in N_*} h_w(F_T(\xi), P_{\theta_\delta}). \quad (6.12)$$

Their boundaries have a positive observation separation from P_0 .

Theorem 6.4 (Weight-wall transition). *Under (6.1) and $B > 0$,*

$$d_{\text{rem}}(\delta) = \mu_E \delta + O(\delta^2) \quad (\delta \downarrow 0). \quad (6.13)$$

The minimum is independent of the sufficiently small neighbourhood N_ for all sufficiently small δ . The base local constant $C_{\text{loc}}(\theta_\delta)$ has the finite positive limit $C_{\text{loc}}(\theta_0)$; neither κ_x nor κ_y vanishes at this wall. For fixed $\lambda > 0$ the whole-model modulus satisfies*

$$0 < \lambda < \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = C_{\text{loc}}(\theta_0)\sqrt{\lambda\delta} + O_\lambda(\delta), \quad (6.14)$$

$$\lambda > \mu_E : \quad \omega_{\theta_\delta}(\lambda\delta) = D_* + O_\lambda(\sqrt{\delta}). \quad (6.15)$$

The exact entrance curve for the closed ball is $t = d_{\text{rem}}(\delta)$, including equality. The layer not decided by the first-order threshold alone has width $O(\delta^2)$ in t .

On the nonidentified side, for small $\varepsilon > 0$,

$$\omega_{\theta_{-\varepsilon}}(0) = \max \left\{ r - s, \ b - r + \sqrt{\frac{\chi\varepsilon}{\pi}} + O(\varepsilon) \right\}, \quad \chi = \frac{f(b) - g(b)}{b - a} > 0. \quad (6.16)$$

If $b - r < r - s$, the exact diameter stays equal to $r - s$ initially; if $b - r \geq r - s$, it opens above D_ at the displayed square-root rate.*

Proof. Theorem 5.1 gives exactly the two representatives at $\delta = 0$. Compactness gives a positive observation gap outside $N_0 \cup N_*$. Since ξ_* itself is at distance $O(\delta)$ from P_{θ_δ} , a minimizer in (6.12) has observation $O(\delta)$ from P_0 and lies away from the boundary of N_* . Lemma 6.3 bounds its coefficient and stochastic displacement by $O(\delta)$. Its exact chart coordinates are therefore δv for bounded $v \in \mathcal{T}_*$. Expansion at the common positive probability array and the reverse triangle inequality give

$$h_w(F_T(\xi), P_{\theta_\delta}) = \frac{\delta}{2} \|S_* v - B_0\|_{I,0} + O(\delta^2), \quad (6.17)$$

uniformly on those bounded coordinates. The lower bound in (6.13) follows by minimizing the leading norm. Conversely a minimizer of (6.9), realized by the exact chart, gives the upper bound. The error is quadratic because of (6.7)–(6.8), not merely an appeal to a generic root-continuity bound. This also implies the weaker $O(\delta^{3/2})$ entrance estimate. The same localization puts all minimizers inside any fixed smaller remote neighbourhood for small δ , proving independence of its choice.

Proposition 5.5 and Theorem 5.6 apply on a compact base path segment including θ_0 ; this pair stays locally coercive there. Thus C_{loc} is positive and locally Lipschitz through the wall. If $\lambda < \mu_E$, the remote neighbourhood is excluded from the radius- $\lambda\delta$ ball for small δ . The exterior gap excludes every other competitor, so the whole ball is the base local ball. Formula (5.14) gives (6.14). Replacing $C_{\text{loc}}(\theta_\delta)$ by $C_{\text{loc}}(\theta_0)$ costs only $O(\delta^{3/2})$.

If $\lambda > \mu_E$, a remote minimizer and the base centre belong to the ball. Every competitor belongs to one of the two neighbourhoods. The two coefficient inverses put its roots within $O_\lambda(\sqrt{\delta})$ of the corresponding exact spectrum at the wall. The bottleneck triangle inequality gives both bounds in (6.15). Compact attainment makes remote membership exactly $t \geq d_{\text{rem}}(\delta)$, including equality. The width of the unresolved first-order layer follows from (6.13); it does not decide the sign of the difference on the critical ray.

For $\theta_{-\varepsilon}$ the extra exact pencil interval is $[c_*, c_* + \varepsilon/\pi]$, since $c_D > c_*$ is fixed. At $c = c_* + \sigma$, expansion at $z = b + u$ gives

$$h_c(b + u) = (b - a)u^2 + u^3 + \sigma(g(b) - f(b)) + O(\sigma|u|).$$

Since $R(b) > 1$, $\chi > 0$ and the two roots near b are $b \pm \sqrt{\chi\sigma} + O(\sigma)$. The third root remains in (s, r) . The larger root increases with c by the monotonic branch of R above its maximum. Comparing sorted lists, the distance to the base spectrum is exactly $\max(r - s, b_+(c) - r)$. Two remote spectra are only $O(\sqrt{\varepsilon})$ apart, less than D_* for small ε . The exact-fibre diameter is therefore attained against the base at the upper pencil endpoint. This proves (6.16) and its two cases. $\square$

Remark 6.5 (The critical ray). At $t = \mu_E\delta$, remote membership is decided by the sign of $d_{\text{rem}}(\delta) - \mu_E\delta$. For algebraic input, (6.12) has a compact first-order minimum formula. Elimination and real algebraic branch isolation determine its Puiseux expansion. The first nonzero higher coefficient decides the critical ray; if the difference is identically zero, the closed ball includes the remote branch. No strict regime is assigned to that ray by convention.

Remark 6.6 (The geometric wall). At $B = 0$ with $E > 0$, the remote double root is $b = D$, and the additional pencil interval again consists of one point at positive spectral distance. On the nonidentified side put $\ell_0 = D - b$. Expansion at the strict maximum of R gives

$$c_D - c_* = \frac{-R''(b)}{2(R(b) - 1)^2} \ell_0^2 + O(\ell_0^3), \quad \ell_0 = \frac{B}{3s - 2r}.$$

The coefficient is positive and the weight floor is inactive. This interval is born with quadratic length in geometric clearance, rather than linear length in the weight parameter. The base local constants remain finite on compact subsets of $\mathfrak{F}$. A two-scale entrance law at that endpoint

requires a different cone with one-sided root coordinates. The complete entrance theorem here is for the weight wall.

Example 6.7 (An exact wall datum). Take $D = 1$, $r = 3/5$, $s = 11/20$, $\pi = 1/10$, the rank-one columns and clocks in Section 5, and equal design weights. Then

$$\alpha_0 = \frac{547}{675}, \quad c_* = \frac{256}{135}, \quad a = \frac{44}{75}, \quad b = \frac{11}{15}, \quad B = \frac{3}{25}, \quad D_* = \frac{2}{15}, \quad \chi = \frac{3}{64},$$

and $v_f^* = (1457, 3403)^\top / 4860$. Exact rational evaluation gives

$$5975.429249985072 < C_{\text{loc}}(\theta_0) < 5975.429249985073,$$

$$2.131024914 \cdot 10^{-6} < \mu_E < 2.131024915 \cdot 10^{-6}.$$

The empty positive support is admissible in each base program and in the remote program. The ancillary exact records retain the rational Gram matrices, squared entrance constant, fourth power of the local constant and isolating intervals. The decimals follow from rational power comparisons, not optimization tolerances. They illustrate two different, poorly conditioned scales; they do not test the universal quantifiers of the family theorems.

Three stability questions. Multiplicity and one-sided boundary constraints determine the weights in Theorem 3.2. Exact identification in the cubic family is instead determined by (5.2). Conditioning of a fixed local law is measured by the positive Fisher programs. At the weight wall, identification fails while those programs stay coercive and the local square-root exponent persists. In Example 3.3, by contrast, the exponent changes and the linear constant diverges. The joint atlas retains these distinct limiting operations rather than identifying their strata.

7. FINITE REMOTE ENTRANCES AND THE CRITICAL WINDOW

The preceding weight wall is an instance of a more general phenomenon. A regular feasible corner, rather than invertibility of the ambient score, is the relevant local hypothesis. We formulate the result in the square-root probability embedding, so that the distance in its statement is exactly the Hellinger distance and not a quadratic approximation to it.

Let X be compact and semialgebraic, let F be a continuous probability map, and let T be a continuous monic-polynomial root-multiset target of fixed degree. Quotient a finite representation symmetry when necessary. Suppose

$$F^{-1}(P_0) = \{x_0, x_1, \dots, x_m\}. \quad (7.1)$$

The points are representatives in this quotient. Let $x_\delta \rightarrow x_0$ be a feasible centre path, $P_\delta = F(x_\delta)$, and suppose P_0 is strictly positive. The design weights are fixed. Define

$$E(P) = (\sqrt{w_j} \sqrt{P_{je}})_{j,e}, \quad p_0 = E(P_0), \quad p_\delta = E(P_\delta) = p_0 + \delta b + \delta^2 c + o(\delta^2).$$

The Euclidean distance between two embedded arrays is h_w .

For each $j \geq 1$, assume that a full relative neighbourhood of x_j in X has coordinates

$$\psi_j : K_j \cap B_{r_j}(0) \longrightarrow X, \quad \psi_j(0) = x_j,$$

where K_j is a closed convex polyhedral cone and ψ_j is a Nash parametrization, a homeomorphism onto that relative neighbourhood. Its inverse need not be differentiable at the vertex. Assume that $E \circ F \circ \psi_j$ has a C^2 extension to an ambient neighbourhood of zero. Write

$$E(F(\psi_j(z))) = p_0 + J_j z + Q_j(z, z) + o(\|z\|^2), \quad \|J_j v\| \geq a_j \|v\| \quad (v \in K_j), \quad a_j > 0. \quad (7.2)$$

Here $Q_j(z, z)$ denotes one half of the second derivative. The score may have a kernel in the ambient vector space; the last inequality is required only on the feasible cone. Choose small disjoint closed neighbourhoods $N_0, \dots, N_m$, with boundaries avoiding the exact fibre, and put

$$d_j(\delta) = \min_{x \in N_j} h_w(F(x), P_\delta), \quad \mu_j = \min_{v \in K_j} \|J_j v - b\|.$$

Theorem 7.1 (Remote entrance for regular corners). *Assume (7.1)–(7.2) and $\mu_j > 0$ for $1 \leq j \leq m$. Then*

$$d_j(\delta) = \mu_j \delta + O(\delta^2) \quad (\delta \downarrow 0). \quad (7.3)$$

For small δ the minimum is unchanged on replacing N_j by any fixed smaller relative neighbourhood of x_j . If $\lambda > 0$ differs from every μ_j , the target image of the radius- $\lambda\delta$ ball converges in Hausdorff distance to

$$\mathcal{D}_\lambda = \{T(x_0)\} \cup \{T(x_j) : \mu_j < \lambda\}. \quad (7.4)$$

Its diameter consequently tends to $\text{diam } \mathcal{D}_\lambda$. For $\lambda < \min_j \mu_j$, the whole ball is the base local ball for small δ . Any compact-uniform local asymptotic law at the base therefore remains valid in this regime.

Proof. Fix a remote index and omit it from the notation. Taylor's theorem and cone coercivity give, after reducing r ,

$$\|E(F(\psi(z))) - p_0\| \geq a\|z\|/2 \quad (z \in K, \|z\| < r).$$

The competitor $z = 0$ gives an upper bound $O(\delta)$ for $d(\delta)$. Compactness and (7.1) exclude the boundary of N from minimizing sequences. A minimizer thus has $z = O(\delta)$: its distance to p_0 is at most $d(\delta) + \|p_\delta - p_0\|$. Write $z = \delta v$. The vectors $v \in K$ are uniformly bounded, and

$$E(F(\psi(\delta v))) - p_\delta = \delta(Jv - b) + \delta^2\{Q(v, v) - c\} + o(\delta^2)$$

uniformly on bounded sets. Minimizing the leading norm gives the lower bound in (7.3). The function $v \mapsto \|Jv - b\|$ is coercive on K , so its minimum is attained; inserting a minimizer in the exact chart gives the upper bound. The same localization shows independence of the fixed neighbourhood.

Outside the union of the N_j the continuous function $h_w(F(x), P_0)$ has a positive minimum. This remains an exclusion gap for balls of radius $\lambda\delta$ about P_δ . In a remote neighbourhood, membership is possible exactly when $t \geq d_j(\delta)$, by compact attainment. Thus a remote neighbourhood is excluded if $\lambda < \mu_j$ and contains a minimizer if $\lambda > \mu_j$. Every sequence of remaining competitors converges, after subselection, to an exact representative; its target converges to that representative's target. Conversely x_δ supplies $T(x_0)$ in the limit and each included remote minimizer supplies $T(x_j)$. Finiteness gives Hausdorff convergence to (7.4). The diameter conclusion follows from the elementary bound by twice the Hausdorff distance. When no remote neighbourhood is included, all competitors are in N_0 , which proves the last assertion. $\square$

The coefficient of the next term can be determined without assuming a unique minimizer or an invertible ambient score. This is useful at a change of active face of a constrained Fisher program.

Theorem 7.2 (The second-order entrance coefficient). *Under the hypotheses of Theorem 7.1, set*

$$M_j = \operatorname{argmin}_{v \in K_j} \|J_j v - b\|.$$

This is a nonempty compact set. The residual $r_j = J_j v - b$ is the same for all $v \in M_j$, and has norm μ_j . Define

$$k_j = \frac{1}{\mu_j} \min_{v \in M_j} \langle r_j, Q_j(v, v) - c \rangle. \quad (7.5)$$

Then

$$d_j(\delta) = \mu_j \delta + k_j \delta^2 + o(\delta^2). \quad (7.6)$$

At radius $t = \mu_j \delta + s \delta^2$, the j th remote neighbourhood is excluded for $s < k_j$ and included for $s > k_j$, for all sufficiently small positive δ . Equality requires the next nonzero term or the exact minimum inequality; it is not assigned to a regime by convention. For algebraic Nash input, μ_j and k_j are real algebraic numbers computable by real quantifier elimination and real-root isolation.

Proof. Coercivity bounds all minimizers. The image $J_j K_j$ is a closed convex polyhedral cone. The squared distance to b has a unique minimizing image point: if two different image points minimized it, their midpoint would have strictly smaller squared distance. This proves the assertion about r_j without requiring uniqueness of its preimage.

Again suppress j . Localization in the preceding proof bounds all scaled minimizers in one compact subset of K , also containing M . On that set the squared normalized distance has the uniform expansion

$$f_\delta(v) = \|Jv - b\|^2 + 2\delta \langle Jv - b, Q(v, v) - c \rangle + o(\delta).$$

Every limit of minimizers of f_δ belongs to M , by comparison with any point of M . The first term is at least μ^2 everywhere. The lower bound for the first-order increment of $\min f_\delta$ is therefore the minimum of the displayed second term over M . For the upper bound evaluate f_δ at a point of M attaining that minimum. Compactness and continuity justify both extrema. Consequently

$$\min f_\delta = \mu^2 + 2\delta \min_{v \in M} \langle r, Q(v, v) - c \rangle + o(\delta).$$

Taking the positive square root and multiplying by δ proves (7.6). Comparing with the proposed radius gives the strict cases. The minimum defining M and the minimum in (7.5) have first-order polynomial formulas with compact attainment; the latter may equivalently be restricted to a sufficiently large algebraic ball. Elimination and isolation give algebraic outputs. The formula is intrinsic to the distance and the fixed centre path, since its value equals the coefficient in (7.6), independently of the chosen chart. $\square$

Corollary 7.3 (Application to the cubic weight wall). *The remote point of Section 6 satisfies the preceding hypotheses with $m = 1$, $K_1 = \mathcal{T}_*$ and $\mu_1 = \mu_E$. In particular its entrance has the expansion*

$$d_{\text{rem}}(\delta) = \mu_E \delta + k_E \delta^2 + o(\delta^2),$$

where k_E is given by (7.5) in the exact remote chart of Proposition 6.1. This formula does not assert a common sign for k_E on the weight-wall family.

Proof. Theorem 5.1 gives the two exact representatives. The remote chart and its converse feasible realization are Proposition 6.1; its smooth coefficient formulas and positive normalizers give the required C^2 observation embedding. In these coordinates

$$J_1 v = \left(\frac{\sqrt{w_j}}{2\sqrt{P_{0,j_e}}} (S_* v)_{j_e} \right)_{j,e}, \quad b = \left(\frac{\sqrt{w_j}}{2\sqrt{P_{0,j_e}}} (B_0)_{j_e} \right)_{j,e}.$$

Lemma 6.2 is exactly cone coercivity and positive separation in this embedding. Hence the first minimization is (6.9). The two second-order derivatives in (7.5) are taken from the rational coefficient chart and the square-root probability map, not from root coordinates with an unproved differentiability at a multiple root. Theorems 7.1 and 7.2 apply. $\square$

Example 7.4 (Both signs of the critical correction). Let $e_0 = (1, 1)/\sqrt{2}$, $e_1 = (-1, 1)/\sqrt{2}$ and

$$q(s) = \frac{(1 - s^2)e_0 + 2se_1}{1 + s^2}, \quad P(s) = (q_1(s)^2, q_2(s)^2).$$

For sufficiently small $|s|$ both coordinates of $q(s)$ are positive. Take a base interval with observation $P(s)$ and a disjoint remote interval $0 \leq u \leq e$ with observation $P(-u/2)$. Give the two intervals distinct simple-root targets at zero, for example $z - (1/4 + s)$ and $z - (3/4 + u)$, and choose e small enough to keep all roots in $[0, 1]$. The exact fibre at $P(0)$ has two points. Choose the centre $s_\delta = (\delta + a\delta^2)/2$, where a is fixed. At the remote point $K = \mathbb{R}_+$, $Jv = -ve_1$, $b = e_1$, $c = ae_1 - e_0/2$, $M = \{0\}$ and $\mu = 1$. Formula (7.5) gives $k = a$. Indeed the nearest remote point is $u = 0$ for small $\delta > 0$, since on this short circle arc the chord distance increases with the separation of the stereographic parameters. Directly,

$$d_{\text{rem}}(\delta) = \frac{2s_\delta}{\sqrt{1 + s_\delta^2}} = \delta + a\delta^2 + O(\delta^3).$$

Thus either sign of the second-order correction occurs in an algebraic probability experiment satisfying all hypotheses. For $a = 0$ the quadratic coefficient vanishes and the next term is $-\delta^3/8$.

These results apply in any coefficient dimension and to any fixed number of latent components once their stated finite-fibre and feasible-cone hypotheses have been verified. They are not a classification of all rank-deficient singularities. The endpoint double root at the geometric wall requires a different parametrization and a different separation argument, supplied in Section 8.

8. THE GEOMETRIC ENDPOINT WALL

We now complete a second wall mechanism in the same rank-deficient cubic experiment. Fix $D > r > 0$, strict weights $\alpha, \gamma = 1 - \alpha$, positive $U = (u, u)$ and invertible positive V , and the exterior design. Put

$$s_0 = \frac{2rD}{3D - r}, \quad s_\delta = s_0 - \delta, \quad f = z(z - r)^2, \quad g_\delta = (z - s_\delta)^3. \quad (8.1)$$

Then $2r/3 < s_0 < r$. Assume the strict weight margin

$$E_0 = \gamma A(s_0) - 4\pi r^3 > 0, \quad A(s) = (r + 3s)(2r - 3s)^2. \quad (8.2)$$

Let θ_δ be this base centre and P_δ its observation. Here $B(\theta_\delta) = -(3D - r)\delta$. Thus positive δ is the identified side, while the weight constraint stays inactive locally. Write $c_* = 4r^3/A(s_0)$ and

$$a = \frac{4rs_0}{r + 3s_0}, \quad h_* = (1 - c_*)f + c_*g_0 = (z - a)(z - D)^2, \quad \beta = \gamma/c_*.$$

The remote weight on h_* is $\beta > \pi$, and its other weight is $1 - \beta > \pi$. Its first channel is (u, u) and its second channel is

$$v_f^* = \frac{\alpha v_1 + (\gamma - \beta)v_2}{1 - \beta}, \quad v_h^* = v_2. \quad (8.3)$$

Both columns are positive and distinct. Denote this remote point by ξ_B . Theorem 5.1 gives exactly the base and this remote representative at $\delta = 0$.

Proposition 8.1 (The endpoint-root parametrization). *A full relative neighbourhood of ξ_B , modulo the component permutation, has the following exact feasible parametrization:*

$$\begin{aligned} f_v(z) &= (z - a_0)((z - r - b_0/2)^2 - X), \\ h_v(z) &= (z - a - c_0)(z - D - d_1)(z - D - d_2), \\ K_B &= \{a_0, X \geq 0, \quad d_1 \leq d_2 \leq 0\}. \end{aligned} \quad (8.4)$$

The weight on h is $\beta + k$, and the four binary channel coordinates have their unrestricted small increments. The other coordinates b_0, c_0, k are free. This is a polynomial parametrization with

continuous semialgebraic inverse; no differentiability of its inverse at $d_1 = d_2$ is asserted. Its first polynomial jets are

$$p_f = -a_0(z - r)^2 - b_0z(z - r) - Xz, \quad p_h = -c_0(z - D)^2 - (d_1 + d_2)(z - a)(z - D). \quad (8.5)$$

Let S_B be the score at ξ_B applied to these jets and the stochastic increments. Then S_B has no nonzero zero on K_B . If $B_B = \frac{d}{d\delta}P_\delta|_0$, then $B_B \notin S_B K_B$. In particular

$$\mu_B = \frac{1}{2} \min_{v \in K_B} \|S_B v - B_B\|_{I,0} > 0. \quad (8.6)$$

The minimum is attained. After seven free coordinates are eliminated, it is a quadratic program with three nonnegative coordinates and eight independent-support tests.

Proof. Separated simple and double factors give the first component's simple root, quadratic mean and squared half-separation as analytic functions of its coefficients. At the second component the separated simple root is analytic; the two roots near D have a unique increasing ordering and depend continuously and semialgebraically on the coefficients. Their feasibility is exactly $d_1 \leq d_2 \leq 0$. All other root bounds, weights and channel inequalities have strict margins, except $a_0 \geq 0$ and $X \geq 0$. This proves exact coverage and injectivity of the parametrization, not merely containment of tangent directions. Its polynomial formulas give a uniform $O(\tau^2)$ coefficient remainder on substituting τv for bounded $v \in K_B$. Positive normalizers and probabilities give the same remainder for the observation embedding.

We prove the two score assertions directly. Put $\dot{g} = 3(z - s_0)^2$, the derivative of g_δ at zero. Equality of remote and base scores implies equality of their derivatives of q, K by the differentiated normalizer argument in Lemma 2.1. The coefficient second marginal at the remote point is $(1 - \beta)v_f^* f + \beta v_2 h_*$. Its two coefficient columns are independent. Modulo $\text{span}(f, g_0)$, equality of its derivative with $\gamma v_2 \dot{g}$ gives

$$p_f = t_f(g_0 - f), \quad p_h = c_* \dot{g} + t_h(g_0 - f).$$

The polynomials have the stated degrees, so no additional monic direction occurs. Evaluation of the first identity at 0 and r gives $t_f = a_0 r^2 / s_0^3 \geq 0$ and $t_f = -Xr / (r - s_0)^3 \leq 0$. Thus $t_f = 0$ and $p_f = 0$.

To interpret the second identity, consider the algebraic double-root pencil curve, without imposing its endpoint constraint,

$$\bar{h}_\delta = (1 - c_*(s_\delta))f + c_*(s_\delta)g_\delta = (z - a(s_\delta))(z - b(s_\delta))^2, \quad b(s) = \frac{rs}{3s - 2r}.$$

At zero, $b(s_0) = D$ and

$$\dot{b} = \frac{d}{d\delta} b(s_0 - \delta) \Big|_0 = \frac{2r^2}{(3s_0 - 2r)^2} > 0. \quad (8.7)$$

Both $p_h(D)$ and $\dot{\bar{h}}(D)$ vanish. Since $g_0(D) - f(D) \neq 0$, this determines t_h uniquely and implies $p_h = \dot{\bar{h}}$. Differentiation in z at D now gives

$$-(d_1 + d_2)(D - a) = p'_h(D) = -2(D - a)\dot{b}.$$

This would require $d_1 + d_2 = 2\dot{b} > 0$, contrary to feasibility. Therefore $S_B v = B_B$ is impossible.

For a zero score the same second-marginal argument instead gives $p_f = t_f(g_0 - f)$ and $p_h = t_h(g_0 - f)$. The first polynomial again vanishes. Evaluating the second at D forces $t_h = 0$. Its coefficients then give $c_0 = 0$ and $d_1 + d_2 = 0$, whence $d_1 = d_2 = 0$ on K_B . The first polynomial's independent columns give $a_0 = b_0 = X = 0$. Coefficient functionals dual to f, h_* recover zero derivatives of both coefficient matrices, and their entry, row and column sums give zero weight and channel increments. Thus the entire cone vector is zero. Compactness of the feasible unit

sphere yields $\|S_B v\|_{I,0} \geq c\|v\|$ on K_B . The objective in (8.6) is consequently coercive and has a positive attained minimum.

Finally the score depends on d_1, d_2 only through $w = -(d_1 + d_2) \geq 0$. Every $w \geq 0$ is realized, for example by $d_1 = d_2 = -w/2$. The remaining nonnegative coordinates are a_0, X ; there are seven free coordinates. Their columns are independent by the zero-score argument. Fisher-orthogonal projection off those columns reduces the problem to three nonnegative variables. The minimal-positive-support argument of Proposition 5.5 gives the eight support tests, including the empty support and only inverting positive definite supported Gram matrices. This remains valid when the full projected Gram matrix is singular. $\square$

Theorem 8.2 (The geometric-wall two-scale law). *Under (8.1)–(8.2), the remote entrance satisfies*

$$d_{\text{rem},B}(\delta) = \mu_B \delta + k_B \delta^2 + o(\delta^2), \quad \mu_B > 0, \quad (8.8)$$

where k_B is given by Theorem 7.2 in the parametrization (8.4). In particular, for fixed $\lambda > 0$,

$$\lambda < \mu_B : \quad \omega_{\theta_\delta}(\lambda\delta) = C_{\text{loc}}(\theta_0)\sqrt{\lambda\delta} + O_\lambda(\delta), \quad (8.9)$$

$$\lambda > \mu_B : \quad \omega_{\theta_\delta}(\lambda\delta) = D - r + O_\lambda(\sqrt{\delta}). \quad (8.10)$$

The base local Fisher constant stays positive and finite at the wall. The strict cases in the critical window $t = \mu_B \delta + s\delta^2$ are decided by $s - k_B$.

On the nonidentified side $\theta_{-\varepsilon}$, for all sufficiently small $\varepsilon > 0$, the additional exact pencil interval has length

$$c_D(s_0 + \varepsilon) - c_*(s_0 + \varepsilon) = \Lambda_B \varepsilon^2 + O(\varepsilon^3), \quad \Lambda_B = \frac{-R''_{s_0}(D)}{2(R_{s_0}(D) - 1)^2} b^2 > 0, \quad (8.11)$$

where $R_s(z) = f(z)/(z - s)^3$. Nevertheless its exact whole-model spectral diameter is constant:

$$\omega_{\theta_{-\varepsilon}}(0) = D - r. \quad (8.12)$$

Thus the two wall mechanisms differ both in the size of the newborn pencil interval and in the opening of its exact spectral diameter.

Proof. The endpoint parametrization and its score separation satisfy the hypotheses of Theorems 7.1 and 7.2. The derivative of the square-root observation embedding is the score multiplied entrywise by $\sqrt{w_j}/(2\sqrt{P_{0,j\varepsilon}})$, so its entrance minimum is precisely (8.6). This proves (8.8) and the critical-window assertion, including possible nonuniqueness of the minimizing endpoint-root split.

The sorted spectra at the wall are $(0, s_0, s_0, s_0, r, r)$ and $(0, a, r, r, D, D)$. Their distance is $\max(r - s_0, D - r) = D - r$, since $r - s_0 = r(D - r)/(3D - r) < D - r$. The local inverse and local Fisher law of Section 5 are uniform on a compact segment of the base path through zero. Below the entrance threshold all competitors are in that base neighbourhood, proving (8.9). Above it, the localization proof for the remote cone bounds its exact coordinates by $O(\delta)$. The remote endpoint roots move by $O(\delta)$ and its interior double cluster by $O(\sqrt{\delta})$; the base inverse gives the same $O(\sqrt{\delta})$ root bound. A remote minimizer and the centre are both feasible, and the exterior observation gap excludes every other region. The bottleneck triangle inequality proves (8.10).

For the last assertions, $b(s_0 + \varepsilon) = D - b\varepsilon + O(\varepsilon^2)$. The function R_s has a strict nondegenerate maximum at $b(s)$ near the wall. Taylor expansion of $R_s(D)$ there, followed by $x \mapsto x/(x - 1)$, gives

$$c_D(s) - c_*(s) = \frac{-R''_s(b(s))}{2(R_s(b(s)) - 1)^2} (D - b(s))^2 + O((D - b(s))^3).$$

This proves (8.11). The strict weight margin keeps $\gamma/\pi > c_D(s)$ for small ε , so the entire interval $[c_*(s), c_D(s)]$ is feasible. Its upper endpoint has a root exactly at D . A remote spectrum has

increasing ordering $(0, a_c, r, b_-(c), b_+(c))$, with $s < a_c < r < b_-(c) \leq b_+(c) \leq D$. Its distance to the base is exactly $\max(r - s, b_+(c) - r) \leq D - r$, and equality is attained at $c_D(s)$. The remote spectra converge uniformly to one spectrum as $\varepsilon \downarrow 0$; their mutual diameter is therefore less than $D - r$ for small ε . Theorem 5.1 excludes any other exact component. This proves (8.12). $\square$

8.1. Exact critical-ray examples. We record two algebraic specializations of the second-order formula. Use the clocks $(6/5, 7/5, 8/5, 9/5, 2, 5/2, 3)$ with equal weights, $u_1 = u_2 = (1/3, 2/3)^\top$, and $v_1 = (1/4, 3/4)^\top$, $v_2 = (3/4, 1/4)^\top$.

Example 8.3 (An endpoint wall with a positive critical correction). Take $D = 1$, $r = 3/5$, $s_0 = 1/2$, $\alpha = 2/5$, $\pi = 1/10$. The remote coordinates and the interval-opening coefficient are

$$a = \frac{4}{7}, \quad c_* = \frac{32}{7}, \quad \beta = \frac{21}{160}, \quad v_f^* = \frac{1}{556}(289, 267)^\top, \quad \dot{b} = 8, \quad \Lambda_B = \frac{38400}{49}.$$

The projected three-variable program has the empty support as its only successful independent-support test. Its linear vector is strictly negative entrywise, so the minimizer has all three nonnegative coordinates zero. The free score has rank seven; hence the first-order minimizer is unique, including the endpoint split. Exact rational arithmetic and positive-root isolation give

$$0.000063237134157102 < \mu_B < 0.000063237134157103, \\ 0.071857358263070 < k_B < 0.071857358263071.$$

Consequently the remote branch is excluded on $t = \mu_B \delta$ for all sufficiently small positive δ . On the opposite side the exact spectral diameter is $2/5$, despite the nonzero quadratic pencil-interval opening.

Example 8.4 (The weight-wall critical ray). At the weight-wall datum of Example 6.7, the four entries of the projected linear vector g_* are strictly negative. The empty cone support is therefore the unique first-order minimizer, with its seven free coordinates recovered by the inverse free Gram matrix. The second-order formula gives

$$0.011285405417542 < k_E < 0.011285405417543.$$

Thus the critical ray $t = \mu_E \delta$ also excludes the remote branch for this datum when $\delta > 0$ is sufficiently small. This sign is a property of this specialization, not a convention at an unresolved ray and not a sign assertion for the whole wall family.

For reproducibility, the finite arithmetic behind both examples can be specified without numerical differentiation. Write the exact remote and base probability expansions at the selected first-order minimizer as

$$P_{\text{rem}}(\delta) = P + \delta A + \delta^2 C + O(\delta^3), \quad P_{\text{base}}(\delta) = P + \delta B + \delta^2 D + O(\delta^3).$$

The rational coefficients are obtained by multiplying the exact polynomial, weight and channel jets and dividing by the normalizer. The square-root expansion then gives

$$\mu^2 = \frac{1}{4} \sum_{j,e} \frac{w_j (A_{je} - B_{je})^2}{P_{je}}, \\ \mu k = \sum_{j,e} w_j \left\{ \frac{(A_{je} - B_{je})(C_{je} - D_{je})}{4P_{je}} - \frac{(A_{je} - B_{je})(A_{je}^2 - B_{je}^2)}{16P_{je}^2} \right\}. \quad (8.13)$$

Strict empty-support KKT inequalities and positive semidefiniteness of the projected Gram matrix prove uniqueness of the cone optimum even when that matrix is singular: the difference of objectives is $z^\top Q z - 2g^\top z > 0$ for every nonzero $z \geq 0$. The rational values of μ^2 , μk , the Gram data and isolating intervals are retained with the exact regression. The interval tests square

rational endpoints and compare integers after clearing positive denominators. They certify these finite specializations only.

9. REAL MONOMIAL DATA IN PARAMETER FAMILIES

We work first with a compact semialgebraic model X , a continuous semialgebraic probability map F , and a continuous semialgebraic monic target polynomial A_x of fixed degree n . Complex coefficients are represented by real and imaginary parts. Suppose the target is constant on $F^{-1}(P)$, with value $A_0(z) = \prod_r (z - r)^{m_r}$. Disjoint isolating discs and compactness give, on a sufficiently small closed observation neighbourhood, unique monic cluster factors

$$A_{r,x}(r + z) = z^{m_r} + \sum_{a < m_r} c_{r,a}(x) z^a, \quad k_{r,a} = m_r - a. \quad (9.1)$$

Multiplication of coprime monic factors has invertible differential. The cluster factors are consequently analytic functions of the total coefficients. Their graphs with the isolating conditions are semialgebraic.

Definition 9.1 (Input and families). An algebraic input is a finite first-order polynomial presentation of X, F, A, P , design weights, isolating data, denominator exclusions and branch choices. A real algebraic constant is encoded by an integer polynomial and a rational isolating interval; a complex constant is encoded by two real algebraic constants. A fixed-format family has a semialgebraic parameter set Θ , a semialgebraic $X \subset \Theta \times K$ with K compact and compact fibres, continuous semialgebraic maps F, A on X , and a semialgebraic choice $P_\theta \in F_\theta(X_\theta)$. Degrees and numbers of coordinates are fixed. Effectivity refers to algebraic presentations and numerical outputs to algebraic parameter values. No complexity bound is asserted.

Adjoin nonnegative square-root coordinates to the observation graph and write

$$R_{je} = \sqrt{w_j}(\sqrt{F(x)_{je}} - \sqrt{P_{je}}), \quad G = \sum_{j,e} R_{je}^2, \quad H_{r,a} = (\Re c_{r,a})^2 + (\Im c_{r,a})^2.$$

They are polynomial functions on the resulting compact real graph Γ . Zero probability cells are permitted. No passage to a Fisher quadratic form is used in this construction.

Definition 9.2 (Real monomial presentation). A real monomial presentation is a finite collection of smooth real algebraic sources and proper maps, collectively covering a neighbourhood of $G = 0$ in Γ , with finite algebraic etale coordinate presentations. On each source component either $G \equiv 0$ and all $H \equiv 0$, or

$$G = u \prod_i |y_i|^{a_i}, \quad H_{r,a} \equiv 0 \quad \text{or} \quad H_{r,a} = v_{r,a} \prod_i |y_i|^{b_{r,a,i}}. \quad (9.2)$$

The units have positive upper and lower bounds on smaller closed real coordinate boxes above $G = 0$. A contributing divisor has $a_i > 0$ and a real point away from the other divisors, with an admissible transversal on which $G > 0$. An identically zero H has order $+\infty$; a component $G \equiv 0$ is recorded separately and contributes no ratio. Properness belongs to the source maps, not to an open coordinate box. The finite encoding of an adapted coordinate presentation is specified in Appendix A.

Lemma 9.3 (Spreading a finite resolution certificate). *Let $S = \text{Spec } A$ be an integral affine variety over a characteristic-zero field, let $K = \text{Frac}(A)$, and let $Y \rightarrow S$ be of finite type. Suppose that over K there is a finite reduced closed chain*

$$Y_{0,K} = Y_{K,\text{red}} \supset Y_{1,K} \supset \cdots \supset Y_{q,K} = \emptyset,$$

projective maps $Z_{j,K} \rightarrow Y_{j,K}$ with smooth sources, specified inverse isomorphisms over $Y_{j,K} \setminus Y_{j+1,K}$, and finite smooth-centre principalizations of specified functions on those sources. There is a

nonempty open S° on which the following hold on every fibre. The closed chain is flat with geometrically reduced fibres and ends in the empty set; the inverse maps remain inverse over its actual successive complements. Sources and centres are relatively smooth, principalization blow-ups commute with fibre formation, and the final sources have relative normal-crossing divisors with fixed integer monomial orders on a finite adapted étale atlas. Zero identities and inverse units persist. A nonzero monomial cannot become identically zero on a component of a retained fibre. For algebraic input, such finite data and an exceptional set are effectively constructible.

Proof. Lemma A.2 descends the complete diagram, including the actual open complements and both inverse identities, and imposes flatness and geometric reducedness. Lemma A.1 proves the needed base change for all powers of every centre ideal after one localization per finite collection of charts. The explicit bad-locus construction in Appendix A treats relative smoothness, normal-crossing intersections, chart coverage, units and zero flags separately, and proves that each discarded algebraic image has proper closure. The nonzero-specialization assertion follows from relative codimension one and the inverse units, not from generic nonvanishing alone. The constructive certificate specification in Appendix B proves the last assertion. $\square$

Theorem 9.4 (Relative real principalization). *On the exact-target identifying locus of a fixed-format family there is a finite semialgebraic stratification with fixed cluster multiplicities, a fixed finite collection of algebraic source chart presentations, fixed integer orders and fixed real-accessibility flags as in Definition 9.2. The source maps are fibrewise proper real covers. Charts, units and box sizes may vary with the parameter. On compact subsets of a stratum the cover is usable over a common sufficiently small observation neighbourhood. The construction is effective for the algebraic presentations of Definition 9.1.*

Proof. Lemma A.3 algebraizes the proper enlarged graph by square-slack lifts without discarding strict exclusions or real branches. Resolve the reduced generic fibre and its successive reduced singular loci, then principalize the nonzero functions on the smooth sources. Lemma 9.3 spreads this finite certificate. The explicit exceptional algebraic loci in Appendix A are reconstructed on reduced bases of strictly smaller dimension, giving finite termination. Lemma A.4 proves both inclusions in the exact real coverage assertion, including lower-dimensional real loci, and descends the selected label branches. Lemma A.5 supplies the real transversals and the finite real flag partition. That partition is not confused with the algebraic dimension-drop recursion. The final proof in Appendix A supplies the uniform neighbourhood and unit bounds by properness over compact parameter subsets. These arguments prove the asserted cover on the original graph, rather than on an algebraic relaxation of its inequalities. $\square$

Proposition 9.5 (Fixed-centre real cover). *Every graph Γ above has a real monomial presentation. Exact pieces and lower-dimensional real loci are included. Every recorded divisor has a real admissible attaining arc.*

Proof. Apply Theorem 9.4 to a one-point base and use Lemma A.5. $\square$

Corollary 9.6 (Intrinsic scalar orders). *For each nonrigid cluster coefficient put*

$$\beta_{r,a} = \min_{\text{recorded real divisors}} \frac{b_{r,a,i}}{a_i}, \quad \alpha = \min_{r,a} \frac{\beta_{r,a}}{k_{r,a}}. \quad (9.3)$$

Either the target is constant on an observation neighbourhood, or $\alpha > 0$ is rational and $\omega_P(t) \asymp t^\alpha$. Every finite coefficient minimum is attained by an admissible arc. The numbers are independent of the monomial presentation.

Proof. The complete real divisor list and the monomial inequalities in Lemma A.5 give $H \leq CG^\beta$, hence $|c_{r,a}| \leq Ch_w^\beta$. A minimizing transversal satisfies $G \asymp u^{a_i}$ and $H \asymp u^{b_i}$ and attains

the exponent. The inclusion $\{G = 0\} \subseteq \{H = 0\}$ forces positive contributing orders. If all coefficients vanish on an observation neighbourhood, the target is locally rigid. Otherwise, for a monic polynomial of degree m , its greatest root modulus is comparable to $\max_{a < m} |c_a|^{1/(m-a)}$. Elementary symmetric identities give one bound; dominance of z^m outside a fixed multiple of this maximum gives the other. The centre lies in every ball, and the diameter is between its greatest centre displacement and twice that displacement. This proves the order of ω_P . The optimal coefficient inequality is intrinsic, so its attained exponent is independent of the presentation. The scalar divisor-ratio calculation is classical; compare [6]. $\square$

10. JOINT SPECIALIZATION AND UNIFORM SPECTRAL ASYMPTOTICS

On a nonrigid identified stratum write $\alpha = p/q$ in lowest terms. Let $\Psi(\theta, \varepsilon, x, U, V)$ denote graph membership and

$$R_{je}(\theta, x) = \varepsilon^q U_{je}, \quad c_{r,a}(\theta, x) = \varepsilon^{pk_{r,a}} V_{r,a}, \quad \sum_{j,e} U_{je}^2 \leq 1. \quad (10.1)$$

Complex equalities denote two real equalities. Define the joint family by the fixed-centre formula

$$\begin{aligned} (\theta, U, V) \in \mathcal{J} &\iff \theta \in S \text{ and } \forall \rho > 0 \exists \varepsilon, x, U', V' : \\ &0 < \varepsilon < \rho, \quad \Psi(\theta, \varepsilon, x, U', V'), \quad \|(U', V') - (U, V)\|^2 < \rho^2. \end{aligned} \quad (10.2)$$

Put $\mathcal{J}_0(\theta) = \mathcal{J}_\theta$ and $\mathcal{C}_0(\theta) = \text{pr}_V \mathcal{J}_0(\theta)$. There is no quantified θ' in this formula. Closure with a drifting centre is a different operation, and taking independent closures in the coefficient coordinates is also a different operation.

Theorem 10.1 (The definable joint initial family). *The sets $\mathcal{J}_0(\theta)$ and $\mathcal{C}_0(\theta)$ form semialgebraic families of nonempty compact sets. The coefficient images of the radius- t observation ball, scaled by $c_{r,a}/t^{\alpha_{kr,a}}$, converge in Hausdorff distance to $\mathcal{C}_0(\theta)$ for every fixed θ . If*

$$Z_r(V) = Z \left(z^{m_r} + \sum_{a < m_r} V_{r,a} z^a \right), \quad \mathcal{L}_0(\theta) = \{(Z_r(V))_r : V \in \mathcal{C}_0(\theta)\},$$

then the scaled cluster-root images converge to $\mathcal{L}_0(\theta)$, and

$$\omega_\theta(t) = C(\theta)t^\alpha + o(t^\alpha), \quad C(\theta) = \max_{V, V' \in \mathcal{C}_0(\theta)} \max_r d_\infty(Z_r(V), Z_r(V')) \in (0, \infty). \quad (10.3)$$

This is the diameter of a joint feasible image, not that of a Cartesian product of scalar envelopes.

Proof. Corollary 9.6 bounds every scaled coefficient for small t ; U is already bounded. The fixed-centre closure is nonempty and compact. Quantifier elimination proves semialgebraicity with θ as a free parameter. Compactness of the graph fibre lets one pass to convergent subsequences of the unscaled graph variables when projecting an initial point, so projection loses no limiting coefficient point.

Fix θ and a point of the outer coefficient limit. Its distance to the positive-radius coefficient section is a bounded nonnegative semialgebraic function of t , with a subsequence tending to zero. The one-variable monotonicity theorem makes its limit zero. A finite net of the compact outer limit gives the inner Hausdorff inclusion. Failure of the outer inclusion would give a bounded convergent subsequence of scaled coefficients outside the defining limit, a contradiction. This proves convergence of the entire ball. Continuity of monic root multisets on bounded coefficient sets gives the corresponding root-set convergence.

Let Δ be the least separation of distinct base roots. For small t , each cluster lies in its $\Delta/4$ disc. A matching within clusters has error tending to zero; a cross-cluster matching has an edge of length at least $\Delta/2$. An optimal matching therefore stays within clusters. Translation and scaling there are exact. Taking the diameter gives (10.3). An attaining coefficient transversal

prevents the limiting diameter from being zero. This also explains why all inequality faces and joint relations in the closure remain relevant. $\square$

Proposition 10.2 (Semialgebraic decay on strata). *Let $e(\theta, t) \geq 0$ be semialgebraic for $0 < t < \tau(\theta)$, with $\tau > 0$ semialgebraic, and suppose $e(\theta, t) \rightarrow 0$ for each fixed θ . After a finite Nash stratification, on every stratum a rational $\nu > 0$ and a positive continuous semialgebraic $\tau_0(\theta)$ satisfy*

$$0 < t < \tau_0(\theta) \implies e(\theta, t) \leq t^\nu.$$

The estimate is uniform on compact subsets; ν need not be sharp.

Proof. Lemma B.1 gives a finite positive rational candidate list from the polynomial supports of a quantifier-free graph formula, including specialized zero coefficients and the eventual-zero case. Choose ν below its positive candidates. The complete argument in Appendix B selects a genuinely positive semialgebraic threshold, makes it continuous, and takes its minimum only on a compact subset of a stratum. No uniform rate is inferred from pointwise convergence alone. $\square$

Theorem 10.3 (Uniform spectral power law). *The identified nonrigid locus has a finite Nash stratification on which α is fixed, $\mathcal{L}_0$ has constant semialgebraic topological type, and $C(\theta)$ is positive and Nash. For every compact subset H of a stratum, for some $\nu > 0$,*

$$d_H(\mathcal{L}_t(\theta), \mathcal{L}_0(\theta)) = O_H(t^\nu), \quad \omega_\theta(t) = C(\theta)t^\alpha + O_H(t^{\alpha+\nu}). \quad (10.4)$$

Here $\mathcal{L}_t$ is the exact clustered root image scaled by $t^{-\alpha}$ and d_H uses the maximum cluster bottleneck metric.

Proof. Represent each root multiset by all its labelled arrays through elementary symmetric identities. The formula

$$M_d(z, z') = \bigvee_{\sigma \in \mathfrak{S}_m} \bigwedge_{i=1}^m |z_i - z'_{\sigma(i)}|^2 \leq d^2 \quad (10.5)$$

expresses bottleneck distance at most d . Use the conjunction over clusters, and real and imaginary parts for complex roots. The two inclusions

$$(\forall z \in \mathcal{L}_t)(\exists z' \in \mathcal{L}_0)M_d(z, z'), \quad (\forall z' \in \mathcal{L}_0)(\exists z \in \mathcal{L}_t)M_d(z, z')$$

and minimality over $d \geq 0$ define the Hausdorff error. Compactness gives attainment. Rational powers of t are encoded by positive algebraic roots, so this error is semialgebraic. Theorem 10.1 proves its pointwise limit zero; Proposition 10.2 gives the compact-uniform power bound.

The diameter has a first-order formula saying that all pairs have distance at most C and one pair attains it. It is therefore semialgebraic and positive. Lemma B.2 gives its Nash refinement. Refine also by positive cluster separation and observation-radius choices, so that compact minima exclude cross-cluster matching uniformly. Hardt triviality supplies the topological types separately. Finally diameters differ by at most twice Hausdorff distance, which gives the second estimate. Topological triviality has supplied neither the error rate nor the Nash dependence. $\square$

Theorem 10.4 (A finite real spectral atlas). *A fixed-format family admits a finite semialgebraic stratification retaining the following data: the semialgebraic type and dimension of the exact target coefficient fibre and its root image; on the identified locus, rigidity or a positive rational exponent, cluster multiplicities, and a fixed real monomial order list with accessibility and minimizing indices; on its nonrigid part, the joint initial coefficient and root-set types, a Nash leading diameter and the compact-uniform power remainder; and finite algebraic descriptions of these objects. On a subfamily with a proved Fisher-cone representation, the active supports and choices of leading constants can be retained in the same refinement. On a nonidentified stratum the modulus tends to the positive diameter of the exact target root image. No general leading set is asserted to be quadratic, and no local leading constant excludes a remote exact component.*

Proof. The exact coefficient fibre is $\{A_x : F_\theta(x) = P_\theta\}$, not the parameter fibre itself. Apply Hardt triviality to its compact image and root graph. Singleton images are selected by a first-order formula. At fixed θ , compactness shows that decreasing observation balls converge in target image to the exact image, so their diameters converge to its diameter. On the singleton locus apply Theorem 9.4, Corollary 9.6, and Theorems 10.1 and 10.3. These are separate constructions entering one finite refinement. A proved Fisher subfamily contributes only finitely many additional support and comparison conditions. Effective descriptions are provided by the following theorem and Appendix B. $\square$

11. EFFECTIVE OUTPUT AND THE ROLE OF COMPUTATION

Theorem 11.1 (Effective algebraic specialization). *For an algebraic input on the identified locus, finite algorithms compute local rigidity, the nonrigid exponent α , quantifier-free presentations of $\mathcal{I}_0, \mathcal{C}_0, \mathcal{L}_0$, and an integer polynomial with a rational isolating interval for C . Exact identification itself is decidable. In algebraically presented parameter families the corresponding strata and formulas are effectively describable. These assertions are independent of any finite example regression.*

Proof. Appendix B gives the complete input–output procedure. It constructs each maximum $m_H(s) = \max\{H : G \leq s\}$ by elimination, extracts a finite support-pair list, and tests the comparability formula (B.1) for every candidate. The unique true candidate selects the actual positive maximum-function branch, while a separate formula detects eventual zero. No unspecified branch-isolation oracle is used. Substitute the resulting p/q in (10.2), eliminate, adjoin roots and use (10.5) to compute the leading diameter. Real-root isolation and field-norm conversion provide its integer polynomial and isolating interval. With free parameters the same finite tests compute strata, followed by Nash refinement. A second, explicit finite-certificate search for the geometric presentations is given in that appendix; spectral outputs do not depend on executing it. $\square$

The symbolic calculations accompanying the explicit polynomial examples test identities and specified finite optimization problems. They do not implement the universal algorithms above and are not proof certificates for relative principalization, uniform asymptotics, or arbitrary degree.

12. THE CONSTRAINED WEIGHTED DEFORMATION

We identify the object of Section 10 before discussing its metric consequences. The datum, the weight filtration and the observation norm are part of the object. A change of parametrization is not a change of any of these three data.

Fix an identified nonrigid datum. Let S be the germ at the origin of the *image* of the actual observation–coefficient relation

$$x \mapsto (R(x), c(x)).$$

All real and imaginary parts are included. The defining inequalities and selected real branches are retained in this image. Set $w = (q, \dots, q, (pk_{r,a})_{r,a})$ when $\alpha = p/q$, and write $D_\varepsilon y = (\varepsilon^{w_i} y_i)_i$. Define

$$\mathfrak{D}_w^+(S) = \overline{\{(\varepsilon, D_\varepsilon^{-1} y) : \varepsilon > 0, y \in S\}}, \quad T_w^+ S = \{v : (0, v) \in \mathfrak{D}_w^+(S)\}. \quad (12.1)$$

Closure is ordinary real closure, locally in the displayed coordinates. Replacing a representative of the germ by a smaller representative does not change its zero fibre. The superscript emphasizes positive real specialization, not a new algebraic deformation operation.

Theorem 12.1 (Comparison and real liftability). *The fixed-centre joint initial fibre satisfies*

$$\mathcal{I}_0 = T_w^+ S \cap \{(U, V) : \|U\| \leq 1\}, \quad \mathcal{C}_0 = \text{pr}_V \mathcal{I}_0. \quad (12.2)$$

Thus it is the observation-unit section of the positive-real weighted tangent relation. It depends on the relation germ, not on a resolution, a square-slack presentation or a proper surjective presentation of that relation.

Suppose an algebraic presentation of the relation has ideal I over $\mathbb{R}$, with its actual real branch conditions specified separately. Form the algebraic weighted deformation by substitution $y_i = \varepsilon^{w_i} v_i$ and saturation by ε . Its zero-fibre ideal is the weight-initial ideal of I , with the minimum-weight convention. The set $T_w^+ S$ consists exactly of those real points of this algebraic zero fibre which have a positive-real semialgebraic lift satisfying the original branch conditions. Merely taking all real points of the algebraic initial fibre can give a strictly larger set, even when I is real prime.

Proof. The forward inclusion in (12.2) follows from the defining substitutions (10.1). For the reverse inclusion take a sequence $(\varepsilon_n, U_n, V_n)$ in the positive deformation with limit $(0, U, V)$ and $\|U\| \leq 1$. If $\|U\| < 1$, it eventually satisfies the unit constraint. At a boundary point put

$$a_n = \max(1, \|U_n\|^{1/q}), \quad \varepsilon'_n = a_n \varepsilon_n.$$

Then $a_n \rightarrow 1$, and the same unscaled relation point has coordinates $U'_n = a_n^{-q} U_n$, $V'_{r,a,n} = a_n^{-pk_{r,a}} V_{r,a,n}$ at ε'_n . These satisfy the unit constraint and have the same limit. This proves the boundary inclusion, which need not follow by interchanging an arbitrary intersection and closure. The coefficient bounds in Corollary 9.6 make this section compact. Image germs are unchanged by the stated surjective presentations, proving the presentation assertion.

Algebraically, contract the ideal generated by $f(D_\varepsilon v)$, $f \in I$, from $\mathbb{R}[\varepsilon, \varepsilon^{-1}, v]$ to $\mathbb{R}[\varepsilon, v]$. This is precisely saturation by ε . For each f , divide $f(D_\varepsilon v)$ by its lowest power of ε ; its value at zero is the minimum-weight part of f . Taking all $f \in I$, or a weight-adapted Groebner basis, shows that the special-fibre ideal is $\text{in}_w I$. This is the usual associated-graded construction, not an additional resolution theorem.

Every real limiting point satisfies the saturated equations. Conversely, a real point is in the constrained zero fibre precisely when it is a limit from the specified positive real part. Semialgebraic curve selection supplies an arc in that part; its positive first coordinate can be used as parameter after a finite Puiseux reparametrization. This proves the liftability characterization. For strictness, take $S = \{(x, y) : y^2 = x^3\}$ at the origin with weights $(1, 1)$. The saturated equation is $v_y^2 = \varepsilon v_x^3$. Its algebraic zero fibre has real points $v_y = 0$, with either sign of v_x . Positive real liftability instead requires $v_x \geq 0$, and every such point is realized. The ideal $(y^2 - x^3)$ is real prime: its coordinate ring embeds in $\mathbb{R}[t]$ by $(x, y) = (t^2, t^3)$. Thus even real primality does not recover the missing accessibility condition from the initial ideal alone. $\square$

For weight one the algebraic construction is the tangent-cone instance of the normal-cone construction; see [12, 18]. Weighted initial ideals are treated in [15]. These algebraic objects do not, by themselves, specify the positive real liftable part or the observation-unit section. Inner-metric tangent limits, as in [2], retain different data: here one rescales a constrained *outer* relation and subsequently applies the specified root map. No identification with an inner-metric tangent cone is asserted.

Theorem 12.2 (Filtered naturality). *Let ϕ be a semialgebraic homeomorphism between two relation germs, fixing the origin. Suppose, locally uniformly on bounded sets,*

$$\phi_\varepsilon = D_\varepsilon^{-1} \phi D_\varepsilon \longrightarrow \phi_0, \quad (\phi^{-1})_\varepsilon \longrightarrow \phi_0^{-1},$$

where ϕ_0 is a homeomorphism. Then $T_w^+ \phi(S) = \phi_0(T_w^+ S)$. If the observation block of ϕ_0 preserves its norm, the corresponding unit sections are also carried onto each other. The same statement holds with the norm transported by ϕ_0 . With the target decoding transported at the same time, the leading spectral image and its diameter are unchanged.

Proof. For a bounded convergent sequence of rescaled relation points v_n , uniform convergence on a compact neighbourhood gives $\phi_{\varepsilon_n}(v_n) \rightarrow \phi_0(\lim v_n)$. This proves one inclusion of tangent relations. Apply the inverse hypothesis for the other. Theorem 12.1 then identifies their unit sections; the last assertion is equality of the decoded images. For example, weighted analytic coordinate changes whose invertible leading graded map is ϕ_0 and whose remaining terms have strictly larger weight satisfy the stated convergence, as do their filtered inverses. $\square$

The transport of the metric is essential. Equivalent observation norms preserve optimal orders, but need not preserve a leading diameter. Section 15 gives this distinction inside the full binary experiment, without adding a discrete experimental label.

12.1. Exactly which scalar probes recover the joint set. Write C_t for the coefficient section scaled as in Theorem 10.1. For $a \in \mathbb{R}^N$ let

$$m_t(a) = \min_{v \in C_t} \|v - a\|^2. \quad (12.3)$$

The dimension N includes both parts of complex coefficients.

Theorem 12.3 (Quadratic reconstruction and finite witnesses). *The limits $m_0(a) = \lim_{t \downarrow 0} m_t(a)$ exist locally uniformly in a and satisfy*

$$m_0(a) = \text{dist}(a, \mathcal{C}_0)^2, \quad \mathcal{C}_0 = \bigcap_{a \in \mathbb{Q}^N} \{v : \|v - a\|^2 \geq m_0(a)\}. \quad (12.4)$$

In particular these scalar probes, unlike their vanishing orders alone, determine the joint coefficient set and hence its decoded spectral set. For compact nonempty sets C, D in a fixed compact ball B , and a finite η -net $A \subset B$ (with an arbitrarily slightly enlarged ball if rational points are required),

$$\max_{a \in A} |\text{dist}(a, C) - \text{dist}(a, D)| \leq d_H(C, D) \leq \max_{a \in A} |\text{dist}(a, C) - \text{dist}(a, D)| + 2\eta. \quad (12.5)$$

For algebraic input one quantifier-free formula with a free describes $m_0(a)$. Distinct initial fibres therefore admit an effectively searchable rational probe with unequal limiting values.

Proof. For compact sets the distance functions differ uniformly by at most their Hausdorff distance. Squaring is uniformly Lipschitz on bounded distance intervals. Theorem 10.1 therefore proves the first assertion. A point of $\mathcal{C}_0$ satisfies every inequality in (12.4). If $v \notin \mathcal{C}_0$, put $d = \text{dist}(v, \mathcal{C}_0) > 0$ and choose rational a with $\|a - v\| < d/3$. Then $\text{dist}(a, \mathcal{C}_0) > 2d/3 > \|a - v\|$, excluding v .

The left inequality in (12.5) is the distance-function bound. The Hausdorff distance is attained as $\text{dist}(x, D)$ for some $x \in C$, or with C, D exchanged. At such an x it equals $|\text{dist}(x, C) - \text{dist}(x, D)|$. Each distance function is one-Lipschitz; replacing x by a net point proves the other inequality.

Finally, membership in C_t , minimization in (12.3), and the limit are first-order formulas over a real closed field, with a free. Theorem 11.1 and quantifier elimination yield one finite formula for their limit graph. If $C \neq D$, the strict separating inequality above persists on an open set of probe centres. Enumeration of rational points, with exact comparison of the resulting algebraic values, must find a witness. The enumeration concerns a witness only, not an approximation to the whole set by scalar coordinate envelopes. $\square$

All affine support values determine only a closed convex hull. Quadratic distance probes determine the actual compact set, including its nonconvex real faces. The reconstruction theorem thus gives a precise sufficiency statement to accompany, rather than overstate, the insufficiency statements of Sections 14 and 15.

13. REAL RESIDUES AND FILTERED CHANGES OF OBSERVATION

We compare the weighted relation with signed valuation geometry before imposing its observation metric. Throughout, positive-real refers to positive values of the degeneration parameter. It does not restrict the coordinates of a real leading coefficient to the positive orthant.

Let $\mathbb{K} = \mathbb{R}\langle t \rangle$ be the real closed field of real algebraic Puiseux series, ordered by positive leading coefficient, with t positive infinitesimal. Write val for its rational valuation, $\mathcal{O}$ for the valuation ring, and $\text{res} : \mathcal{O} \rightarrow \mathbb{R}$ for its residue map. We use the convention $\text{val}(0) = +\infty$. A semialgebraic germ and its defining inequalities are interpreted over $\mathbb{K}$ by the same real-closed-field formulas.

Let $S \subset \mathbb{R}^n$ be the actual semialgebraic observation-coefficient relation germ at zero and let $w \in \mathbb{Q}_{>0}^n$. Put

$$\mathcal{B}_w(S) = \{y \in S(\mathbb{K}) : \text{val}(y_i) \geq w_i \text{ for all } i\}.$$

Only the germ matters, since every point of this set is infinitesimal. The coefficientwise residue below is taken after multiplication by t^{-w_i} .

Theorem 13.1 (Weighted real residue comparison). *The positive-real weighted tangent relation satisfies*

$$T_w^+ S = \text{res}(t^{-w} \mathcal{B}_w(S)). \quad (13.1)$$

For every sign vector $\sigma \in \{-, +\}^n$, its part with all coordinates nonzero has the exact comparison

$$\begin{aligned} \exists v \in T_w^+ S \cap (\mathbb{R}^*)^n, \quad \text{sgn } v = \sigma \\ \iff \exists y \in S(\mathbb{K}), \quad \text{val } y = w, \quad \text{sgn } y = \sigma. \end{aligned} \quad (13.2)$$

Thus forgetting the nonzero residue magnitudes gives the fixed-weight fibre of the coordinate signed valuation map. Imposing $S \subset \mathbb{R}_{>0}^n$ gives its positive-orthant specialization. Zero coordinates on the left of (13.1) also retain lifts of higher valuation; they must not be discarded by requiring equality of every coordinate valuation.

For the observation and coefficient blocks of Theorem 12.1, the manuscript's joint object is consequently the following operation:

$$\boxed{\mathcal{B}_w(S) \xrightarrow{\text{res}(t^{-w} \cdot)} T_w^+ S \xrightarrow{\|U\| \leq 1} \mathcal{I}_0 \xrightarrow{\text{target projection and roots}} \mathcal{L}_\theta.} \quad (13.3)$$

The observation-unit section and spectral decoder in this formula are additional metric data, not part of the coordinate signed valuation.

Proof. A point on the right of (13.1) is represented by algebraic Puiseux series. After a common positive ramification these series converge on a small positive interval. Transfer of the finite semialgebraic defining formula, or equivalently the eventual signs of its convergent algebraic functions, makes the resulting real points belong to S . Their rescaled limit is their displayed residue, proving one inclusion.

Conversely let $(0, v)$ lie in the closure of $\{(\epsilon, \epsilon^{-w} y) : \epsilon > 0, y \in S\}$. This is a semialgebraic set after clearing the weight denominators. Real curve selection gives a semialgebraic arc with positive first coordinate approaching $(0, v)$. Its first coordinate is nonconstant and eventually strictly monotone. Reparametrize by that coordinate. Each component is an algebraic Puiseux germ in ϵ , so it gives a point of $\mathbb{K}$ after renaming ϵ as t . Boundedness of the rescaled arc says $\text{val } y_i \geq w_i$, and its limit is $\text{res}(t^{-w} y)$. This proves equality. A rescaled coordinate has nonzero residue precisely when its valuation is exactly its prescribed weight; its sign is then the residue sign. This proves (13.2). The observation-unit identity in (13.3) is Theorem 12.1, including its boundary rescaling argument at $\|U\| = 1$; it is not an unproved interchange of closure and intersection with a closed ball. $\square$

Jell–Scheiderer–Yu [19] develop real tropicalization for semialgebraic sets over an ordered nonarchimedean real closed field and identify real analytification with an inverse limit of such real tropicalizations. Under the normalization $|t| = e^{-1}$, the right side of (13.2) is the fibre of their *coordinate* signed valuation map at $(\sigma_i e^{-w_i})_i$. Formula (13.1) retains the full residues before that projection. We make no claim that the full inverse-limit real analytification loses the same information as one coordinate tropicalization. The fixed observation norm and root decoder are specified here independently of that inverse-limit construction.

Positive tropical geometry asks for positive lifts, rather than all real points of a complex initial variety. The initial-ideal and lifting criteria studied by Rose–Telek [20] are directly relevant in that setting. In the present setting positivity applies to the scale; coordinate orthants and all feasibility inequalities must be kept separately. The following familiar nonsingular criterion records exactly when an initial-point calculation alone suffices locally.

Proposition 13.2 (A sufficient real lifting test). *After clearing weight denominators, suppose a local presentation of the weighted deformation is*

$$F(t, z) = 0, \quad g_1(t, z) > 0, \dots, g_s(t, z) > 0,$$

with Nash functions, and suppose these are the complete local equations and inequalities of the actual deformed relation. If $F(0, v) = 0$, $D_z F(0, v)$ has full row rank, and $g_i(0, v) > 0$ for every i , then $v \in T_w^+ S$. This is a sufficient test, not an equivalence at a singular initial point or at an active inequality.

Proof. Choose as many z coordinates as there are equations so that the corresponding Jacobian minor is invertible, and keep the remaining coordinates fixed at their values in v . The Nash implicit-function theorem gives $z(t)$ with $z(0) = v$ and $F(t, z(t)) = 0$. The strict inequalities persist for small positive t . Undoing the weighted substitution gives a real lift in the actual relation. $\square$

13.1. Naturality from finite weighted jets. We next derive the convergence needed for naturality from a condition on the presentation, instead of assuming that convergence. For $w \in \mathbb{Q}_{\geq 0}^n$ and $w' \in \mathbb{Q}_{\geq 0}^n$, call a Nash isomorphism germ $h : (\mathbb{R}^n, 0) \rightarrow (\mathbb{R}^n, 0)$ *bi-filtered* if every nonzero Taylor monomial y^α of h_j has $w \cdot \alpha \geq w'_j$, and the inverse satisfies the corresponding condition with w, w' reversed. Only finitely many monomials have weight below any w'_j ; the vanishing conditions are therefore conditions on finite jets. Let $h_{[w, w']}$ consist of the terms with equality of weights.

Theorem 13.3 (Finite-jet filtered naturality). *A bi-filtered Nash isomorphism satisfies, locally uniformly on bounded sets,*

$$D_{t, w}^{-1} h D_{t, w} = h_{[w, w']} + O(t^\eta), \quad \eta > 0, \tag{13.4}$$

where $D_{t, w} = \text{diag}(t^{w_i})$. The analogous statement holds for h^{-1} , and its weighted initial map is the inverse polynomial map of $h_{[w, w']}$. Consequently

$$T_{w'}^+(h(S)) = h_{[w, w']}(T_w^+ S).$$

Weighted initial maps preserve composition. They commute with the residue comparison (13.1). A positive ramification $t = s^q$ replaces w by qw and leaves the residue set unchanged. When the observation-unit section and spectral decoder are transported, so is the resulting metric fibre. Equality of untransported bottleneck diameters requires that the transported target map be an isometry.

Proof. Choose a common denominator for all the weights. In a finite Taylor polynomial, every term of weight strictly larger than its output weight has a gap bounded below by a positive

reciprocal integer. Such terms are $O(t^n)$ on bounded rescaled sets. Choose the Taylor order N so large that $(N+1) \min_i w_i > \max_j w'_j + \eta$. The analytic Taylor remainder is $O(\|D_{t,w} z\|^{N+1})$ uniformly for bounded z , and hence has the same bound after division by the output weights. This proves (13.4). The same argument applies to the inverse. Substitution in the identities $h^{-1}h = hh^{-1} = \text{id}$ shows that their limiting polynomial maps are mutually inverse; boundedness needed for substitution follows from the uniform estimates just proved. The same argument proves preservation of composition.

Apply these limits to any rescaled arc in S and then apply the inverse to obtain both inclusions of the tangent-set identity. Alternatively, valuation compatibility of the Taylor series and reduction modulo the positive-valuation ideal give

$$\text{res}(t^{-w'} h(y)) = h_{[w,w']}(\text{res}(t^{-w} y)),$$

which proves commutation with (13.1). Ramification follows by the substitution $t = s^q$. Finally a transported section and decoder have the same feasible decoded points by their definitions; the metric is preserved exactly when its transport is respected. $\square$

For example, at weights $(1, 2)$ the map $h(y_1, y_2) = (y_1, y_2 + y_1^2)$ is bi-filtered and its initial map is nonlinear. The theorem thus includes weighted changes beyond ordinary linear similarities, while stating precisely which metric data must be transported. Neither the residue comparison nor the implicit-function lifting test is presented as a new tropical fundamental theorem. The observation-specific finite determination is Theorem 4.1; the comparison specifies the real object to which that theorem applies.

14. SCALAR ENVELOPES DO NOT DETERMINE THE SPECTRAL INVARIANT

The distinction between the scalar orders and the joint initial fibre can be made exact. The following examples have not only the same optimal orders but the same individual coefficient envelopes at every sufficiently small radius. Their leading spectral diameters are nevertheless different. The assertion concerns the scalar data of Corollary 9.6 and separate coefficient envelopes; it does not concern the collection of value functions of all possible combinations of the coefficients.

Theorem 14.1 (Separation from marginal scalar data). *There are two compact semialgebraic probability experiments with a unique exact target, real quadratic targets with roots in $[0, 1]$, and identical coefficient orders and individual coefficient envelopes, for which*

$$\omega_A(t) = (1 + \sqrt{5})t, \quad \omega_B(t) = \frac{1 + \sqrt{17}}{2}t \quad (0 \leq t \leq 1/20).$$

In both experiments the coefficient orders are $(\beta_1, \beta_2) = (1, 2)$ and the spectral exponent is one. On each source arc the orders of (G, H_1, H_2) are $(2, 2, 4)$. Thus the order lists and the separate optimal coefficient envelopes do not determine the joint initial root set or its diameter.

Proof. Put $r = 1/2$ and $e = 1/20$. In each experiment the source is the disjoint union of two closed intervals $0 \leq \rho \leq e$. Its probability map and datum are

$$F(\rho) = ((1 - \rho^2/2)^2, \rho^2(1 - \rho^2/4)), \quad P_0 = (1, 0).$$

The entries are nonnegative and sum to one. Since $\sqrt{F_1} = 1 - \rho^2/2$ on this interval,

$$h(F(\rho), P_0)^2 = 2 - 2\sqrt{F_1} = \rho^2.$$

Associate with an interval labelled (u, v) the target

$$A_{u,v,\rho}(z) = (z - r)^2 + u\rho(z - r) + v\rho^2.$$

Use the label sets

$$\mathcal{E}_A = \{(1, -1), (2, -4)\}, \quad \mathcal{E}_B = \{(1, -4), (2, -1)\}.$$

These are compact semialgebraic experiments in the sense of Definition 9.1; the discrete label is simply an additional source coordinate. All exact targets are $(z - r)^2$. The two root increments on an interval are

$$\rho \ell(u, v), \quad \rho q(u, v), \quad \ell(u, v) = \frac{-u - \sqrt{u^2 - 4v}}{2} < 0 < q(u, v) = \frac{-u + \sqrt{u^2 - 4v}}{2}. \quad (14.1)$$

They lie in $[0, 1]$ after translation by r , since the greatest absolute increment is at most $(1 + \sqrt{5})/20 < 1/2$.

Write $c_1 = u\rho$ and $c_0 = v\rho^2$. In either experiment the exact marginal images over the radius- t ball are

$$\{c_1\} = [0, 2t], \quad \{c_0\} = [-4t^2, 0].$$

In particular $\max |c_1| = 2t$ and $\max |c_0| = 4t^2$, not just to leading order. Moreover $G = \rho^2$, $H_1 = u^2\rho^2$, and $H_2 = v^2\rho^4$. The order assertions follow. Even the unordered lists of individual squared units agree: they are $\{1, 4\}$ and $\{1, 16\}$. Their association across the two coordinates is different.

For $t > 0$, put $\lambda = \rho/t$. The scaled coefficient image is already independent of t :

$$\mathcal{C}_{0,E} = \bigcup_{(u,v) \in \mathcal{E}_E} \{(u\lambda, v\lambda^2) : 0 \leq \lambda \leq 1\}, \quad E \in \{A, B\}. \quad (14.2)$$

This is its joint initial coefficient set. The ordered root image consists of $(\lambda \ell(u, v), \lambda q(u, v))$. Its first coordinate is nonpositive, its second nonnegative, and the zero vector belongs to it. The diameter in the maximum metric is the larger of its two coordinate widths. Since $u > 0$, $-\ell(u, v) > q(u, v)$ on each interval. Therefore

$$C_E = \max_{(u,v) \in \mathcal{E}_E} \frac{u + \sqrt{u^2 - 4v}}{2}.$$

For A the maximum is $1 + \sqrt{5}$. For B it is $(1 + \sqrt{17})/2$, which exceeds $1 + \sqrt{2}$. The inequalities are strict and these two constants differ. Translation and multiplication by t in (14.1) prove the asserted exact diameter identities. No product approximation of (14.2) has been used. $\square$

The example isolates the information carried by the joint specialization: relations between weighted coefficients survive both real closure and the root map. A finite list of scalar exponents cannot recover these relations, even when augmented by the separate sharp envelopes. Conversely the full joint set determines the diameter by Theorem 10.1; no new scalar resolution theorem is required for this distinction.

15. IDENTICAL SCALAR ORDERS IN THE BINARY EXPERIMENT

The example in Section 14 keeps sharp marginal ranges fixed. We now keep a different, much larger valuative object fixed inside the full experiment of Section 2: all real analytic arc orders of the observation discrepancy and every polynomial scalar function of the target coefficients. Leading units are not included in this datum. The distinction is necessary, since Theorem 12.3 shows how richer scalar values recover the joint set.

Take degree one, $D = 1$, $\pi = 1/10$, clocks $(2, 3, 4)$ and weights $1/3$. Consider the two strictly interior full-rank centres θ_e , with $e \in \{1/4, 1/8\}$, given by

$$f = z - \frac{1}{4}, \quad g = z - \frac{3}{4}, \quad \alpha = \frac{1}{2}, \quad U_e = \begin{pmatrix} \frac{1}{2} + e & \frac{1}{2} - e \\ \frac{1}{2} - e & \frac{1}{2} + e \end{pmatrix}, \quad V = \begin{pmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{3}{4} & \frac{1}{4} \end{pmatrix}. \quad (15.1)$$

All competitors still range over the original closed binary model; no parameter is held known. In a common small open coordinate ball write

$$z = (\dot{r}, \dot{s}, \dot{\alpha}, \dot{u}_1, \dot{u}_2, \dot{v}_1, \dot{v}_2).$$

The first entries of the binary channel columns are the independent channel coordinates. Let $G_e(z)$ be squared Hellinger discrepancy from θ_e , and let $\Delta c(z)$ be the two target coefficient increments. The latter polynomial map is the same at both centres.

Theorem 15.1 (Valuative equivalence without metric equivalence). *For every nonconstant real analytic arc $z(\tau)$ through zero,*

$$\text{ord}_\tau G_{1/4}(z(\tau)) = \text{ord}_\tau G_{1/8}(z(\tau)) = 2 \min_i \text{ord}_\tau z_i(\tau). \quad (15.2)$$

For every polynomial H in the target coefficient increments, including all mixed polynomial combinations, the orders of $H(\Delta c(z(\tau)))$ agree at the two centres, with the same convention for identically zero functions. The corresponding accessible real divisorial order data can be computed on one common real monomial presentation and agree as well.

Nevertheless the two whole-model leading spectral sets are distinct. Their exponents are one, and their squared leading diameters are

$$C_{1/4}^2 = \frac{129609775953}{25280}, \quad C_{1/8}^2 = \frac{10058351786055}{1396384}. \quad (15.3)$$

Thus the complete scalar arc-order information just specified does not determine the joint metric initial fibre, even within the principal binary observation model.

Proof. Both channels are invertible and the component polynomials are coprime. Lemma 2.2 implies that the observation derivative in these seven coordinates is injective. Since all probability cells are positive, the square-root embedding is analytic and

$$G_e(z) = \frac{1}{4} z^\top H_e z + O(\|z\|^3), \quad H_e > 0.$$

It follows either by this expansion or by two-sided comparison with $\|z\|^2$ that (15.2) holds. The coefficient functions are literally identical in the common coordinates, proving their order assertion for every H .

For the divisorial assertion, first blow up the maximal ideal at zero. On every real affine chart the pullback of G_e is the square of an exceptional coordinate times a strictly positive analytic unit near the real exceptional locus. Indeed the leading factor is the positive quadratic form H_e evaluated on a nonzero real projective direction. These units remain nonzero under every further real modification. Principalize any finite specified collection of the common coefficient functions on this same space. The exceptional orders and their real accessibility are the same for both experiments. Alternatively every accessible divisor has a real transversal, so its observation order already follows from (15.2). The assertion concerns real accessible data; no equality of complex zero divisors of the two positive quadratic forms is used. Every additional finite collection of polynomial combinations is handled in the same way.

We compute the metric sets. Let S_e be the 12-by-7 score matrix from (2.3), at the three clocks, in the coordinate order above. Set

$$H_e = S_e^\top \text{diag}((3P_{e,je})^{-1}) S_e, \quad M_e = (H_e^{-1})_{\{1,2\},\{1,2\}}.$$

Profiling the five free nuisance coordinates in the positive definite quadratic form shows that the leading pair of root increments is exactly

$$L_e = \{x \in \mathbb{R}^2 : x^\top M_e^{-1} x \leq 4\}. \quad (15.4)$$

For completeness, the lower bound on G_e bounds all competitor coordinates by $O(t)$. Division by t and Taylor expansion give the outer inclusion in (15.4). Every vector in its interior is realized

by its linear parameter arc, which is feasible because the centre is interior. Boundary vectors are limits of their radial contractions. This proves Hausdorff convergence. Exact uniqueness and compactness, from Lemma 2.2, exclude competitors outside this coordinate neighbourhood at sufficiently small radii. The limit is consequently a whole-model limit.

Here the entries of the matrix are obtained by differentiating the explicit rational functions

$$P_{j,ab} = \frac{\alpha(T_j - r)u_{1a}v_{1b} + (1 - \alpha)(T_j - s)u_{2a}v_{2b}}{\alpha(T_j - r) + (1 - \alpha)(T_j - s)}.$$

Rational elimination gives

$$M_{1/4} = \frac{1}{404480} \begin{pmatrix} 129609775953 & 79361335413 \\ 79361335413 & 48691215153 \end{pmatrix}, \quad (15.5)$$

$$M_{1/8} = \frac{1}{22342144} \begin{pmatrix} 10058351786055 & 6157532670525 \\ 6157532670525 & 3781629625095 \end{pmatrix}. \quad (15.6)$$

These equalities can be checked solely by forming H_e from the preceding rational display and multiplying its inverse; no fitted numerical covariance is involved. For separated real roots the bottleneck metric is the maximum norm of their ordered increments. A centrally symmetric ellipsoid (15.4) has maximum-norm diameter $4 \max_i \sqrt{(M_e)_{ii}}$. The first diagonal entry is the larger in each displayed matrix. This proves (15.3). The values are unequal, so neither the leading root sets nor their coefficient images can be equal. $\square$

The example fixes all orders of the target scalar functions, not their sharp leading values. Those values contain the positive units erased by valuation. It therefore complements the sharp-marginal example without claiming that arbitrary scalar information is insufficient. The relative real atlas remains useful for finding the common weights; its units and its joint relation are necessary for recovering the metric section at those weights.

16. SINGULAR ENTRANCES IN SEMIALGEBRAIC FAMILIES

The regular-corner calculation of Section 7 does not exhaust remote identification. We now describe entrance orders without a corner parametrization or an injective observation derivative. The required finiteness is finiteness of the exact parameter fibre, modulo any already specified finite symmetry, not smoothness of its representatives.

Let $X \subset \Theta \times K$ be semialgebraic with compact fibres in a fixed compact set K . Let $f_\theta : X_\theta \rightarrow \mathbb{R}^M$ and $A_\theta : X_\theta \rightarrow \mathbb{C}[z]$ be continuous semialgebraic maps, the latter monic of fixed degree. For the probability experiment, f is its weighted square-root embedding. Let

$$p_\theta(\delta) = f_\theta(x_\theta(\delta)), \quad x_\theta(\delta) \longrightarrow x_{0,\theta}, \quad \delta \downarrow 0,$$

be a semialgebraic centre path with p_θ extending C^1 to zero. All data, including this path, have fixed format. Assume $f_\theta^{-1}(p_\theta(0))$ is finite. After a finite semialgebraic partition its elements can be listed by continuous semialgebraic sections $x_{0,\theta}, \dots, x_{N,\theta}$; subsequent refinements will be understood. Choose pairwise disjoint small closed relative neighbourhoods $N_{j,\theta}$ of these points, and put

$$d_j(\theta, \delta) = \min_{x \in N_{j,\theta}} \|f_\theta(x) - p_\theta(\delta)\|. \quad (16.1)$$

The neighbourhoods can be chosen semialgebraically, for example using closed balls of radius smaller than a quarter of the least separation of the listed fibre points.

Theorem 16.1 (Finite entrance orders and exact membership). *There is a finite Nash stratification of the parameter set with the following properties. For each j either d_j is identically zero for*

all sufficiently small positive δ , or there are a rational $\rho_j \geq 1$ and a positive Nash function a_j such that, for every compact subset H of the stratum,

$$d_j(\theta, \delta) = a_j(\theta)\delta^{\rho_j} + O_H(\delta^{\rho_j + \nu_j}), \quad \nu_j > 0. \quad (16.2)$$

The nonzero exponents belong to a finite rational list determined by the fixed-format presentation. The germ of d_j is independent of the sufficiently small isolating neighbourhood.

For any specified positive semialgebraic threshold $t_\theta(\delta) \rightarrow 0$, a further finite partition decides, for each j , whether $t_\theta - d_j$ is eventually positive, negative, or identically zero. The remote component is included precisely in the first and third cases. At every fixed parameter the feasible target root sets converge in Hausdorff distance to

$$\bigcup_{j: d_j \leq t_\theta \text{ eventually}} \{Z(A_\theta(x_{j,\theta}))\}. \quad (16.3)$$

All the strata, exponents, leading-coefficient graphs and membership decisions are effectively describable for algebraic input. No regular-corner assumption is made in these assertions.

Proof. First fix a parameter. The boundary of an isolating neighbourhood contains no point of the exact fibre. Its observation distance from $p(0)$ has a positive minimum. The same is true of the complement of smaller neighbourhoods of all the exact points. Minima over these compact sets are semialgebraic functions of the parameter. Refinement makes them continuous and positive, so the resulting separation bounds are uniform on compact subsets of a stratum. Radii and the finitely many fibre sections admit the same refinement. Finite semialgebraic choice of the fibre points follows by cell decomposition of their finite-fibre graph.

The competitor x_j gives $d_j \leq \|p(\delta) - p(0)\| = O(\delta)$. If a sequence of minimizers did not tend to x_j , compactness would give a different point of the exact fibre in N_j . This is impossible. The same argument for the annulus between two smaller isolating neighbourhoods shows that their minima are equal for all sufficiently small δ . Their distance germs, not just their leading powers, are therefore independent of this choice.

The graph of (16.1) is a first-order formula: there is a point attaining the displayed squared distance, and every point of N_j has at least that squared distance. Quantifier elimination makes the graph semialgebraic. An eventually nonzero nonnegative one-variable semialgebraic function has an expansion $a\delta^\rho + o(\delta^\rho)$ with $a > 0$ and rational ρ . For clarity about finiteness in families, apply Lemma B.1 to a quantifier-free graph formula for d_j . Among the finitely many nonzero specialized polynomial supports, two lowest-order monomials must balance on a positive graph branch. Their exponent differences give a finite rational candidate list. Eventual vanishing is tested separately. The comparability test used in (B.1) selects the actual exponent by a first-order formula. The inequality $d_j = O(\delta)$ forces $\rho \geq 1$.

The limit $a = \lim d_j/\delta^\rho$ also has a first-order graph. Refine until it is positive and Nash. On such a stratum the error $|d_j/(a\delta^\rho) - 1|$ tends to zero on each fibre. Proposition 10.2 gives a positive rational power bound and a positive continuous radius. Taking their bounds and the bounds on a on compact subsets proves (16.2). Thus neither a metric rate nor uniformity has been deduced from topological triviality.

For the threshold assertion, a one-variable semialgebraic function has constant sign or is zero on some terminal interval $(0, \eta)$. Each of these alternatives has a first-order formula with the parameter free, so a finite partition decides them. The sign may be determined by a higher Puiseux coefficient when leading terms coincide; no finite truncation is declared sufficient before the sign test has been made. If the difference is identically zero, compact attainment in (16.1) puts a point of N_j in the closed ball. Thus equality is inclusion, not an assigned side of a critical ray.

The exterior separation already proved forces every small-radius competitor into one of the N_j . Within an included neighbourhood, all feasible points tend to its unique exact point, while a minimizer provides at least one feasible point for every sufficiently small δ . Continuity of the target and of finite-degree roots proves both Hausdorff inclusions in (16.3). The centre path ensures the union is nonempty. Finally every selection, limiting graph and sign test used above is a finite real-algebraic formula. The finite-support algorithm and real-root isolation in Theorem 11.1 prove the stated effectivity. $\square$

This theorem describes singular *entrance orders*, not a classification of singularities of the source. Its proof uses the relative semialgebraic machinery for a conclusion not supplied by a regular tangent-cone projection: entrances tangent to the centre path may occur at strictly higher rational orders.

Theorem 16.2 (Realization of every rational contact order). *Every rational $\rho \geq 1$ occurs as a positive remote entrance exponent in a compact semialgebraic model with strictly positive probability observations, a two-point exact fibre and distinct polynomial targets on its two representatives. For $\rho = n/m > 1$ the probability maps can be chosen polynomial in their respective branch parameters, of degree at most n . Hence there is no universal finite list independent of format, although Theorem 16.1 gives a finite list for every fixed-format family.*

Proof. Choose a strictly positive probability vector P_0 with at least three cells and linearly independent zero-sum vectors A, B such that $\langle A, B \rangle_I = 0$, where $\langle H, H' \rangle_I = \sum_e H_e H'_e / P_{0,e}$. Let the model be the disjoint union of two sufficiently short closed intervals, with maps

$$P_{\text{base}}(s) = P_0 + As, \quad P_{\text{rem}}(u) = P_0 + Au^m + Bu^n, \quad s, u \geq 0.$$

All probabilities are positive after shortening the intervals. Independence of A, B makes the exact fibre at P_0 precisely the two origins. Give the two intervals different constant monic degree-one targets. Take the centre path $s = \delta$ and use Hellinger distance.

Put $x = u^m$ and $\rho = n/m > 1$. The trial $x = \delta$ gives an entrance upper bound $O(\delta^\rho)$. Hellinger distance is locally comparable to Euclidean probability distance, and independence gives

$$|x - \delta| + x^\rho \leq C \|A(x - \delta) + Bx^\rho\|.$$

Every minimizer thus has $x = \delta + O(\delta^\rho)$. Write $x = \delta + v\delta^\rho$; the relevant v are bounded. The square-root embedding, expanded uniformly at P_0 , gives

$$\delta^{-\rho} h(P_{\text{rem}}(u), P_{\text{base}}(\delta)) = \frac{1}{2} \|vA + B\|_I + o(1).$$

Every bounded v is feasible for small δ since $\rho > 1$. The minimum of the right side is $\|B\|_I/2 > 0$ by the chosen orthogonality. This proves the desired exponent and leading coefficient. For $\rho = 1$, use instead $P_{\text{rem}}(u) = P_0 + Bu$, with the same orthogonal A, B . Projection onto the nonnegative B ray gives the positive coefficient $\|A\|_I/2$. These constructions concern the stated general class of probability models; they do not assert that every exponent occurs in the cubic binary family. $\square$

16.1. An algebraic test for the regular regime. The following criterion derives the corner hypotheses from a local presentation. It also separates two ways for that regime to fail: singular feasible geometry and tangency of the centre path to a regular observation image.

Proposition 16.3 (A checkable corner criterion). *At a remote representative x_* suppose the model is locally*

$$X = \{x : h(x) = 0, \ g_i(x) \geq 0\},$$

with Nash functions. Assume $Dh(x_)$ has full row rank and the differentials of the active g_i , restricted to $\ker Dh(x_*)$, are linearly independent. Put*

$$K_* = \{v : Dh(x_*)v = 0, \ Dg_i(x_*)v \geq 0 \text{ for active } i\}.$$

Suppose f has a C^2 Nash extension near x_* , the centre admits $p(\delta) = p(0) + \delta b + \delta^2 c + o(\delta^2)$, and

$$\ker Df(x_*) \cap K_* = \{0\}, \quad p'(0) \notin Df(x_*)K_*. \quad (16.4)$$

Then a full Nash corner chart exists, its observation derivative is cone-coercive, and the entrance has exponent one. The first and second coefficients are exactly those of Theorems 7.1 and 7.2. For algebraic input all these hypotheses are finite algebraic or first-order tests.

Proof. The Nash implicit-function theorem first supplies coordinates on $h = 0$. The independent active functions can then be used as the first coordinates on this manifold, completed by free coordinates. In these coordinates the feasible germ is exactly an orthant times a Euclidean space; inactive inequalities have a strict margin. The derivative of the inverse chart identifies this orthant cone with K_* . The first condition in (16.4), compactness of its unit section and continuity give a positive coercivity constant. Its linear image is a closed polyhedral cone, so the second condition gives a strictly positive attained distance from $p'(0)$ to that image. The existing regular-corner theorems now apply, including nonunique preimages of the first projected point. Matrix ranks are tested by minors; both conditions in (16.4) are first-order statements in linear equations and inequalities. Nash branch specifications and their derivatives at algebraic points are encoded by their polynomial equations and isolating conditions. $\square$

Metric projection, constrained value functions and second-order sensitivity have a substantial existing theory [16, 8, 9]. The regular-corner expansion is used here in that role, with its proof and normalization retained. The additional statements are the fixed-format classification and exact membership rule without its regularity hypotheses, the realization theorem, and the algebraic verification of the regular regime. The two cubic walls of Sections 6 and 8 are explicit realizations of the linear regime; they are not offered as an exhaustive list of singular entrance geometries.

17. DIVISORIAL GEOMETRY OF A REMOTE ENTRANCE

We now identify the singularity datum behind a nonzero entrance order. The datum is a pair: the ideal of the observation residual and the ideal of the centre-path parameter, together with the accessible real sectors. It is not the singularity of the remote image alone. The norm on the residual supplies additional unit data for the leading coefficient.

Fix a remote representative and a sufficiently small compact isolating neighbourhood N . Write

$$R(x, \delta) = f(x) - p(\delta), \quad \mathbf{a} = (R_1, \dots, R_M), \quad \mathbf{t} = (\delta), \quad d(\delta) = \min_{x \in N} \|R(x, \delta)\|.$$

In this section N is semialgebraic, f is Nash on a neighbourhood of N , and p is Nash near zero. Assume that $R(x, 0) = 0$ has just one solution in N , that it lies in the relative interior of N , and that $d(\delta) > 0$ for all sufficiently small positive δ . We also assume $p(0) = f(x_*)$ at that solution. This is the nonexact alternative of Theorem 16.1 in the Nash category; its broader continuous-semialgebraic statement is retained unchanged. A Nash path after a positive ramification $\delta = s^q$ is treated by applying the following construction to s and dividing its order by q .

We use simultaneous real principalization of the *finite* pair $(\mathbf{a}, \mathbf{t})$, with rectilinearization of the inequalities of $N \times [0, \delta_0]$. The standard resolution and rectilinearization results [3, 4] give, after shrinking and covering a compact preimage, finitely many closed real sectors with coordinates (r, z) , $r_i \geq 0$, on which

$$\delta = u(r, z) \prod_i r_i^{a_i}, \quad \|R\| = v(r, z) \prod_i r_i^{b_i}, \quad a_i, b_i \in \mathbb{N}, \quad u, v > 0. \quad (17.1)$$

The units and their inverses are bounded on these sectors. To obtain v , principalize the residual ideal, write its pulled-back generators as $r^b U_1, \dots, r^b U_M$ with no common zero, and take $v = (\sum U_i^2)^{1/2}$. The charts parametrize the actual closed feasible set, not the complex zero locus

of its equations. Only sectors in the closure of their positive-time interiors are retained. We call their coordinate faces *accessible real divisors*. If a chart has a lower-dimensional source component, coordinates and faces are taken in that component; identically zero-time components are discarded. These operations are on the finite diagram just specified.

Theorem 17.1 (Divisorial contact and the metric unit). *In any such finite real monomial presentation, an accessible index with $a_i = 0$ has $b_i = 0$. Define*

$$\rho = \max_{\text{accessible } i, a_i > 0} \frac{b_i}{a_i}. \quad (17.2)$$

Then ρ is positive and independent of the presentation. It has the intrinsic real-arc characterization

$$\rho = \sup_{\gamma} \frac{\text{ord}(\mathbf{a} \circ \gamma)}{\text{ord}(\delta \circ \gamma)}, \quad (17.3)$$

where γ ranges over feasible positive-time semialgebraic Puiseux arcs tending to $(x_, 0)$, and the order of an ideal is the minimum order of its displayed generators. The supremum is attained by an arc transverse to a suitable accessible divisor.*

On each resolved sector the function $\delta^\rho / \|R\|$ extends continuously to time zero as

$$\Lambda(r, z) = \frac{u(r, z)^\rho}{v(r, z)} \prod_i r_i^{\rho a_i - b_i}. \quad (17.4)$$

Let M be its maximum on the compact accessible zero-time preimage, allowing all the sectors. Then $0 < M < \infty$ and

$$d(\delta) = M^{-1} \delta^\rho + o(\delta^\rho). \quad (17.5)$$

Equivalently, ρ is the least real number c for which $\delta^c \leq C \|R(x, \delta)\|$ throughout the feasible germ, for some $C > 0$. Thus the entrance exponent is a relative real Łojasiewicz contact exponent of $(\mathbf{a}, \mathbf{t})$, and the entrance coefficient is the reciprocal of a resolved unit maximum.

Proof. All assertions concern the finite real sectors of (17.1). If $a_i = 0 < b_i$, set $r_i = 0$ while keeping the other boundary coordinates small and positive. The time is positive and the residual vanishes. The image is feasible, which contradicts $d(\delta) > 0$. Thus $b_i = 0$ whenever $a_i = 0$. There are positive-time points with both time and residual tending to zero, for instance the fixed competitor x_* . Consequently at least one accessible ratio b_i/a_i is positive.

For ρ in (17.2), every exponent $\rho a_i - b_i$ is nonnegative. The bounded positive units therefore make (17.4) continuous and bounded on each compact sector. Choose an index attaining the maximal ratio and approach a generic point of its face, with all other boundary coordinates positive. The ratio has a strictly positive value there. Its image at time zero has zero residual and therefore is $(x_*, 0)$. It follows that its maximum M over the zero-time preimage is positive and finite.

For an arbitrary feasible Puiseux arc, properness and semialgebraic selection give a Puiseux lift after further ramification. On a sector containing a final subarc, write $\beta_i = \text{ord } r_i \geq 0$; coordinates tending to a nonzero value have order zero. Coordinates with positive a_i cannot vanish identically along a positive-time arc. Coordinates with positive b_i cannot vanish identically either, since the residual is nonzero there. The units have order zero. Hence

$$\frac{\text{ord}(\mathbf{a} \circ \gamma)}{\text{ord}(\delta \circ \gamma)} = \frac{\sum_i b_i \beta_i}{\sum_i a_i \beta_i} \leq \max_{a_i > 0} \frac{b_i}{a_i}.$$

A transverse arc at a generic maximizing face gives equality. This proves (17.3). It proves independence of the presentation without identifying a real sector with all real points of a complex exceptional divisor.

For $\delta > 0$, surjectivity onto the feasible set and compact attainment give the exact identity

$$\max_{\text{resolved time}=\delta} \Lambda = \frac{\delta^\rho}{d(\delta)}. \quad (17.6)$$

Every sequence of maximizers on its left has a subsequence converging to the compact zero-time preimage, so its upper limit is at most M . For the reverse bound, take a zero-time point attaining M . It belongs to the closure of the positive-time part. Curve selection supplies a positive-time semialgebraic arc tending to it. Its time coordinate is nonconstant and, after shortening the arc, strictly monotone, with image containing every sufficiently small positive time. Evaluation along this arc gives a lower limit at least M in (17.6). This proves (17.5). Finally the bounded extension of Λ gives $\delta^\rho \leq C\|R\|$. On a transverse maximizing arc the ratio $\delta^c/\|R\|$ is unbounded for $c < \rho$. This proves the last characterization. $\square$

Corollary 17.2 (Integral closure and the role of the norm). *Replacing the residual ideal by an ideal with the same integral closure leaves (17.3) unchanged, with the same feasible real sectors and time parameter. The leading coefficient need not be preserved by this replacement. If the centre path extends C^1 to zero, then $\rho \geq 1$.*

Proof. If h is integral over an ideal I , an integral equation has the form $h^n + a_1 h^{n-1} + \dots + a_n = 0$, with $a_j \in I^j$. Along an arc, $\text{ord } h < \text{ord } I$ would make h^n the unique lowest-order term, an impossibility. Ideals with the same integral closure thus have the same minimum order on every arc, and (17.3) applies. This is the valuative mechanism behind the classical connection with Łojasiewicz exponents [21]. Multiplication of every residual generator by a positive constant preserves the ideal but multiplies its norm and the entrance coefficient by that constant. The latter requires the unit in (17.4), not just the unnormed ideal. Finally x_* is an admissible competitor and $\|R(x_*, \delta)\| = \|p(0) - p(\delta)\| = O(\delta)$. Together with (17.5), this forces $\rho \geq 1$. $\square$

17.1. How the contact pair recovers the existing examples. The singular realization in Theorem 16.2 has, in probability coordinates, the residual

$$A(u^m - \delta) + Bu^n, \quad n > m.$$

The sector following the tangent contact can be parametrized by

$$\delta = u^m - u^n z = u^m(1 - u^{n-m} z).$$

There its two independent residual coordinates have common order n , whereas time has order m . Thus $(a, b) = (m, n)$ and the accessible contact ratio is n/m . No larger ratio is possible: if $u^m < \delta/2$ then $|u^m - \delta| \geq \delta/2$, whereas if $u^m \geq \delta/2$ then $u^n \geq (\delta/2)^{n/m}$. Independence of A, B and local equivalence of the probability and Hellinger metrics give a lower bound by a positive multiple of $\delta^{n/m}$ in both cases. In the tangent sector the resolved Hellinger unit at $u = 0$ is

$$\frac{1}{2}\|zA + B\|_I.$$

Its reciprocal maximum is $2/\|B\|_I$ when $\langle A, B \rangle_I = 0$, recovering the exact coefficient $\|B\|_I/2$. This computation identifies the divisor and its metric unit; it does not merely read a Puiseux exponent from an already minimized scalar function.

In the regular regime of Proposition 16.3, cone coercivity localizes minimizing displacements to order δ . The ordinary blow-up $x - x_* = \delta v$ has residual $\delta(Jv - b) + O(\delta^2)$ on the feasible cone. Separation makes its leading norm positive; the relevant contact is $(a, b) = (1, 1)$, and the unit optimization gives $\min_v \|Jv - b\| = \mu$. This recovers the first coefficient of Theorem 7.1. The signed second coefficient and the exact equality-shell test still require the second-order and exact feasible-chart arguments there. In particular, (17.5) does not assign a side when a threshold has the same leading term as $d(\delta)$.

For the two cubic identification walls, the direct feasible charts prove precisely this regular localization, even where the channel rank drops. Their entrance exponents are therefore one, while their metric units and nonidentified-side openings differ. The divisorial formula explains what the exponent measures; it does not replace the exact cubic fibre classification, divide by a vanishing determinant, or suppress either wall mechanism.

APPENDIX A. FINITE ALGEBRAIC CERTIFICATES AND REAL DESCENT

This appendix supplies the full family argument used in Lemma 9.3 and Theorem 9.4. A variety in the algebraic construction is a reduced separated scheme of finite type over a characteristic-zero field. Smoothness is geometric smoothness. For an effective input the ground constants belong to a specified finite real algebraic extension of $\mathbb{Q}$; generic fibres are presented over its finitely generated function fields. No algorithmic assertion about an arbitrary computable field is intended. Real realizations are subsequently taken in $\mathbb{R}$, with the original sign and branch conditions retained.

An algebraic coordinate presentation consists of finitely many affine coordinate rings, localized by specified nonzero elements, transition morphisms and their inverse identities, and a coordinate map whose relative Jacobian is invertible. An adapted presentation also lists each divisor equation, its expression as a coordinate times a unit, and the monomial identities for the functions. Projective maps are recorded by homogeneous equations or by their finite blow-up sequences. These are finite polynomial data, not a numerical collection of coordinate boxes.

A.1. The Rees algebra and a finite diagram.

Lemma A.1 (One localization for all powers of an ideal). *Let A be a noetherian domain, B an algebra of finite type over A , and $I \subset B$ a finitely generated ideal. There is $a \in A \setminus \{0\}$ such that, for every A_a -algebra A' , with $B' = B_a \otimes_{A_a} A'$,*

$$I_a^n \otimes_{A_a} A' \xrightarrow{\sim} (I_a B')^n \quad (n \geq 0), \quad \left(\bigoplus_{n \geq 0} I_a^n \right) \otimes_{A_a} A' \cong \bigoplus_{n \geq 0} (I_a B')^n.$$

The first isomorphism identifies both sides as submodules of B' . Consequently the blow-up in I_a commutes with this base change. One localization suffices for any specified finite sequence of centre ideals on finitely many affine charts of successive blow-ups.

Proof. The associated graded algebra $\mathrm{gr}_I B = \bigoplus_{n \geq 0} I^n / I^{n+1}$ is of finite type over A : it is generated by B/I in degree zero and the images of a finite set of generators of I in degree one. Since A is noetherian, these algebras are finitely presented. Apply generic freeness [17, Tag 051S] to B as a module over itself and to $\mathrm{gr}_I B$ as a module over itself. After inverting the product of the two resulting elements, both are flat over A_a . Each graded piece is a direct summand as an A_a -module, hence flat. In particular B_a / I_a is flat. Induction using

$$0 \longrightarrow I_a^{n-1} / I_a^n \longrightarrow B_a / I_a^n \longrightarrow B_a / I_a^{n-1} \longrightarrow 0$$

shows that B_a / I_a^n is flat for every $n \geq 1$. Tensoring $0 \rightarrow I_a^n \rightarrow B_a \rightarrow B_a / I_a^n \rightarrow 0$ with A' is therefore exact on the left. Its image in B' is the ideal generated by products of n images of elements of I_a , namely $(I_a B')^n$. The isomorphisms respect multiplication and grading, proving the Rees algebra assertion. Relative Proj commutes with base change; this proves the blow-up assertion. For a finite sequence, apply the argument to the actual centre ideal on every affine chart at each stage and multiply the finitely many localizing elements. Previously proved identities survive further localization. No countable intersection of open sets is used. $\square$

Lemma A.2 (Spreading the entire stratifying diagram). *Let A be a finitely generated integral algebra over a characteristic-zero field k , and let $K = \mathrm{Frac}(A)$. Suppose $Y \rightarrow \mathrm{Spec} A$ is of finite type and a generic certificate consists of a reduced closed chain $Y_{0,K} = Y_{K,\mathrm{red}} \supset \cdots \supset Y_{q,K} = \emptyset$,*

projective $Z_{j,K} \rightarrow Y_{j,K}$, and specified inverse isomorphisms above $E_{j,K} = Y_{j,K} \setminus Y_{j+1,K}$. Then, after replacing A by A_a for one nonzero a , the chain can be chosen with $Y_0 = Y_{\text{red}}$, all its members flat with geometrically reduced fibres, and the stated isomorphisms hold above the actual complements $E_j = Y_j \setminus Y_{j+1}$. In particular they hold above $(E_j)_s$ on every fibre. The union of these complements is the whole reduced fibre; no constancy of the number of irreducible components is required.

Proof. First replace Y by its reduction. Take reduced closures of the generic $Y_{j,K}$ inside Y and use $Y_q = \emptyset$. Clear denominators in the projective equations and affine presentations of the generic maps. All schemes involved are of finite presentation over the noetherian base. The finite-presentation limit lemma [17, Tag 01ZM, Lemma 32.10.1(1)–(3)] descends objects, morphisms, and equalities of morphisms. Apply it to the inverse maps on the generic complements, their inclusions, and both inverse identities. The complements used on the model are $Y_j \setminus Y_{j+1}$; their generic fibres are the prescribed complements. The descended inverse maps and equalities therefore give isomorphisms on these actual open subschemes after a further localization, not merely on unspecified subsets with the same generic points. Finite affine covers and their intersection diagrams are included in this application of the limit lemma. Thus chart coverage, as well as the formula for each map, descends.

Each reduced generic member is geometrically reduced because K has characteristic zero. Generic freeness makes each member flat on a dense open set. The generic-fibre reducedness theorem [17, Section 37.26, Lemma 37.26.4] then permits a further common shrink so that all fibres of these finitely many members are geometrically reduced. This is an imposed property, not an optional replacement of scheme structure after specializing. Finally, on every fibre the sets $Y_{j,s} \setminus Y_{j+1,s}$ form a disjoint partition of $Y_{0,s}$ because the chain ends in the empty set. For any point there is a unique index at which it leaves this finite descending chain. The corresponding isomorphism covers that point, including points on a special fibre component. This argument neither identifies fibre components with generic components nor assumes their number to be constant. $\square$

A.2. Relative smoothness, normal crossings, and exceptional sets. We now prove Lemma 9.3 with a precise list of the further conditions. Separate the finitely many open-and-closed components of each smooth generic source according to the identically zero functions. These components and their idempotent identities are part of the finite diagram. On a component on which a function is declared zero, descend that zero identity. On every other component principalize the product of the remaining functions, recording all factors and units.

The absolute input is characteristic-zero resolution and principalization [4, 5]. More precisely, the variety resolution of [5, Theorem 1.1] supplies the smooth sources; the marked-ideal resolution of [5, Theorem 1.3], applied with mark one, supplies principalization together with the normal-crossing exceptional boundary. A product which is a monomial times a unit makes each of its factors a monomial times a unit on the regular local rings. This use is over the field K . Compatibility with arbitrary special fibres is supplied by Lemma A.1, not by a claim that functoriality for smooth maps implies arbitrary base-change compatibility.

Spread the finite sequence of centres and its adapted charts. The following finite conditions are imposed before retaining the open base. The source and centre smooth loci are the loci on which their finite presentations satisfy the relative smoothness criterion. On a smooth source the intersections of the designated divisor equations must be smooth of their prescribed relative codimensions; in adapted coordinates this says that the appropriate relative differentials are independent. This condition is imposed for every subset of the finite divisor list. Each coordinate chart must be etale, each declared inverse unit must multiply its unit to one, and the finite charts must cover their source. All scheme-theoretic identities in these statements are spread

by the finite-diagram lemma. Their remaining rank conditions are open algebraic conditions; complements of chart unions are constructible algebraic sets.

For clarity, the discarded algebraic parameter set is the union of the closures of images of the following bad sets, together with the localizing hypersurfaces already used:

- (1) the original irreducible components not dominating the current reduced integral base;
- (2) nonsmooth points of a source or centre, and failures of the specified relative dimension or centre codimension;
- (3) failures of smoothness or codimension of any designated divisor intersection, or failure of an adapted etale Jacobian;
- (4) the complement of the chosen finite source charts, and any zero of a function declared invertible;
- (5) failures of a descended zero identity, a chart transition, or an inverse isomorphism on an actual complement in the closed chain.

For items (4)–(5) the bad set is empty once the relevant identities and coverage have descended; it is harmless to include it explicitly in the certificate. Every bad set in this list has empty generic fibre: this is respectively dominance, generic smoothness in the given certificate, generic normal crossings, generic coverage and invertibility, or the specified generic identities. Images are constructible by Chevalley’s theorem [17, Tag 054K]. A constructible subset of an integral noetherian space which is dense contains a nonempty open subset and hence the generic point. Therefore the closure of each of these images is proper. This justification applies also when $Y \rightarrow S$ is not proper.

After their removal, apply Lemma A.1 on every spread blow-up chart. The resulting smooth sources and centres, relative normal crossings, unit identities and finite chart covers hold on every retained fibre. A nonzero monomial cannot specialize to the zero function on a fibre component: its unit is invertible and its finitely many divisors each have relative codimension one. Their union cannot contain a component of the smooth fibre. An identically zero generic function remains zero by its descended identity. The open-and-closed source decomposition prevents a change of these flags through an unlabelled component intersection. Flatness of the closed chain and smoothness of the sources exclude additional vertical components on the retained open set. Coverage of any special fibre components is, more directly, the partition argument in Lemma A.2. This proves all fibrewise assertions of Lemma 9.3.

Each discarded algebraic subset has strictly smaller dimension than its current integral base. Replace it by its reduced irreducible components and reconstruct the entire generic certificate on each component. There are finitely many components at each step and depth at most $\dim S + 1$. This gives finite termination. The exceptional list is not required to be minimal. Importantly, a change of a *real* accessibility flag need not lie in a proper algebraic subset: two full-dimensional real chambers can have different flags. Such changes are treated by finite real quantifier-elimination partitions below, not by this dimension argument.

A.3. Exact real algebraization.

Lemma A.3 (The square-slack cover with all branch conditions). *Let S be a semialgebraic base and $\Gamma \rightarrow S$ a semialgebraic proper map realized as a graph in $S \times \mathbb{R}^N$. The graph includes all auxiliary functions, sign choices and denominator exclusions. After a finite base refinement, it admits finitely many algebraic square-slack lifts whose real points over S project exactly onto Γ . Projection is proper over compact subsets of S . Every later map factoring through these lifts projects only to points of Γ .*

Proof. Let C be the ordinary semialgebraic closure of Γ in the ambient real algebraic space. Properness implies that Γ is relatively closed in $S \times \mathbb{R}^N$: a convergent sequence over a convergent

base sequence lies over a compact subset of S and has a graph limit there. It follows that $C \cap (S \times \mathbb{R}^N) = \Gamma$. A closed semialgebraic set is a finite union of basic closed semialgebraic sets [7, Theorem 2.7.2]. For a piece

$$C_l = \{x : f_{l,a}(x) = 0, p_{l,b}(x) \geq 0\}$$

use the algebraic set

$$Y_l = \{(x, s) : f_{l,a}(x) = 0, p_{l,b}(x) = s_b^2\}.$$

Every real point of Y_l projects into C_l , and each point of C_l has a real lift. Over S , the union therefore projects exactly to Γ . Over a compact base subset the graph is compact. The slack coordinates are bounded by the square roots of maxima of the finitely many polynomials there, and the lift is closed; hence the lifted real set is compact. This proves properness and the last assertion.

Strict exclusions are encoded in the complete graph formula before this construction. For a denominator excluded everywhere on that compact graph, $d \neq 0$ may also be retained by adjoining w with $dw = 1$, together with any required sign of d . Nonnegative square-root branches are specified in the graph, not selected after taking an algebraic closure. If adjoining such coordinates destroys properness on the initial base, first apply Hardt triviality to the enlarged graph, whose individual fibres are compact, and refine the base. A trivialization with compact fibre makes the restricted graph map proper. Closure is taken only *after* this step. Consequently no point with a vanishing excluded denominator is reintroduced over the retained stratum. General Boolean branch conditions are instead kept in the complete graph formula: one does not split a compact fibre into nonclosed pieces, adjoin unbounded reciprocals on those pieces, and then assume their fibres are compact. The equality $C \cap (S \times \mathbb{R}^N) = \Gamma$ already preserves all such conditions without that operation. $\square$

Lemma A.4 (Real branch selection and simultaneous descent). *The root labels and Nash graph functions used in the cluster construction can be carried on finitely many selected real Nash branches of algebraic base covers. On a finite semialgebraic refinement, the proper resolved square-slack sources cover exactly the original graph in every fibre. Descent by root permutations preserves the entire joint coefficient and root relation, not merely its scalar order list.*

Proof. Refine a Nash stratum by the regular locus of each integral component of its algebraic closure. Its lower-dimensional singular parts are treated recursively. A Nash function is algebraic over the coordinate function field. Adjoin the finitely many graph coordinates needed here to that field. The extension is finite and separable in characteristic zero. Choose equations and invert their leading coefficients and discriminants; a further localization gives a finite etale algebraic cover. Include the original graph formula to select the intended real branches. Finite semialgebraic cell decomposition makes each selected graph a Nash section and separates its alternatives. Complex sheets and real sheets failing the original formula are not retained. Thus monodromy is handled by base refinement, rather than by globally labelling a nontrivial cover. Exceptional leading coefficients and discriminants are handled by the reduced lower-dimensional base recursion. A finite union of selected sections has compact inverse image over compact subsets of a stratum.

Apply Lemma A.3 on each selected branch. For a real point y of a lifted fibre, choose the index with $y \in Y_{j,s} \setminus Y_{j+1,s}$ in the spread singular-locus chain. The inverse isomorphism of Lemma A.2 supplies a real point of its smooth source above y . A blow-up of a smooth real variety in a smooth centre of positive codimension c has fibre $\mathbb{RP}^{c-1}$ at a real centre point and is an isomorphism away from the centre. In local smooth coordinates the centre is a coordinate subspace, which proves both assertions directly. Such a blow-up therefore does not lose real points. Composing the finitely many principalizations and taking the finite union over j proves real surjectivity. A real singular point with no top-dimensional real branch is covered by a later member of the chain.

Projectivity of these maps and properness of the slack projection prove properness of the real cover.

Conversely every real resolved point factors through a real slack point on the selected branch, and hence projects to the original graph by Lemma A.3. This proves equality, not just inclusion, of the covered real fibres. A root-label permutation simultaneously permutes cluster indices, their coefficient coordinates and root arrays. The equations defining Ψ and the fixed-centre formula (10.2) are equivariant for this simultaneous action. A finite union commutes with taking a fibrewise closure; projection to the finite quotient therefore preserves the joint relation and its root image. Descent never pairs one coefficient from one sheet with another coefficient from a different sheet. The bottleneck metric is invariant under these permutations. $\square$

A.4. Real accessibility and completeness of the order list.

Lemma A.5 (Real-accessible divisors). *On a resolved square-slack source with the above relative adapted coordinates, accessibility of a divisor is a first-order real condition on the parameter. Every recorded divisor has an admissible real transversal attaining its monomial orders. Every real degeneration relevant to a finite coefficient order is controlled by one of the recorded divisors. Thus the minimum in Corollary 9.6 is both an upper-inequality exponent and an attained exponent.*

Proof. In a chart with source equations $E(\theta, y) = 0$, chart exclusions $D(\theta, y) \neq 0$, and adapted divisor coordinates $y_1, \dots, y_l$, the flag for i with $a_i > 0$ is the formula

$$\exists y : E(\theta, y) = 0, \quad D(\theta, y) \neq 0, \quad y_i = 0, \quad \prod_{h \neq i, h \leq l} y_h \neq 0, \quad \det J_{\text{et}}(\theta, y) \neq 0. \quad (\text{A.1})$$

Inverse units and the selected real base formula are included in E, D . Charts are the ones meeting the inverse image of $G = 0$. Since the full real slack fibre already represents the admissible graph, there is no additional unrecorded sign restriction on source directions. At a point satisfying (A.1), the real inverse function theorem gives a transversal with $y_i = t$ and other adapted coordinates fixed. For small nonzero t , $G > 0$, and every point projects into the original graph. The unit bounds give $G \asymp |t|^{a_i}$ and $H \asymp |t|^{b_i}$ for each nonzero H on that source component.

A real degeneration with $G \rightarrow 0$ has a lifted accumulation point above $G = 0$, by properness. Some adapted chart contains that point. If it lies on several divisors, relative normal crossings allows a small real perturbation *on any one of them* avoiding all the others. Hence each divisor of positive G order meeting this real chart has a point of the form (A.1). In a smaller box, with β the minimum of its recorded ratios, $b_i \geq \beta a_i$ for $a_i > 0$ and $b_i \geq 0$ otherwise. The monomial identities and positive unit bounds imply $H \leq CG^\beta$ throughout the box, including its intersections. Equivalently, along a lifted arc the order ratio is a weighted average of the b_i/a_i with nonnegative additional terms from indices with $a_i = 0$; it cannot be smaller than their minimum. A finite cover near the compact exact fibre gives this inequality for all competitors. The transversal for a minimizing divisor attains it. If G is identically zero on a source with real points, the exact-target condition gives $H = 0$ there. A smooth fibre component with a real point has a Zariski-dense real open subset. Thus a nonzero monomial H cannot occur on such an exact real component; real-empty components contribute no order. This handles zero flags as well as the nonzero ratios.

Finally (A.1) is a finite polynomial formula. Quantifier elimination partitions the base by all its truth values. These semialgebraic pieces may have full dimension. Nash refinement makes the choices compatible with the selected base branches. This partition is finite without an algebraic dimension-drop assertion. $\square$

Completion of Theorem 9.4. Stratify the enlarged graph and the exact cluster data as in Lemma A.3, and carry labels as in Lemma A.4. Over an integral algebraic base resolve the

reduced generic fibre, then its reduced singular locus, and continue until the empty set. Dimension decreases in this chain. Principalize on its smooth sources the product of the nonzero functions. Lemmas A.1 and A.2, followed by the explicit bad-locus construction, spread this certificate over a nonempty open base with all fibrewise properties. Repeat on the finitely many reduced exceptional components. Lemma A.4 proves exact real coverage after descent, and Lemma A.5 proves the accessibility assertions and the completeness of the orders.

For a fixed fibre, cover the compact inverse image of $G = 0$ by finitely many smaller real coordinate boxes with positive lower and finite upper unit bounds. Properness implies that a sufficiently small observation neighbourhood is covered: otherwise a sequence with $G \rightarrow 0$ outside the chosen source neighbourhood would have a lifted accumulation point above $G = 0$ outside it. The same argument works above a compact base subset, since the selected label graph, the slack lift and the projective resolved maps are proper there. Alternatively their positive radii and unit bounds can be selected semialgebraically and made continuous by finite refinement. Their extrema on a compact parameter subset give common neighbourhoods and bounds. Fibrewise proper source maps must not be confused with the smaller, nonproper coordinate boxes. This proves all existence and compact-uniform assertions. The effective construction for algebraic input is detailed at the end of Appendix B. $\square$

APPENDIX B. FINITE SUPPORT ORDERS, UNIFORM DECAY, AND EFFECTIVE OUTPUT

The metric and effective parts can be proved without making a parameter-dependent choice of a Puiseux branch. Only the ordinary one-variable algebraic-branch theorem is used to prove that a finite candidate list is exhaustive. Selection from that list is then a first-order decision problem.

Lemma B.1 (A finite support test for the order). *Let $e(\theta, t) \geq 0$ be a semialgebraic function, defined for $0 < t < \tau(\theta)$, where $\tau > 0$ is semialgebraic, and suppose $e(\theta, t) \rightarrow 0$ for fixed θ . A quantifier-free graph formula supplies a finite computable set $\mathcal{B} \subset \mathbb{Q}_{>0}$ such that for every θ either $e(\theta, t)$ is eventually zero or exactly one $b \in \mathcal{B}$ satisfies*

$$\exists c, C, a > 0 \forall t : 0 < t < a \implies ct^b \leq e(\theta, t) \leq Ct^b, \quad a < \tau(\theta). \quad (\text{B.1})$$

Each of these alternatives is an effective semialgebraic condition for algebraic input. No real exponent is quantified over.

Proof. Eliminate quantifiers in the graph formula and let $P_1, \dots, P_N$ be its finite list of polynomials in (θ, t, y) . Fix θ and discard those specializing to the zero polynomial in (t, y) . At a graph point some remaining polynomial must vanish. Otherwise all nonzero polynomial signs are unchanged on a small two-dimensional neighbourhood; the zero specialized polynomials impose constant truth values there. The graph formula would then contain that neighbourhood, contradicting single-valuedness. For each remaining P_i , the set $\{t : P_i(\theta, t, e(\theta, t)) = 0\}$ is semialgebraic in one variable. These finitely many sets cover a punctured interval. After their finitely many endpoints are removed, at least one vanishing equation holds on an entire interval $(0, a)$. A polynomial independent of y cannot vanish there unless it was one of the discarded zero polynomials.

By one-variable semialgebraic decomposition, on a smaller interval the function is either identically zero or positive, continuous, and a single real algebraic branch of such an equation. The ordinary algebraic Puiseux theorem gives, in the positive case, $e(\theta, t) = a_\theta t^{b_\theta} + o(t^{b_\theta})$ with $a_\theta > 0$ and rational $b_\theta > 0$; see [7, 1]. Write its vanishing equation as $\sum p_{ij}(\theta) t^i y^j = 0$. Inserting the first term shows that $\min(i + b_\theta j)$ is achieved at least twice among nonzero specialized coefficients. Otherwise its unique lowest term could not cancel. Therefore

$$b_\theta = (i - i')/(j' - j) > 0$$

for two distinct support points of one of the original P_i . Take $\mathcal{B}$ to be all such positive ratios. Specialization may remove support points but cannot add a pair to this list. This proves finiteness uniformly in θ , without following algebraic branches through parameter specializations.

Eventual zero is the formula $\exists a > 0 \forall t (0 < t < a \Rightarrow e(\theta, t) = 0)$, with $a < \tau(\theta)$. For each candidate, (B.1) is a first-order polynomial formula: replace $t^{p/q}$ by its unique positive root, or put $t = s^q$ throughout. Quantifier elimination decides it. The actual branch expansion supplies one true candidate; two distinct powers cannot be bounded above and below by positive multiples of each other as $t \downarrow 0$. Exactly one candidate is therefore true in the nonzero case. This also tracks the branch selected by the original Boolean graph formula, including all specialized coefficients and inequality faces, rather than selecting a root of a polynomial without checking graph membership. $\square$

Full proof of Proposition 10.2. Use Lemma B.1. If the candidate set is nonempty, choose a positive rational $\nu < \min \mathcal{B}$; if it is empty use $\nu = 1$. On every parameter fibre $e(\theta, t) \leq t^\nu$ eventually, including the eventual-zero case. The set of pairs (θ, a) with $0 < a \leq \min(1, \tau(\theta))$ and this inequality for all $0 < t < a$ is semialgebraic and has a nonempty positive fibre at every parameter. Semialgebraic choice selects a positive $a = \tau_0(\theta)$, not a possibly vanishing limit value at the boundary of a stratum. A finite Nash stratification makes this selected function continuous. Its minimum on a compact subset of a stratum is strictly positive; the estimate is uniform there. The exponent may be smaller than the sharp order, which is exactly what is needed for the uniform remainder. $\square$

Lemma B.2 (Nash refinement of graph functions). *Finitely many semialgebraic functions on a semialgebraic set admit a common finite Nash stratification on which each function is Nash. For algebraic presentations this refinement is effective.*

Proof. Take their combined graph. Semialgebraic Nash stratification, refined by the ranks of the coordinate projection and by a compatible stratification of its image, is finite; these are the usual graph and rank-stratification constructions in [7, 1]. On a graph stratum the projection is injective. If its constant rank were less than the dimension of that stratum, the constant-rank theorem would give a positive-dimensional local fibre, contradicting injectivity. The projection thus has full rank onto its image stratum. Its local inverse is analytic by the inverse function theorem and is semialgebraic because its graph is semialgebraic; hence it is Nash. Finitely refine images to separate overlapping strata and repeat on their lower-dimensional boundaries. This proves the claimed finite partition and gives the precise reason that graph stratification, rather than topological triviality alone, yields Nash functions. Effective semialgebraic decomposition and rank tests give the algebraic version of the construction. $\square$

The application to Theorem 10.3 is now direct. The polynomial matching formula (10.5), with two quantified inclusions and minimality over the distance, defines the Hausdorff error as a semialgebraic function of (θ, t) . Theorem 10.1 proves its pointwise convergence to zero. Proposition 10.2 gives a compact-uniform power estimate. Lemma B.2 applied to the diameter and the positive separation and radius functions gives the asserted Nash refinement. Hardt triviality is used separately for the topological type of the leading fibres. The inequality $|\text{diam } A - \text{diam } B| \leq 2d_H(A, B)$ proves the diameter remainder.

B.1. An explicit terminating spectral algorithm. Here is a proof of Theorem 11.1 with its data representations and branch-selection step made explicit. The input is the polynomial formula of Definition 9.1; each algebraic constant has its integer polynomial and rational isolating interval. Arithmetic is in a primitive-element representation of their finite real algebraic field. All quantified formulas below are subjected to real quantifier elimination as in [1]. Finite root permutations are literal finite disjunctions, and complex coordinates are pairs of real coordinates.

First decide exact identification by the formula that all x, x' with $F(x) = F(x') = P$ have equal target coefficients. On this locus construct the cluster graph using its polynomial factorization equations and the specified isolating conditions. For each $H = |c|^2$, form the graph of

$$m_H(s) = \max\{H(x) : x \in \Gamma, G(x) \leq s\}$$

by the conjunction of the universal upper-bound formula and an existential attainment formula. Compactness gives the maximum; compactness and $\{G = 0\} \subset \{H = 0\}$ give $m_H(s) \rightarrow 0$. Apply quantifier elimination to this graph. Construct its finite support-pair list as in Lemma B.1, test eventual zero, and test (B.1) for each rational candidate. The unique true candidate is β_H . This is a finite algorithm for selecting the *actual maximum-function branch*. It needs no unspecified command to isolate an algebraic branch at the boundary. If all maxima are eventually zero the target is locally rigid; otherwise set $\alpha = \min_H \beta_H/k_H$ over the nonzero maxima.

For a family, keep θ free in exactly these tests. Their finitely many truth sets give the parameter pieces and include all zero specializations. Nash-refine them using Lemma B.2. The supports are finite before specialization, so this procedure still terminates without a bound on a real-valued exponent.

Next substitute $\alpha = p/q$ in the fixed-centre formula (10.2). Elimination computes a quantifier-free presentation of $\mathcal{J}_0$ and its projection $\mathcal{C}_0$. Adjoin the labelled roots by elementary symmetric identities to obtain $\mathcal{L}_0$. Use (10.5) to express that all pairs of elements have distance at most C and that some pair attains C . Compactness makes this a formula for one positive real number. Univariate real-root isolation applied to its eliminated formula selects that number over the input field. Taking a field norm gives a nonzero integer polynomial vanishing at C ; renewed root isolation and the same formula select a rational isolating interval for the intended root. The same graph formulas with free parameters compute the family of constants, and the preceding Nash and decay constructions compute a compatible refinement and an admissible rational remainder exponent. No complexity bound is claimed.

B.2. Effective geometric certificates without an implicit oracle. The geometric output can also be given a terminating, though generally inefficient, specification. Over the field just described, enumerate finite algebraic diagrams and finite semialgebraic base partitions in increasing finite encoding length, refining a fixed semialgebraic triviality partition of the enlarged graph with compact fibres. Properness on those base pieces follows from that trivialization and is preserved under further base restriction. A candidate consists of square-slack presentations, reduced closed chains with inverse-map identities, smooth sources with specified projective maps, finite centre ideals and adapted etale charts, integer monomial orders, inverse units and selected real branch formulas. Verify polynomial identities and centre blow-up charts by ideal arithmetic; verify relative smoothness, chart ranks and divisor intersections by their Jacobian criteria. Properness is certified by the specified projective construction and the square-slack argument, not by numerical bounds on open charts. Verify partition coverage, fibrewise real surjectivity, zero flags and accessibility by real quantifier elimination. These are finite formulas: for example coverage is $\forall \theta \forall x \in \Gamma_\theta \exists y$ in one of the finitely many source charts with the stated projection. Each positive neighbourhood and unit bound can likewise be represented by a semialgebraic choice graph and its universal verification formula.

The existence proof in Appendix A, performed over finite algebraic and rational-function extensions of the input field, produces a certificate with such a finite encoding. An exhaustive search therefore eventually finds a verified certificate. Invalid candidates are rejected by the finite tests. This is an explicit input–output existence algorithm; it is not a claim that the accompanying regression script implements a practical resolution package. The direct maximum-function algorithm above computes the spectral outputs without this enumeration. Thus neither

effectivity route depends on an unrepresented branch oracle, and neither is inferred from a finite list of example calculations.

APPENDIX C. FORMAT, FINITE DIAGRAMS, AND THEOREM DEPENDENCIES

We make explicit two conventions used in the general semialgebraic statements. Neither convention alters the whole-model quantifiers in the concrete binary theorems.

Definition C.1 (Geometric and arithmetic format). A fixed-format family is specified by a fixed finite list of formulas for the feasible-set graph, observation graph, monic target-coefficient graph, centre-path graph, and any isolating neighbourhoods or threshold functions used in the assertion. Its geometric format is

$$\mathcal{F} = (n_\Theta, n_X, n_{\text{aux}}, M, d_A, s, D, L).$$

Here the first three numbers bound the parameter, source and auxiliary variables in the entire list; M is the observation dimension; d_A is the target degree; s bounds the number of polynomial atoms; D bounds their total degrees; and L bounds the length of their Boolean and quantifier syntax trees, with the quantifier blocks included. Rational functions are encoded by graph equations and denominator sign conditions. A Nash branch is encoded by its polynomial equations and real branch inequalities. Square roots, root constraints, path ramifications and neighbourhood choices are included as auxiliary graph data, not treated as zero-cost primitive operations.

Coefficients may range over real parameters for geometric finiteness. For effective algebraic input, each fixed coefficient is specified by a square-free integer polynomial and a rational isolating interval, with a consistent common real algebraic embedding for joint arithmetic. Bounds on their degrees and binary encoding lengths may be appended as the arithmetic format. No bit-complexity estimate is claimed here. A new threshold or a selected scalar polynomial must be included in the finite list before a uniform bound is asserted for it.

Proposition C.2 (An explicit meaning of the finite order list). *For the format in Definition C.1, there is a computable integer $B(\mathcal{F})$ such that every nonzero exponent in Theorem 16.1 belongs to*

$$\{p/q : 1 \leq p, q \leq B(\mathcal{F})\}.$$

When the centre path is C^1 , only the members at least one occur. The integer can be obtained from a real quantifier-elimination degree bound applied to the actual minimum graph formula; it is not an unquantified appeal to a notion of bounded format.

Proof. The graph formula for the attained minimum uses one copy of the source variables for an attaining point, another for comparison with all competitors, the already specified graph formulas, and an equation for the squared norm. Its numbers of variables, atoms and degrees are bounded by computable functions of $\mathcal{F}$. Apply real quantifier elimination [1] and let B bound the degrees of all output polynomials, increasing it if necessary to a positive integer. Fix a parameter. On a final positive interval the graph of a nonzero minimum lies on the zero set of a nonzero specialized output polynomial $P(\delta, z)$: otherwise a sign cell containing the graph would have interior in the (δ, z) plane, contrary to its being a graph. There are only finitely many cells and polynomials, so one final branch and a nonzero polynomial can be selected. Identically zero specialized polynomials are discarded first.

Substitute $z = a\delta^\rho + o(\delta^\rho)$, $a > 0$, using the one-variable semialgebraic Puiseux theorem. At least two distinct monomials $\delta^i z^j$ have the lowest order; a unique lowest term could not cancel. They must have different j , since otherwise equality of orders would force equal i . Thus $\rho = (i - i')/(j' - j)$, with nonzero integer denominator and numerator of absolute value at most B . Positivity of ρ gives the asserted finite list after reduction. Vanishing coefficients only remove monomials and do not enlarge this bound. The C^1 lower bound follows from the fixed exact competitor as in Theorem 16.1. $\square$

C.1. Finite scalar families, not one modification for all polynomials. A common monomial presentation is asserted for every *finite selected* family of functions, ideals and inequalities, after a common refinement depending on that family. It is not asserted that one fixed modification simultaneously monomializes the infinite ring of all polynomial scalar functions of the target coefficients. This convention applies in particular to the divisorial-order interpretation of Theorem 15.1. The theorem’s comparison along every analytic arc and for every scalar polynomial remains valid pointwise for each polynomial; it does not require a simultaneous principalization for all of them.

There is a simple reason for the distinction. In a plane, the infinitely many germs $x - cy$, $c \in \mathbb{R}$, have distinct strict transforms. No fixed finite normal-crossings divisor can contain the zero divisors of all of them. A finite chosen list can be included in a resolution diagram; its size and degrees then enter $\mathcal{F}$. The contact formula in Theorem 17.1 uses only the finite residual/time pair and feasibility inequalities. The finite calibration theorem uses exactly the probes listed in (4.4).

C.2. Dependency of the concrete results. The normalizer recovery and full-rank coefficient inverse are direct interpolation and matrix-pencil arguments. The multiplicity and boundary laws use these inverses, real-root cluster expansions and finite Fisher projection. The finite calibration theorem uses the same inverse to prove the convergence of *actual* constrained minima and their whole-model recovery; its stability bound is finite-dimensional. None of these proofs needs the relative resolution atlas. The semialgebraic refinement of their algebraic output functions is the separate organizing use of real algebraic geometry.

The exact cubic fibre and the inverse through rank loss are direct affine-pencil and second-marginal arguments. The weight and endpoint walls require their different exact feasible charts and the regular conic expansion. Those proofs, including equality-shell membership and the signed second correction, do not depend on a general singular resolution theorem. They have therefore been placed before the general weighted framework in the principal article.

The general atlas is essential to the uniform weighted output theorem for arbitrary specified semialgebraic observation relations, beyond the explicit binary normal forms. Real curve selection and algebraic Puiseux germs give the residue comparison. Simultaneous principalization of the residual/time pair supplies the divisorial entrance formula; the scalar semialgebraic entrance theorem supplies the broader continuous-semialgebraic statement and uniform family refinements. These are distinct inputs and conclusions. We do not claim that the existence of a general atlas is necessary for a computation already proved by the concrete pencil or Fisher arguments.

APPENDIX D. EXACT OUTPUTS AND FINITE ARITHMETIC

Three different assertions should not be conflated. The existence of the relative real atlas is a geometric theorem. The first-order procedures produce spectral or entrance outputs from algebraic presentations. The rational calculations of specific Fisher programs are finite specializations, not implementations or proofs of the entire atlas.

For the direct procedures the input is a finite list of polynomials with rational or real algebraic coefficients, a Boolean formula of polynomial sign conditions and the quantifiers specifying the model and its maps. A real algebraic coefficient is represented by its integer polynomial and a rational isolating interval. Rational powers use a nonnegative auxiliary variable and its integer-power equation. Compact minimum formulas have the form

$$\exists x [\Phi(x) \wedge Q(x) = z] \wedge \forall y [\Phi(y) \Rightarrow Q(y) \geq z].$$

The sign or limit formulas retain the centre as a free fixed parameter. Elimination produces a finite output formula; at algebraic parameter values, root isolation gives the numerical algebraic

outputs. In particular, the quadratic-probe graph and the entrance minimum graph fit this encoding without any projective blow-up representation.

The separate certificate search in Appendix B concerns geometric presentations. Its existence-by-enumeration role is supplementary: the direct outputs above do not wait for that search, and no run of the finite example regression is described as having constructed such a certificate. This separation leaves the geometric proofs and the direct algorithmic conclusions intact.

For the two cubic critical examples, use the coordinate orders

$$(a_0, b_0, X, c_0, w, k, u_f, u_h, v_f, v_h) \quad \text{and} \quad (a_0, b_0, X, c_0, d_0, Y, k, u_f, u_h, v_f, v_h),$$

respectively. At the endpoint, $w = -(d_1 + d_2) \geq 0$; for a first-order representative choose $d_1 = d_2 = -w/2$. The nonnegative coordinates are (a_0, X, w) and (a_0, X, Y, k) . Both displayed specializations have the empty cone support, so $w = 0$ at their endpoint optimum and the endpoint split is uniquely zero. Thus the representative choice introduces no second-order ambiguity into those exact examples.

Let L and D be respectively the seven free score columns and the cone columns, $W = \text{diag}(w_j/P_{j_e})$, and let B be the base probability velocity. All entries are rational at the stated examples. Set

$$G_L = L^\top W L, \quad b_f = G_L^{-1} L^\top W B, \quad \tilde{D} = D - L G_L^{-1} L^\top W D, \\ Q = \tilde{D}^\top W \tilde{D}, \quad g = \tilde{D}^\top W (B - L b_f), \quad b_* = (B - L b_f)^\top W (B - L b_f).$$

The exact checks give $\text{rank } L = 7$, $\text{rank } Q = 2$, $g < 0$ entrywise and nonnegative principal minors of Q . Every nonempty independent-support KKT test fails, whereas the empty support succeeds. Moreover $z^\top Q z - 2g^\top z > 0$ for every nonzero $z \geq 0$. This proves uniqueness of the cone optimum despite the singular projected Gram matrix. The free optimum is b_f and $\mu^2 = b_*/4$. Multiplication and division of the exact probability jets then give μk by (8.13). For positive rational interval endpoints l, u , the tests for μ and k are respectively

$$l^2 < \mu^2 < u^2, \quad l^2 \mu^2 < (\mu k)^2 < u^2 \mu^2, \quad \mu k > 0.$$

After clearing positive denominators these are integer comparisons. The same exact arithmetic forms and inverts the two Fisher matrices in (15.5)–(15.6). The ancillary record gives the full rational Gram matrices, linear vectors, optimizers and interval tests; reproducibility of these finite operations is separate from the mathematical arguments above.

REFERENCES

- [1] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second edition, Algorithms and Computation in Mathematics 10, Springer, Berlin, 2006.
- [2] A. Bernig and A. Lytchak, Tangent spaces and Gromov–Hausdorff limits of subanalytic spaces, *Journal fuer die reine und angewandte Mathematik* **608** (2007), 1–15.
- [3] E. Bierstone and P. D. Milman, Semianalytic and subanalytic sets, *Publications Mathématiques de l’IHÉS* **67** (1988), 5–42.
- [4] E. Bierstone and P. D. Milman, Canonical desingularization in characteristic zero by blowing up the maximum strata of a local invariant, *Inventiones Mathematicae* **128** (1997), 207–302.
- [5] E. Bierstone and P. D. Milman, Functoriality in resolution of singularities, *Publications of the Research Institute for Mathematical Sciences* **44** (2008), no. 2, 609–639; arXiv:math/0702375.
- [6] C. Bivià-Ausina and S. Encinas, Łojasiewicz exponents and resolution of singularities, *Archiv der Mathematik* **93** (2009), 225–234; arXiv:0903.1932.
- [7] J. Bochnak, M. Coste, and M.-F. Roy, *Real Algebraic Geometry*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) 36, Springer, Berlin, 1998.
- [8] J. F. Bonnans and A. Shapiro, Optimization problems with perturbations: a guided tour, *SIAM Review* **40** (1998), no. 2, 228–264.
- [9] J. F. Bonnans and A. Shapiro, *Perturbation Analysis of Optimization Problems*, Springer Series in Operations Research, Springer, New York, 2000.

- [10] J. Chen, Optimal rate of convergence for finite mixture models, *The Annals of Statistics* **23** (1995), no. 1, 221–233.
- [11] S. Encinas, A. Nobile, and O. Villamayor, On algorithmic equiresolution and stratification of Hilbert schemes, *Proceedings of the London Mathematical Society* (3) **86** (2003), no. 3, 607–648; arXiv:math/0010228.
- [12] W. Fulton, *Intersection Theory*, second edition, *Ergebnisse der Mathematik und ihrer Grenzgebiete* (3) 2, Springer, Berlin, 1998, Chapter 5.
- [13] T. H. Hà, An algebraic theory of Łojasiewicz exponents, preprint arXiv:2602.18410v2, March 14, 2026.
- [14] R. M. Hardt, Semi-algebraic local-triviality in semi-algebraic mappings, *American Journal of Mathematics* **102** (1980), no. 2, 291–302.
- [15] D. Maclagan and B. Sturmfels, *Introduction to Tropical Geometry*, Graduate Studies in Mathematics 161, American Mathematical Society, Providence, RI, 2015.
- [16] R. T. Rockafellar and R. J.-B. Wets, *Variational Analysis*, Grundlehren der mathematischen Wissenschaften 317, Springer, Berlin, 1998, Chapters 4, 6 and 13.
- [17] The Stacks Project Authors, *The Stacks Project*, Tags 01ZM (finite-presentation limits), 051S (generic freeness), 054K (Chevalley’s theorem), and Section 37.26, Lemma 37.26.4 (geometrically reduced generic fibres), consulted September 20, 2026.
- [18] The Stacks Project Authors, *The Stacks Project*, Tag 062Z, Section 31.20, The normal cone of an immersion, especially Definitions 31.20.1 and 31.20.5 and Lemmas 31.20.3–31.20.4, consulted September 20, 2026.
- [19] P. Jell, C. Scheiderer, and J. Yu, *Real tropicalization and analytification of semialgebraic sets*, *Int. Math. Res. Not. IMRN* **2022** (2022), no. 2, 928–958. DOI: 10.1093/imrn/rnaa112.
- [20] K. Rose and M. L. Telek, *Computing positive tropical varieties and lower bounds on the number of positive roots*, *J. Symbolic Comput.* **132** (2026), 102477. DOI: 10.1016/j.jsc.2025.102477.
- [21] M. Lejeune-Jalabert and B. Teissier, *Clôture intégrale des idéaux et équisingularité*, *Ann. Fac. Sci. Toulouse Math.* (6) **17** (2008), no. 4, 781–859, with an appendix by J.-J. Risler. DOI: 10.5802/afst.1203.